Prime-factor densities: gap spectra, zeta-zero criteria, and generalized Robin inequalities

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Abstract

We study reversals in the density of the kth smallest distinct prime factor and the finer arithmetic error in its consecutive-coefficient ratio. For fixed local laws with $\eta_p^{-1} = p/\nu + \beta + o(1)$, the finite maximal separated reversal count has exact rate $\limsup_k k^{-1} \log\log(3 + N_\eta(k)) = \sup_{H\geq 2} \gamma_H/(H+\nu)$, where γ_H records bounded-gap frequency. This spectrum has a finite phase profile and arithmetic realizations by affine prime sums. A uniform comparison with a full entire-function reference transfers the classical Nicolas error to the actual moving-rank coefficient ratio throughout a class of local laws. Its eventual sign gives RH criteria with exact forced-prime rank shifts and reference constants. Persistent excursions yield positive-area criteria; explicit short-interval sampling hypotheses also give localized reversal-count criteria. Hardy–Littlewood-shaped gap counts give separate conditional results on the last ascent. For consecutive-root polynomials plus a prime, we identify the exact complete correction and prove an unconditional negative sign when the degree is $1 + \lfloor N/(\log N)^j \rfloor$, for each fixed $j \geq 1$. If instead $d(N) \leq CN^\alpha$ for fixed $1/2 \leq \alpha < 1$, an eventual nonpositive complete profile excludes zeta zeros to the right of α; the signed premise remains unproved. A separate application to generalized divisor sums constructs explicit modified least common multiples with a positive normalized logarithm, uniformly for $1 < \kappa \leq 3/2 + 2/(5\log\log M_N)$ at unbounded integer cutoffs. Under RH the same window holds eventually at every integer cutoff; consequently, every fixed $1 < \kappa \leq 3/2$ has arbitrarily large counterexamples to the generalized upper inequality. An RH-conditional harmonic second moment measures excess zero multiplicity. We also prove prime-coordinate and residue-rational inverse theorems and construct distinct laws with identical entire profiles and sharp cubic finite-cutoff defects. The spectral and local compari-

son components have maintained Lean proofs, as does the full prime-coordinate inverse classification. The optimal cutoff distortion equals the least recurring prime gap; its adaptive characterization and full inverse are also Lean-formalized. The residue-rational classification and continuum-cardinality assertion remain ordinary proofs. Classical inputs and unproved arithmetic premises are distinguished; no new prime-gap bound or proof of RH is claimed.

1 Introduction and historical context

Let $d_k(p)$ be the natural density of positive integers whose kth smallest *distinct* prime factor is p. The question whether $p \mapsto d_k(p)$ first rises and then falls is now called Erdős Problem 690. Our subject is the finer structure left after unimodality fails: the number of separated reversals, their possible locations, and the arithmetic information encoded by their growth.

Erdős formulated the question in 1979 [1, p. 75]. The same paper discusses the typical scale $\log\log p_k(n) \sim k$ for prime factors of integers. This concerns where probability mass lies, not the mode of the individual-prime density. Erdős–Tenenbaum [2, Corollary 5] later corrected the earlier assertion about the mode: a maximizing prime satisfies $\log p \sim k/\log k$, rather than $\log p \sim k$. Neither statement counts reversals. The occurrence of a double exponential in classical prime-factor distribution does not by itself supply the theorem below.

Cambie [3, Theorem 5] proved unimodality for $k = 1, 2, 3$ and verified its failure for $4 \le k \le 20$. His Theorem 3 gives superpolynomially many local maxima for the different density $\delta_1(n, m)$ in Erdős Problem 692: the density of integers having exactly one divisor in $\{n+1, \dots, m-1\}$. Its moving variable is m, with n fixed. It is not a reversal count for $d_k(p)$.

Wang–Crapis [4, Theorem 1.1] proved non-unimodality for every $k \ge 4$. Their proof compares a symmetric-polynomial ratio with prime gaps and produces a descent followed by an ascent. We extend the comparison to local laws and make the construction quantitative. The question here is how the maximal number of witnesses grows with k. Its exact rate also requires a converse: a sufficiently late ascent must come from an appropriately short gap.

The first contribution is the exact reversal frequency spectrum, its arithmetic realization, and its inverse late-ascent interpretation. Quantitative bounded-gap results are existing analytic inputs, not new sieve results. The second is a uniform transfer of a classical RH-sensitive error into the actual moving-rank coefficient ratio, preserving its sign at a scale much smaller

than any fixed inverse power of $\log\log x$. The transfer holds throughout an explicit class of local probability laws, rather than only for the ordinary integers: each family contributes its full reference profile and the same centered arithmetic signal.

Nicolas's primorial criterion and its signed error [7, 8] supply the analytic source. The finite generating product and its saddle-point analysis already occur in Erdős–Tenenbaum [2]; the normalized Euler product also appears in mod-Poisson theory [9, 10]. We claim neither the generating function nor the Nicolas spectral expansion as new. The potentially new statements are the exact gap-frequency spectrum and the quantitative comparison with a full theta-centered coefficient reference, together with their stated applications. Priority remains provisional pending independent review.

There is also a separate divisor-sum application. Fan–Kobayashi–Molnar [13, Theorem 6.1 and §8] prove an eventual generalized Robin upper bound under RH for each fixed real exponent $\kappa > 3/2$ and ask about smaller exponents. Section 18 gives unconditional unbounded violations for every $1 < \kappa \leq 3/2$, and a shrinking window above the endpoint. The additional growing-family calculations in Section 16 retain complete arithmetic corrections. Their independent signs and their RH-sensitive conditional implications have different degree ranges. Section 17 states the exact N^{α} zero-free implication and its still-open signed hypothesis. The endpoint/window result and these family applications are potentially new; independent priority assessment remains outstanding.

Sections 4–8 prove the unconditional spectrum. Section 9 treats recurrence and the separate Hardy–Littlewood conditional location result. Sections 10–13 establish the coefficient transfer, pointwise RH criteria and reference capacities. Sections 14–15 treat persistent positive area and the localized signed-count criteria. Sections 16 and 18 give the growing arithmetic families and the generalized-divisor-sum application; Section 17 gives the power-degree criterion. The public repository contains maintained proofs and a provenance ledger. This is an AI-assisted research draft; independent expert review and assessment of novelty remain outstanding.

2 The argument in plain English

Choose a position in an integer's ordered list of prime factors, say the tenth distinct prime factor. For each prime, ask what fraction of sampled integers put that prime in the chosen position. These fractions form a sequence. A reversal means that the sequence drops and later rises again. We count

occurrences separated far enough that they do not reuse the same stretch of the sequence.

Two effects compete. Moving to a larger prime lowers its own chance of dividing the integer. At the same time, the additional smaller primes make it easier to have the required number of preceding factors. An exact formula measures this competition. On the relevant scale, a short gap between consecutive primes favors a rise, whereas a long gap favors a drop.

The useful measuring scale is the logarithm of the logarithm of the prime. On that scale the available rise threshold is approximately the rank divided by the scale. This explains the double-exponential locations in the argument. Eventually even the shortest possible prime gap is too large for a rise, so every fixed rank still has a final decreasing tail.

Many short gaps alone are not enough to count reversals. We must put a long gap before each selected short gap and keep the witnesses distinct. Periodic blocks of composite integers provide barriers that consecutive primes must jump. Grouping short gaps between the barriers and taking alternate occupied groups produces separated witnesses, losing only a fixed factor in their number.

Finally, different short-gap sizes can have different frequencies. Each size contributes its frequency divided by a threshold cost. The greatest contribution is exactly the reversal growth rate. For products of affine prime sums, the number of distinct factors changes that cost, while finitely many exceptional primes change only the initial details.

A second, much finer comparison uses the entire prime-product profile as a reference. The ratio that decides an ascent differs from this reference by a small signed error. We show that this error carries the same RH information as Nicolas's primorial inequality. Under RH it is eventually negative, with a bias from prime squares. If RH is false it has larger excursions of both signs. The rank moves outward with the prime cutoff in this statement; it does not contradict eventual decline at each fixed rank.

Changing the sampled family changes its local factor probabilities and its reference profile. We prove that a quadratic change in the local odds preserves the small RH-sensitive error after this exact reference has been subtracted. Forced primes shift the rank and also change the reference constant; both effects are retained. Thus one independently proved centered inequality in any qualifying family would imply RH.

This determines how the prefix changes the threshold for a possible ascent. An actual ascent also needs an actual prime gap below that threshold. RH controls the fine reference error, while a Hardy–Littlewood twin-count assumption supplies twins near the potential endpoint. Their roles and their

different precision are kept separate throughout.

Two further calculations test what arithmetic structure can actually supply. A polynomial with many prescribed roots makes divisibility by small primes less likely. We calculate its entire correction, rather than just its leading local probability, and prove a complete negative sign when the degree is only logarithmically smaller than the prime cutoff. This range must be distinguished from the much smaller power-sized degrees relevant to excluding zeta zeros.

For generalized divisor sums, increasing selected prime exponents in a least common multiple balances an extra height cost against a smaller local-factor loss. The full balance gives explicit counterexamples in a shrinking parameter window around 3/2. The proof works whether RH is true or false. It answers a separate inequality question; it does not resolve the signed premise needed for an independent proof of RH.

The finer error can also be detected by counting. We choose actual short prime gaps in many separate cells, place a long-gap descent before each, and test a finely spaced range of local laws. If RH fails, its positive error persists across a whole interval. Every selected cell then adds more actual ascents than the full reference predicts. A sufficiently accurate upper bound on this signed count would therefore imply RH. This uses known short-interval bounded gaps at some fixed width; it does not require twin primes, and it does not establish that upper bound.

3 Definitions and main results

All logarithms are natural. Index the primes by $p_0 = 2, p_1 = 3, \ldots$, and write $\mathcal{L}(x) = \log\log x$ for $x > 1$ and $\mathcal{E}(t) = \exp(\exp t)$. Ranks k are positive integers. A fixed *local law* consists of probabilities $0 \le \eta_p \le 1$. Define the finite products

$$M_r(p) = [z^r] \prod_{q<p} (1 - \eta_q + \eta_q z), \qquad D_{\eta,k}(p) = \eta_p M_{k-1}(p), \tag{1}$$

with $M_r = 0$ for $r < 0$. For the ordinary law $\eta_p = 1/p$, the Chinese remainder theorem identifies $D_{\eta,k}$ with d_k. The second canonical law is

$$\eta_2^{\mathrm{odd}} = 0, \qquad \eta_p^{\mathrm{odd}} = \frac{p-2}{(p-1)^2} \quad (p > 2). \tag{2}$$

Definition 3.1. A reversal at rank k consists of indices $d < a$ with

$$D_{\eta,k}(p_{d+1}) < D_{\eta,k}(p_d), \qquad D_{\eta,k}(p_a) < D_{\eta,k}(p_{a+1}).$$

A family of m such witnesses is separated if $a_i + 1 < d_j$ for $i < j$. Its maximal attainable size, when finite, is $N_\eta(k)$.

This counts raw-density descent–ascent witnesses, not oscillations of a centered error or a smoothed approximation.

For integers $H \geq 0$, set

$$\begin{aligned} G_H(X) &= \#\{i : p_i \leq X,\ p_{i+1} - p_i \leq H\}, \\ \gamma_H &= \limsup_{\substack{X\to\infty \\ X\in\mathbb{N}}} \frac{\log\log(3 + G_H(X))}{\mathcal{L}(X)}, \\ \Lambda(\nu) &= \sup_{H\geq 2} \frac{\gamma_H}{H+\nu} \quad (\nu > 0). \end{aligned} \tag{3}$$

The 3 keeps the logarithms defined when counts vanish. We have $0 \leq \gamma_H \leq 1$ and $\gamma_0 = \gamma_1 = 0$.

Theorem 3.2 (Exact reversal spectrum)**.** *Suppose $\nu > 0$, $\beta \in \mathbb{R}$, and, along sufficiently large primes,*

$$0 < \eta_p < 1, \qquad \eta_p^{-1} = p/\nu + \beta + \rho(p), \qquad \rho(p) \longrightarrow 0. \tag{4}$$

Every fixed-rank sequence $D_{\eta,k}$ is eventually nonincreasing, its reversal maximum is finite and attained, and

$$\limsup_{k\to\infty} \frac{\log\log(3 + N_\eta(k))}{k} = \Lambda(\nu) > 0. \tag{5}$$

The law, its parameters and its exceptional coordinates are fixed before k tends to infinity.

The weaker estimate $\eta_p = \nu/p + O(p^{-2})$ alone does not supply the vanishing difference of reciprocal errors used in this theorem.

Theorem 3.3 (Canonical bounds)**.** *For either canonical law and every $\varepsilon > 0$, for all sufficiently large k,*

$$\mathcal{E}(k/602) \leq N_\eta(k) \leq \mathcal{E}((1/3 + \varepsilon)k).$$

Each fixed-rank canonical sequence is eventually strictly decreasing.

The positive constant uses a conservative quantitative width-600 input; it is not a claim about the best known numerical prime-gap bound.

Theorem 3.4 (Affine arithmetic realization)**.** *Fix $m \geq 1$, positive integers c_j, and integers d_j, with $c_i d_j \neq c_j d_i$ for $i \neq j$. Put*

$$F(p,q) = \prod_{j=1}^{m} (c_j(p+q) + d_j).$$

Sample all ordered prime pairs $p, q \leq X$, including the diagonal, with denominator $\pi(X)^2$. At every fixed rank k and prime ℓ, the probability that ℓ is the kth smallest distinct prime factor of $|F(p,q)|$ has a limit $D_{F,k}(\ell)$; zero outputs are excluded from the event. The maximal separated reversal count $N_F(k)$ is finite and attained, and

$$\limsup_{k\to\infty} \frac{\log\log(3 + N_F(k))}{k} = \Lambda(m) > 0.$$

The size limit comes first, with family, rank and prime fixed. Outputs are weighted by their prime-pair representations.

The second main result, Theorem 11.1, concerns the ordinary coefficient ratio at rank $r = \lfloor \alpha(\gamma + \log\log\theta(x)) \rfloor$, for each fixed $\alpha > 0$. Its precise full-profile normalization is defined in Section 10. Eventual negativity of that centered ratio is equivalent to RH. This statement is separate from Theorem 3.2, whose hypotheses contain no RH assumption.

4 The exact comparison and uniform estimates

4.1 First differences

For consecutive primes $p < q$ with interior probabilities,

$$M_r(q) = (1 - \eta_p) M_r(p) + \eta_p M_{r-1}(p).$$

Direct subtraction gives

$$D_{\eta,k}(q) - D_{\eta,k}(p) = \eta_p \eta_q \{M_{k-2}(p) - T_\eta(p,q) M_{k-1}(p)\}, \tag{6}$$

where $T_\eta(p,q) = \eta_q^{-1} - \eta_p^{-1} + 1$. When $M_{k-1}(p) > 0$, an ascent is equivalent to $M_{k-2}(p)/M_{k-1}(p) > T_\eta(p,q)$. For the ordinary law, writing $R = M_{k-2}/M_{k-1}$ and $g = q - p$, the exact real-valued gap threshold is $R - 1$: a strict ascent occurs if and only if $g < R - 1$. Equality gives a flat step. This threshold does not constrain the actual gap to be small. The ordinary comparison level is $q - p + 1$; for the odd law at $p > 2$ it is

$$q - p + 1 - \frac{q-p}{(p-2)(q-2)}.$$

4.2 Finite exceptions and symmetric polynomials

Only finitely many coordinates in (4) can be forced ($\eta_p = 1$) or excluded ($\eta_p = 0$). Let f count the forced coordinates. Past them, each contributes a factor z to the generating polynomial; excluded coordinates contribute 1. Retain all interior coordinates, including the exceptional ones, and set

$$w_p = \frac{\eta_p}{1-\eta_p}, \qquad S(p) = \sum_{\substack{q<p \\ 0<\eta_q<1}} w_q.$$

For $k > f$, put $r = k-1-f$. The common positive product of $1-\eta_q$ cancels in the ratio, leaving $E_{r-1}(p)/E_r(p)$, where E_r is the elementary symmetric polynomial of degree r in the finite list of interior odds.

Lemma 4.1. *For at least $r \geq 1$ positive weights, let W_{r-1} be the largest sum of $r-1$ distinct weights. Then*

$$\frac{r}{S} \leq \frac{E_{r-1}}{E_r} \leq \frac{r}{S - W_{r-1}}, \tag{7}$$

and the right denominator is positive.

Proof. The terms of SE_{r-1} which choose a new index total rE_r. The remaining terms are

$$\sum_{|A|=r-1} \left(\sum_{j \in A} w_j \right) \prod_{j \in A} w_j,$$

between zero and $W_{r-1}E_{r-1}$. Rearrangement proves the bounds. At least one positive weight remains after any $r-1$ are removed. □

This specializes to Wang–Crapis [4, Lemmas 3.1–3.2]; the present use also retains the exact forced-rank shift.

Expanding (4) gives $w_p = \nu/p + O(p^{-2})$. The reciprocal-prime estimate and the convergent quadratic tail give

$$S(p) = \nu\mathcal{L}(p-1) + O_\eta(1), \qquad W_{r-1} = O_\eta(1 + \log(r+1)). \tag{8}$$

For the second estimate, dominate all but finitely many odds by C/p and use $p_j \geq j+2$. The finitely many exceptional odds contribute a fixed constant; their order need not be monotone.

Lemma 4.2 (Uniform sign margins)**.** *Fix $0 < u \le v$. Uniformly in the band $uk \le \mathcal{L}(p-1) \le vk$, as $k \to \infty$,*

$$\frac{E_{r-1}(p)}{E_r(p)} = \frac{k}{\nu \mathcal{L}(p-1)} + o(1), \qquad r = k - 1 - f.$$

Gaps at most H give strict ascents throughout the band if $(H+\nu)v < 1$. Gaps at least D give strict descents there if $Du > 2$, for sufficiently large rank and lower prime.

Proof. The prime number theorem ensures more than r interior primes below p uniformly in this band. Equations (7) and (8) have numerator $k + O_\eta(1)$ and denominator $\nu\mathcal{L}(p-1) + O_\eta(\log(k+1))$, bounded below by a positive multiple of k. This proves the uniform ratio statement.

Writing $g = q - p$, the threshold is $g/\nu + 1 + \rho(q) - \rho(p)$. Its error is bounded by twice the supremum of $|\rho|$ beyond p, even for large g. For $g \le H$ the limiting lower ratio is $1/(\nu v) > (H+\nu)/\nu$. For $g \ge D$ the upper ratio is $1/(\nu u) + o(1) < D/\nu$. Both margins are strict. □

Without an upper band endpoint, the same estimates give a one-sided bound: on $\mathcal{L}(p-1) \ge bk$ the ratio is at most $1/(\nu b) + o(1)$, uniformly for fixed $b > 0$.

4.3 Every fixed rank

For fixed $r \ge 1$, the ratio tends to zero by (7)–(8). Consecutive sufficiently large primes have gap at least two, so their threshold is uniformly positive. Equation (6) gives strict descent for supported degrees. If $k \le f$, the tail is zero. If $k = f + 1$, the remaining degree is zero, its predecessor coefficient vanishes, and the same identity again gives descent. Thus every fixed-rank tail is nonincreasing. For the ordinary law, eventual decrease was already noted by Cambie [3, §3]. The extension here retains the forced and excluded coordinates.

Once all ascent indices lie below an integer I, a separated family has size at most I. Attainable sizes form a nonempty bounded subset of $\mathbb{N}$ (zero is attainable), so their maximum is attained. The canonical laws have no forced coordinates and have positive supported tails, giving strict eventual decrease.

5 From short gaps to separated reversals

The following finite argument supplies the counting step and its loss. It turns actual small-gap supply into separated raw reversals. The later RH coefficient correction is not an input to this construction.

Lemma 5.1 (Periodic composite barriers)**.** *Fix integers $D \geq H+1$ and $M = D!$. Let $\mathcal{S}$ be a nonempty finite set of integer starts such that*

$$X \leq n, \quad n+H \leq Y, \quad \#([n, n+H] \cap \{primes\}) \geq 2, \quad X \geq M+2.$$

There are m separated pairs of consecutive-prime gaps, a gap at least D followed by one at most H, all endpoints in $[X, Y]$, with

$$|\mathcal{S}| \leq 2(M+H)(m+1). \tag{9}$$

Proof. Choose consecutive primes $p < q$ in each window, and assign them to $c = \lfloor (p-2)/M \rfloor$. Since $cM + j$ is composite for $2 \leq j \leq D$, and $c \geq 1$,

$$cM + D + 1 \leq p < q \leq (c+1)M + 1.$$

Indeed a prime q beyond the last endpoint but within $H < D$ of p would lie in the next composite block. A short pair cannot cross that barrier.

Starts assigned to cell c lie between $cM+D+1-H$ and $(c+1)M+1$, so there are at most $M+H$ of them. Order the occupied cells $c_0 < \cdots < c_{t-1}$ and choose one pair in each, with lower-prime index u_j. Between chosen primes in two successive occupied cells, a consecutive-prime gap crosses the destination cell's whole composite block. Its length is at least D, and it precedes the chosen short gap there.

Take destination cells $c_2, c_4, \ldots, c_{2m}$, where $m = \lfloor (t-1)/2 \rfloor$. Choose each crossing after the lower prime selected in the preceding occupied cell. The unused occupied cell between successive destinations ensures $a_i + 1 < d_{i+1}$. Every crossing lies between selected endpoints, so remains in $[X, Y]$. Finally $t \leq 2(m+1)$ and $|\mathcal{S}| \leq (M+H)t$, proving (9). □

No uniform distribution among cells is needed. Empty stretches are harmless, since a crossing still exists between selected occupied cells.

Fix $u, v > 0$ with $(H+\nu)v < 1$ and choose an integer $D \geq H+1$ with $D > 2/u$. If all primes from X through $Y+H$ lie in the sign band, the two lemmas imply, at sufficiently large k,

$$G_H(Y) \leq X + B(N_\eta(k)+1), \qquad B = 2(D!+H). \tag{10}$$

At most X lower primes are deleted below X; each retained short gap supplies a window starting at its lower prime. The upper endpoint is $Y + H$, since a counted gap may end above Y. All constants are fixed before k grows.

Corollary 5.2 (Positive-power transfer)**.** *Suppose that for fixed H and $\delta > 0$, for every sufficiently large X at least X^δ integer starts in $[X, 2X]$ have two primes in their width-H windows. For every $0 < a < 1/(H + \nu)$,*

$$N_\eta(k) \geq \mathcal{E}(ak) \quad \textit{for all sufficiently large } k.$$

Proof. Choose $a < b < 1/(H + \nu)$ and $X = \lceil \mathcal{E}(bk) \rceil$. Every endpoint in $[X, 2X + H]$ has $\mathcal{L}(p - 1) = bk + o(1)$. Fixed $u < b < v < 1/(H + \nu)$ therefore provide the sign band. The cell lemma gives $N_\eta(k) + 1 \geq X^\delta/B$. Since $\log(X^\delta/B) = \delta \exp(bk) + O(1)$ and $a < b$, the right side eventually exceeds $1 + \mathcal{E}(ak)$. □

We use the existing quantitative input

$$\#\{n \in [X, 2X] \cap \mathbb{N} : \#([n, n + 600] \cap \{\text{primes}\}) \geq 2\} \geq X^{1/3} \tag{11}$$

for all sufficiently large X. Its maintained declaration is `PrimeGaps.eventually_many_prime_windows_600`. Section 20 records its provenance. Quantitative prime-cluster theory, including Maynard's work [5], is the analytic background. We use this count, not merely infinitude of bounded gaps. With $\nu = 1$ the corollary applies at $a = 1/602 < 1/601$.

6 Proof of the exact spectrum

The two directions preserve the frequency subsequences and each individual gap's cutoff. This matching rate, rather than the presence of a double exponential alone, is the principal unconditional result.

6.1 Lower bound: arbitrary frequency subsequences

Fix H and $0 < a < \gamma_H/(H + \nu)$. Choose

$$0 < g < \gamma_H, \qquad 0 < v_0 < v < \frac{1}{H + \nu}, \qquad a < gv_0,$$

then $0 < u < a$ and $a < b < gv_0$. There are arbitrarily large integers Y with $3 + G_H(Y) > \mathcal{E}(g\mathcal{L}(Y))$. Assign $k = \lceil \mathcal{L}(Y)/v_0 \rceil$. The bounds

$v_0k - v_0 < \mathcal{L}(Y) \le v_0k$ imply $3 + G_H(Y) > \mathcal{E}(bk)$ at arbitrarily large assigned ranks.

Take $X = \lceil \mathcal{E}(uk) \rceil + 1$. The primes between X and $Y + H$ lie in a band with positive lower coefficient and upper coefficient v. Equation (10) applies. If $N_\eta(k) < \mathcal{E}(ak)$, then

$$3 + G_H(Y) \le 3 + X + B(1 + \mathcal{E}(ak)) < \mathcal{E}(bk)$$

eventually, a contradiction. Therefore $N_\eta(k) \ge \mathcal{E}(ak)$ for arbitrarily large k. Supremizing over H proves the lower bound $\Lambda(\nu)$ in (5). The argument keeps the actual frequency subsequence; it does not replace it by an all-large-X hypothesis.

6.2 Upper bound: each gap retains its own cutoff

For fixed integer $g \ge 2$ and $\varepsilon > 0$, the one-sided ratio bound implies that an ascent across a gap of size g must satisfy, apart from fixed initial exceptions,

$$\mathcal{L}(p - 1) < \left(\frac{1}{g + \nu} + \varepsilon \right) k. \tag{12}$$

The same estimate excludes all gaps greater than J above $\mathcal{L}(p - 1) = uk$ if $1/(J + 1 + \nu) < u$. In both statements, the reciprocal error is uniformly small beyond the lower prime.

Consequently

$$N_\eta(k) \le T_k + \sum_{2 \le g \le J} G_g(U_g(k)), \tag{13}$$

where we may take

$$T_k = \lceil \mathcal{E}(uk) \rceil + O(1), \qquad U_g(k) = \left\lceil \mathcal{E}\left(\left(\frac{1}{g + \nu} + \varepsilon \right) k \right) \right\rceil + 1.$$

Each witness has a distinct ascent index. There are at most T_k initial ones; all others have an actual gap $g \le J$ and are counted in that gap's term. Overcounting by the cumulative G_g only strengthens the upper bound.

Let $A > \Lambda(\nu)$. Choose $0 < u < A$ and then J as above. For each fixed g and $\delta > 0$, eventually

$$\log\log(3 + G_g(U)) \le (\gamma_g + \delta)\mathcal{L}(U).$$

Since $\gamma_g/(g + \nu) \le \Lambda(\nu)$, choose $\varepsilon, \delta > 0$ small enough that

$$(\gamma_g + \delta)\left(\frac{1}{g + \nu} + \varepsilon \right) < A \quad (2 \le g \le J).$$

There are only finitely many such comparisons. Rounding changes their double-logarithmic cutoffs by $o(1)$. Each term of (13), including T_k, has rate strictly below A. A fixed finite sum has rate at most the greatest term rate. Letting A decrease to $\Lambda(\nu)$ proves the matching bound.

The input (11) implies $\gamma_{600} = 1$. A given consecutive pair belongs to at most 601 integer-start windows of width 600, so the associated cumulative gap count has a positive-power lower bound along large cutoffs. The trivial bound $G_H(X) \le X$ gives the reverse inequality on its exponent. Hence $\Lambda(\nu) \ge 1/(600+\nu) > 0$. Together with the fixed-rank proof this completes Theorem 3.2.

For the canonical upper bound use the minimum eventual gap, two, and $\nu = 1$ in (12). Reducing ε slightly absorbs the rounding, leaving all ascents below $\mathcal{E}((1/3+\varepsilon)k)$. Their number bounds $N_\eta(k)$. Together with Corollary 5.2, this proves Theorem 3.3.

7 A finite phase profile

Theorem 7.1. *Let H_* be the least $H \ge 2$ for which $\gamma_H = 1$. This integer exists, and for every fixed $\nu > 0$,*

$$\Lambda(\nu) = \max_{2 \le H \le H_*} \frac{\gamma_H}{H+\nu}.$$

For all sufficiently large ν,

$$\Lambda(\nu) = \frac{1}{H_*+\nu}, \qquad \frac{1}{\Lambda(\nu)} - \nu = H_*. \tag{14}$$

Proof. Existence follows from $\gamma_{600} = 1$. If $H > H_*$, then $\gamma_H/(H+\nu) \le 1/(H+\nu) < 1/(H_*+\nu)$. For each $H < H_*$ put $\gamma = \gamma_H < 1$. With positive denominators, the inequality $\gamma/(H+\nu) < 1/(H_*+\nu)$ is equivalent to $(1-\gamma)\nu > \gamma H_* - H$. It holds beyond a finite threshold. Take the greatest threshold over the finitely many smaller classes. □

Here full frequency means only $\gamma_H = 1$. It need not imply positive relative density or a fixed power lower bound at every large cutoff. The theorem identifies a finite maximum without determining its entries numerically. The rate is an upper envelope of the gap-frequency data; the result recovers H_*, not every individual value of γ_H.

8 Arithmetic realization by affine prime sums

We prove Theorem 3.4 without discarding exceptional primes.

8.1 Fixed-modulus limits and independence

For fixed Q, the prime number theorem in arithmetic progressions gives

$$\frac{\#\{p \le X : p \text{ prime},\ p \equiv a \pmod{Q}\}}{\pi(X)} \longrightarrow \frac{1}{\varphi(Q)} \quad ((a,Q)=1).$$

Nonunit prime classes contain only primes dividing Q, so their normalized contribution vanishes. Full Cartesian sampling of the ordered pairs makes each fixed pair of unit classes have limiting probability $\varphi(Q)^{-2}$.

Take Q to be the product of any fixed finite set of primes. The Chinese remainder theorem identifies the unit-pair space with the product of the local unit-pair spaces. Divisibility of F at a prime depends only on the local pair, so the limiting divisibility indicators are independent, with exact law

$$\eta_\ell = \frac{\#\{(u,v) \in (\mathbb{F}_\ell^\times)^2 : \prod_j (c_j(u+v)+d_j) = 0\}}{(\ell-1)^2}. \tag{15}$$

For fixed k, ℓ the rank event uses only the primes at most ℓ. Summing their joint atoms gives (1).

If $F(p,q)=0$, then some $p+q = -d_j/c_j \le |d_j| \le \sum_j |d_j|$. Only finitely many such pairs exist, independently of X. Their normalized contribution is zero. The diagonal was included throughout; its $\pi(X)$ pairs are also negligible against $\pi(X)^2$. No modulus or rank was allowed to grow with X.

8.2 Exact local counts

Set $T_\ell = \{t \in \mathbb{F}_\ell : \prod_j (c_j t + d_j) = 0\}$. There are $\ell - 1$ unit pairs with sum zero, and $\ell - 2$ with any specified nonzero sum. Thus at every prime,

$$\eta_\ell = \frac{(\ell-2)|T_\ell| + \mathbf{1}_{0 \in T_\ell}}{(\ell-1)^2}. \tag{16}$$

If a factor is identically zero modulo ℓ, $T_\ell = \mathbb{F}_\ell$ and $\eta_\ell = 1$. An empty root set gives zero, and coincident roots are counted once. The formula covers all coefficient, shift, determinant, and small-prime exceptions.

Outside the finite set dividing the nonzero coefficients, nonzero shifts, or determinants $c_i d_j - c_j d_i$, there are m distinct roots. At most one shift is zero; let z_0 be one if it is and zero otherwise. Then

$$\eta_\ell = \frac{m(\ell-2)+z_0}{(\ell-1)^2}, \qquad \eta_\ell^{-1} = \frac{\ell}{m} - \frac{z_0}{m^2} + O_F(\ell^{-1}). \tag{17}$$

The law satisfies (4) with $\nu = m$. Theorem 3.2 proves the arithmetic theorem.

The simplest instances illustrate the exceptional coordinate:

$F(p,q)$	η_2	$\eta_\ell \quad (\ell > 2)$
$p+q$	1	$1/(\ell-1)$
$p+q+1$	0	$(\ell-2)/(\ell-1)^2$
$2(p+q)-1$	0	$(\ell-2)/(\ell-1)^2$.

The forced factor 2 in $p+q$ shifts subsequent ranks by one. Both odd examples have exactly the canonical odd law. All three have rate $\Lambda(1)$, despite different finite-rank behavior. Equations (5) and (14) also show how sufficiently large fixed numbers of distinct affine factors recover H_*. There is no uniformity claim when the family varies with the rank. The arithmetic realization uses fixed-modulus prime progression limits. Section 12 supplies the separate full-profile coefficient theorem that transfers the RH criterion to every one of these fixed families, with dimension $\nu = m$ and exact forced-coordinate rank shifts.

9 Late ascents and bounded prime gaps

For a canonical law, a *late ascent at coefficient* b is an ascent across consecutive primes $p < q$ with $\mathcal{L}(p-1) \geq bk$. Unbounded supply requires existence beyond every prescribed lower bound on both k and p.

Theorem 9.1 (Recurrence equivalence)**.** *Let $H \geq 1$ be an integer and $1/(H+2) < b < 1/(H+1)$. Gaps at most H recur arbitrarily far out if and only if for every K, Q there exist $k \geq K$ and consecutive primes $p < q$, with $p \geq Q$ and $\mathcal{L}(p-1) \geq bk$, at which both canonical densities ascend.*

Proof. Choose $b < a < 1/(H+1)$. To a sufficiently late gap at most H assign $k = \lfloor \mathcal{L}(p-1)/a \rfloor$. Then $k \to \infty$ with p, $\mathcal{L}(p-1) \geq ak \geq bk$, and $\mathcal{L}(p-1)/k \to a$. A small fixed band around a in Lemma 4.2 gives both ascents.

Conversely, $b > 1/(H+2)$ and the one-sided ratio bound exclude gaps at least $H+1$ from sufficiently late ascents. In the odd law the correction to the ordinary threshold has absolute value at most $1/(p-2)$, absorbed by the fixed strict margin. Unbounded lower primes therefore give recurring gaps at most H. □

Corollary 9.2 (Critical location)**.** *Suppose gaps at most $h \geq 1$ recur arbitrarily far out and every sufficiently late consecutive gap is at least h. The critical canonical late-ascent coefficient is $1/(h+1)$: every smaller positive coefficient has unbounded simultaneous ascent supply; every larger coefficient forces both densities to descend for all sufficiently large ranks and lower primes in that region. No assertion at equality is made.*

Proof. Use the forward rank assignment on gaps at most h and the one-sided upper ratio bound on gaps at least h. □

For every fixed $b > 1/5$, an ascent in either canonical law at sufficiently large k, p with $\mathcal{L}(p-1) \geq bk$ must cross a gap strictly below four. Gaps between odd primes are positive even integers, so it must be two. An independently proved unbounded supply would therefore imply infinitely many twin primes. That supply is not established here. A large finite ascent certifies a finite prime pair, not recurrence.

9.1 Infinitude, frequency and the last ascent

For either canonical family, let P_k denote the prime starting its last ascent, which exists for all sufficiently large k. If h is the least recurring prime gap, Corollary 9.2 and the upper cutoff imply

$$\limsup_{k\to\infty} \frac{\mathcal{L}(P_k)}{k} = \frac{1}{h+1}.$$

Indeed every smaller coefficient occurs at unbounded ranks and endpoints, whereas every larger coefficient is eventually excluded. Bounded-gap infinitude at width H therefore gives the lower location coefficient $1/(H+1)$. For example, taking width 186 as an input gives a coefficient at least $1/187$; it does not identify the least recurring gap. Twin-prime infinitude would give the location limsup exactly $1/3$. These implications concern location. A frequency hypothesis is needed for the corresponding reversal count.

Theorem 9.3 (Conditional last-ascent location)**.** *For either canonical law, let X_k be the largest supported prime p at which a hypothetical twin step satisfies the strict ascent criterion. Suppose that for some $C > 0$,*

$$\pi_2(X) \sim CX/(\log X)^2, \qquad \pi_2(X) = \#\{p \leq X : p,\ p+2 \text{ are prime}\}.$$

Then, for every sufficiently large k, P_k is exactly the last lower twin member at or below X_k, and

$$\frac{P_k}{X_k} \longrightarrow 1, \qquad \frac{\mathcal{L}(P_k)}{k} \longrightarrow \frac{1}{3}.$$

The same hypothesis and the count transfer give $\log\log(3+N_\eta(k))/k \to 1/3$ *for the canonical reversal counts.*

Proof. At a supported prefix, adding a positive weight decreases e_{r-1}/e_r by the elementary symmetric log-concavity inequality. The ordinary hypothetical twin threshold is 3; the odd threshold is $3-2/(p(p-2))$, which increases with $p > 2$. Thus the supported eligible primes form an initial segment. The uniform ratio estimates and Bertrand's theorem show that its last prime X_k exists for large k and $\mathcal{L}(X_k)/k \to 1/3$.

For every fixed $\varepsilon \in (0,1)$, subtracting the two assumed count asymptotics gives

$$\pi_2(X) - \pi_2((1-\varepsilon)X) \sim C\varepsilon X/(\log X)^2 > 0.$$

First use dyadic coverage in a fixed scale band strictly between 1/5 and 1/3. It supplies a twin ascent whose supported prefix serves as an anchor. Every later ascent must be a twin by the 1/5 exclusion, and lies at most at X_k. Conversely the last twin at or below X_k follows the anchor, has positive support and satisfies the strict criterion. This proves the exact identification. Now the displayed proportional coverage gives $(1-\varepsilon)X_k < P_k \le X_k$ eventually, for each fixed ε, and hence both location limits.

The global twin asymptotic also supplies a positive-power number of width-two windows in every sufficiently large dyadic interval. Corollary 5.2 for every $a < 1/3$, together with the canonical upper bound, proves the count limit. □

The usual Hardy–Littlewood twin conjecture supplies a particular positive constant C; the proof uses no value of that constant. The fixed-tuple asymptotic and the stronger uniform tuple assumptions used for short-interval distribution should be distinguished [14, (1) and Theorem 1]. Bare twin-prime infinitude does not supply the proportional coverage used here. Nor does the displayed asymptotic give a polylogarithmic or square-root bound for $X_k - P_k$ along these particular cutoffs. A limit on $\mathcal{L}(P_k)/k$ also does not imply $P_k/\mathcal{E}(k/3) \to 1$: a constant shift in the inner exponent already changes the original endpoint by a large multiplicative factor.

10 Prime-product errors and the full coefficient reference

We first develop the finer comparison for the ordinary integer law. Section 12 then extends the product and coefficient criteria to qualifying local laws. The

later arithmetic capacity, area and localized-count results retain their stated ordinary-law scope. We keep the arithmetic cutoff x and the rank r distinct: x tends to infinity, while r is comparable to $\log\log x$. All prime products in these sections include $p \le x$. For the prefix before an ascent prime p, the corresponding cutoff is $x = p - 1$.

10.1 Classical analytic quantities

Write

$$\theta(x) = \sum_{p \le x} \log p, \qquad \psi(x) = \sum_{p^a \le x} \log p, \qquad \kappa(t) = \frac{1/\log t + 1/(\log t)^2}{t^2}.$$

For sufficiently large x, set

$$\begin{aligned} J(x) &= \int_x^\infty (\psi(t) - t)\kappa(t)\, dt, \\ K(x) &= \int_x^\infty (\theta(t) - t)\kappa(t)\, dt, \\ P(x) &= J(x) - K(x) = \int_x^\infty (\psi(t) - \theta(t))\kappa(t)\, dt, \\ f(x) &= e^\gamma \log\theta(x) \prod_{p \le x} (1 - 1/p), \qquad F(x) = \log f(x). \end{aligned} \tag{18}$$

The quantitative prime number theorem gives convergence of these integrals. These are Nicolas's quantities [8, (1.9), (1.15)–(1.19)]; P retains every prime power. Let

$$A(x) = \sum_{p > x} \big(-\log(1 - 1/p) - 1/p\big),$$

and, when $\theta(x) > 1$, let

$$D(x) = \frac{\theta(x) - x}{x \log x} - \big(\log\log\theta(x) - \log\log x\big).$$

Partial summation and the convergent logarithmic prime-power series give the exact identity

$$F(x) = K(x) - D(x) + A(x) = J(x) - P(x) - D(x) + A(x). \tag{19}$$

Here $A(x) \ge 0$, $A(x) = O(1/x)$, and $D(x) \ge 0$ by concavity of $\log\log t$ on $t > 1$. No sign is assigned to $J(x)$.

We use the following classical facts, with their hypotheses retained. Nicolas's criterion is equivalent to eventual $F(x) < 0$. Under RH,

$$F(x) = -\frac{2 + W(x)}{\sqrt{x}\log x} + O\left(\frac{1}{\sqrt{x}(\log x)^2}\right), \qquad |W(x)| \le \beta_0, \tag{20}$$

where

$$W(x) = \operatorname{Re}\sum_{\rho} \frac{x^{i\operatorname{Im}\rho}}{\rho(1-\rho)}, \qquad \beta_0 = 2 + \gamma - \log(4\pi) < 2.$$

The zeros in this absolutely convergent series are counted with multiplicity. Defining the real projection explicitly matches the maintained formal statement. The expansion and its constant are classical [8, Proposition 2.1]. In the decomposition (19), the zero wave comes from J, while

$$P(x) \sim \frac{2}{\sqrt{x}\log x}$$

holds unconditionally and gives the leading prime-square contribution. Under RH the standard bounds $|\psi(x) - x| + |\theta(x) - x| = O(\sqrt{x}(\log x)^2)$ imply

$$D(x) = O((\log x)^3/x), \qquad |F(x) - K(x)| = O((\log x)^3/x). \tag{21}$$

Indeed the second derivative of $\log\log t$ is $-(\log t + 1)/(t^2(\log t)^2)$; Taylor's integral remainder, with $\theta(x)/x \to 1$, proves the first estimate, and (19) proves the second.

If RH fails, F has excursions of both signs of order x^{-b} for some $0 < b < 1/2$ [7, 8]. Section 11.1 records the zero-dependent version used in the coefficient transfer. In particular, the estimate for D in (21) is never used under the assumption that RH is false.

10.2 An entire profile and two different clocks

Define

$$\begin{aligned} G_x(z) &= \prod_{p \le x} \left(1 + \frac{z}{p-1}\right), & a_r(x) &= [z^r]G_x(z), \\ \mathcal{H}(z) &= \prod_{p} \left(1 + \frac{z}{p-1}\right)(1 - 1/p)^z, & C_r(u) &= [z^r]\mathcal{H}(z)e^{uz}. \end{aligned} \tag{22}$$

The power $(1-1/p)^z$ uses the real logarithm of its positive base. The product for $\mathcal{H}$ converges uniformly on compact subsets of $\mathbb{C}$: the logarithmic first-order terms cancel and the remaining tails are bounded by a constant times

$\sum_p p^{-2}$. It defines an entire function, positive on the nonnegative real axis, with $\mathcal{H}(0) = \mathcal{H}(1) = 1$.

The finite generating product is classical in this problem [2, (1.6)]; the normalized Euler factor also appears in the mod-Poisson description of prime divisibility [9, §4]. Our reference retains all its Taylor coefficients. It is not a finite-order truncation of a saddle-point expansion.

Let $\mathcal{H}_x$ be the finite product in (22), and put

$$B(x) = \log G_x(1), \qquad L(x) = \gamma + \log\log\theta(x). \tag{23}$$

Then $F(x) = L(x) - B(x)$ and, exactly,

$$G_x(z) = \mathcal{H}_x(z)e^{B(x)z}. \tag{24}$$

For positive supported degrees define

$$R_r(x) = \frac{a_{r-1}(x)}{a_r(x)}, \qquad \mathcal{R}_r(u) = \frac{C_{r-1}(u)}{C_r(u)}, \qquad \Delta_r(x) = R_r(x) - \mathcal{R}_r(L(x)).$$

The denominator conditions below ensure that these ratios have their ordinary real-valued meaning.

Theorem 10.1 (Uniform coefficient transfer)**.** *Fix $0 < a \le b < \infty$. There exist X and $K > 0$ such that, for every real $x \ge X$ and integer r with $a \le r/L(x) \le b$, both denominators are positive and*

$$\left|\Delta_r(x) - \frac{r}{L(x)^2}F(x)\right| \le K\left(\frac{|F(x)|}{L(x)^2} + \frac{1}{rx}\right). \tag{25}$$

The constants may depend on a, b, but not on r or x.

Proof. Write $\mathcal{H}(z) = \sum_{j\ge0} h_j z^j$ and

$$Q_r(u) = \frac{r!C_r(u)}{u^r} = \sum_{j=0}^{r} h_j \frac{(r)_j}{u^j}.$$

For $t = r/u$ in a fixed positive compact interval, the falling-factorial estimate

$$\left|\frac{(r)_j}{r^j} - 1 + \frac{j(j-1)}{2r}\right| \le \frac{j^4}{r^2}$$

holds also for $j > r$ if $(r)_j$ is extended by zero. For $j \le r$, apply the first two inclusion-exclusion bounds to $\prod_{i=0}^{j-1}(1 - i/r)$; the other case is immediate.

Cauchy coefficient bounds on a circle of radius larger than the compact interval justify summation with the factor j^4. Hence

$$Q_r(u) = \mathcal{H}(t) - \frac{t^2\mathcal{H}''(t)}{2r} + O(r^{-2}). \tag{26}$$

Since $\mathcal{H}(t) > 0$, the normalized denominators are bounded away from zero. Using $C_r' = C_{r-1}$ and the same expansion for $r-1, r-2$ gives

$$\mathcal{R}_r'(u) = \frac{C_{r-2}(u)}{C_r(u)} - \mathcal{R}_r(u)^2 = -\frac{r}{u^2} + O(r/u^3). \tag{27}$$

The first-order profile terms cancel in this paired difference.

On any fixed larger circle, $\mathcal{H}_x - \mathcal{H} = O(1/x)$ by the quadratic prime tail. To compare ratios without losing the cancellation, use

$$\frac{C_{r-1}(u)}{C_r(u)} = \frac{r}{u}\left(1 + \frac{Q_{r-1}(u) - Q_r(u)}{Q_r(u)}\right).$$

The identity $(r)_j - (r-1)_j = j(r-1)_{j-1}$ supplies a factor $1/u$ in the numerator difference. Replacing $\mathcal{H}$ by $\mathcal{H}_x$ therefore changes this ratio by $O(1/(rx))$, uniformly in the compact rank range. Equation (24) yields

$$R_r(x) = \mathcal{R}_r(B(x)) + O(1/(rx)).$$

Unconditionally $F(x) = o(1)$ by PNT and Mertens's theorem. Integrate (27) between L and $B = L - F$. The main term is rF/L^2; its error is $O(|F|/L^2)$ since r is comparable to L and $F = o(1)$. This proves (25). □

The coefficient expansion belongs to the saddle-point and mod-Poisson tradition [2, 9, 10]. The sign-preserving comparison (25), its theta centering, and its application to the actual ascent ratio are the potentially new statements here. Any fixed error $O(L^{-m})$ would be larger than the RH signal $1/(L\sqrt{x}\log x)$.

11 RH criteria and quantitative excursions

Theorem 11.1 (Moving-rank RH criterion)**.** *Fix $\alpha > 0$ and set $r(x) = \lfloor \alpha L(x) \rfloor$. Then RH is equivalent to*

$$R_{r(x)}(x) < \mathcal{R}_{r(x)}(L(x)) \quad \textit{for every sufficiently large } x.$$

Under RH, more precisely,

$$\frac{L(x)^2}{r(x)}\sqrt{x}\log x\,\Delta_{r(x)}(x) = -2 - W(x) + o(1). \tag{28}$$

If RH is false, there exists one $b \in (0, 1/2)$ such that, for every fixed $\alpha > 0$,

$$\limsup_{x\to\infty} x^b L(x)\Delta_{r(x)}(x) = +\infty, \qquad \liminf_{x\to\infty} x^b L(x)\Delta_{r(x)}(x) = -\infty. \tag{29}$$

The witnessing endpoints may depend on α and on the sign.

Proof. The floor rank is eventually in a fixed positive compact band. Under RH, substitute (20) into (25). After multiplication by $(L^2/r)\sqrt{x}\log x$, the two errors are respectively $O(1/L)$ and $O((\log x)/\sqrt{x})$. Both vanish, giving (28). Since $2 - \beta_0 > 0$, the sign is eventually negative.

Under false RH, Theorem 11.2 below supplies b and excursions of F with arbitrarily large amplitudes. In a band $a \le r/L \le b_1$, (25) gives, eventually,

$$\begin{aligned} F \ge 0 &\implies L\Delta_r \ge (a/2)F - K/(ax), \\ F \le 0 &\implies L\Delta_r \le (a/2)F + K/(ax). \end{aligned}$$

Here b_1 denotes the fixed band endpoint, not the excursion exponent. For the floor rank one may take $a = \alpha/2$. Since $x^{-1} = o(x^{-b})$, both excursions transfer with arbitrarily large amplitudes. This proves (29) and the reverse implication. □

The equality (28) is an ordinary corollary of the compiled linearization and spectral estimates. The maintained criterion and the quantitative false-RH consumer are separate named declarations; the ledger distinguishes these verification scopes.

11.1 The corrected logarithm at power-profile peaks

Theorem 11.2 (Zero-dependent excursions)**.** *Suppose $\zeta(\rho) = 0$ with $1/2 < \operatorname{Re}\rho < 1$. For every*

$$1 - \operatorname{Re}\rho < b \le 1/2$$

and each $U \in \{J, K, F\}$,

$$\limsup_{x\to\infty} x^b U(x) = +\infty, \qquad \liminf_{x\to\infty} x^b U(x) = -\infty.$$

Consequently, false RH supplies a common exponent $0 < b < 1/2$.

Proof. We give the analytic mechanism and the endpoint correction explicitly. For $a = \log 3$, initially in $\operatorname{Re} s > 0$, put

$$\widehat{J}(s) = \int_a^\infty J(e^u)e^{-su}\,du, \qquad T(z) = \int_3^\infty (\psi(t) - t)t^{-z-1}\,dt.$$

The Mellin identity is $T(z) = -\zeta'(z)/(z\zeta(z)) - 1/(z-1)$ minus the entire contribution of the finite interval from 1 to 3. Integration by parts gives

$$s\widehat{J}''(s) + 2\widehat{J}'(s) = T'(s+1) - T(s+1) + a^2 J(3)e^{-as}. \tag{30}$$

The cancellation at $z = 1$ and the quantitative PNT extend $\widehat{J}$ across $s = 0$ and along the real interval $s > -1/2$. At $s = \rho - 1$, the double pole from T' cannot cancel with the simple pole from T. Thus $\widehat{J}$ cannot be holomorphic there.

An eventual bound $J(x) \geq -Cx^{-b}$ would make $J(e^u) + Ce^{-bu}$ eventually nonnegative. Its Laplace transform has the preceding nonreal singularity, but is analytic on the real interval $s > -b$. Landau's singularity principle contradicts this: its abscissa of convergence lies between $\operatorname{Re}\rho - 1$ and 0, hence strictly above $-b$. Applying the same argument to $-J$ rules out the opposite bound. This is the classical Landau method; see [11, Theorem 3]. Since $P(x) = O(x^{-1/2}/\log x) = o(x^{-b})$, $K = J - P$ inherits both signs.

For negative excursions, $F \leq K + A$ and $A = O(1/x)$ suffice. For positive excursions one must control D without RH. Fix $A_0 > 0$, and choose an integer local maximum $x = n + 1$, $n \geq 2$, of the sequence $K(n) - A_0 n^{-b}$ between two lower values bracketing a high positive value. Such maxima exist arbitrarily far out because of the two unbounded excursions of K; rounding real endpoints to integers costs $O((\log x)/x) = o(x^{-b})$.

The two adjacent comparisons are

$$K(n) - A_0 n^{-b} \leq K(n+1) - A_0(n+1)^{-b},$$

$$K(n+2) - A_0(n+2)^{-b} \leq K(n+1) - A_0(n+1)^{-b}.$$

On each open unit interval θ is constant. Using $K(v) - K(w) = \int_v^w (\theta(t) - t)\kappa(t)\,dt$ and the possible endpoint jump of at most $\log x$ gives

$$0 < \theta(x) - x \leq (2A_0 + 1)x^{1-b}\log x.$$

Taylor's integral remainder for $\log\log$ and $A(x) \geq 0$ now yield

$$F(x) \geq K(x) - \frac{(2A_0+1)^2}{2}(\log x + 1)x^{-2b}.$$

The loss is $o(x^{-b})$. Choosing A_0 larger than any prescribed amplitude proves the positive excursions of F. Finally, if RH fails, symmetry of the zeros supplies such a ρ; choose any b strictly between $1 - \operatorname{Re}\rho$ and $1/2$. □

The existence of false-RH oscillations of F is classical [7, 8]. The displayed proof records the zero-dependent exponent and the exact nonlinear correction used by the formal coefficient consumer.

11.2 Integral, clock and positive-tilt criteria

Define

$$\mathcal{C}(x) = \exp\bigl(e^{-\gamma} G_x(1)\bigr).$$

The exact identity $\mathcal{C}(x) = \theta(x)^{1/f(x)}$ shows that eventual $\theta(x) < \mathcal{C}(x)$ is another form of Nicolas's criterion. Combining the preceding results gives

$$\text{RH} \iff K(x) < 0 \text{ eventually} \iff \theta(x) < \mathcal{C}(x) \text{ eventually}. \tag{31}$$

Under RH,

$$\frac{\mathcal{C}(x) - \theta(x)}{\sqrt{x}} = 2 + W(x) + o(1). \tag{32}$$

Indeed $\log \mathcal{C} - \log\theta = (1/f - 1)\log\theta$; (20), $\theta(x)/x \to 1$, and $e^t = 1 + t + O(t^2)$ prove (32). This is a change of variables in the classical Nicolas estimate.

For fixed $z > 0$, put

$$T_z(x) = \log\left(\frac{e^{\gamma z}\mathcal{H}(z)(\log\theta(x))^z}{G_x(z)}\right).$$

Then, exactly,

$$T_z(x) = zF(x) + \sum_{p>x}\left\{\log\left(1 + \frac{z}{p-1}\right) - z\log\left(1 + \frac{1}{p-1}\right)\right\}. \tag{33}$$

For fixed z the tail is $O_z(1/x)$. It is nonpositive for $z \geq 1$, and lies between 0 and Cz/x for $0 < z \leq 1$, with an absolute C. Concavity of $\log(1 + zw)$ proves the signs; the upper bound follows from $\log(1 + zw) \leq zw$ and $w - \log(1 + w) \leq w^2/2$. Therefore

$$\text{RH} \iff T_z(x) < 0 \text{ eventually}$$

for each fixed $z > 0$. Under RH a *single* sufficiently large cutoff works simultaneously for all $z > 0$, by splitting at $z = 1$ in (33); the converse follows from $T_1 = F$.

The corresponding positive kernel is

$$\kappa_z(t) = \frac{z}{(t-1)(t-1+z)\log t} + \frac{\log(1+z/(t-1))}{t(\log t)^2},$$
$$I_z(x) = \int_x^\infty (\theta(t)-t)\kappa_z(t)\,dt.$$

For $t \geq e$, $|\kappa_z(t) - z\kappa(t)| \leq 14(z+z^2)/(t^3 \log t)$. Chebyshev's bound gives $I_z = zK + O_z(1/(x\log x))$. It follows that eventual $I_z < 0$ is equivalent to RH and that

$$\sqrt{x}\log x\, I_z(x) = -z(2+W(x)) + o(1)$$

under RH. Under false RH every fixed positive tilt inherits the zero-dependent two-sided integral excursions. These tilt criteria transport the same leading error; they do not supply independent sign information by themselves.

12 RH transfer across qualifying local laws

12.1 Qualification and the exact family constants

Fix an independent local law with probabilities $0 \leq \eta_p \leq 1$. For an arithmetic ensemble, identifying this model with its densities also requires independence of its limiting finite-prime divisibility laws. Marginal probabilities alone do not establish that identification. Put

$$w_p = \begin{cases} \eta_p/(1-\eta_p), & \eta_p < 1, \\ 0, & \eta_p = 1. \end{cases}$$

A forced coordinate is removed from the odds product and retained in the rank shift below. The concrete qualification condition is

$$\nu > 0, \quad D \geq 0, \quad w_p \geq 0, \qquad \left| w_p - \frac{\nu}{p-1} \right| \leq \frac{D}{(p-1)^2} \quad \text{for every prime } p. \tag{34}$$

All parameters and the family are fixed before the cutoff tends to infinity. The condition follows from $\eta_p = \nu/p + O(p^{-2})$ along the primes, including arbitrary finite exceptions. Indeed, eventually $\eta_p \leq 1/2$, so

$$\frac{\eta_p}{1-\eta_p} = \eta_p + \frac{\eta_p^2}{1-\eta_p} = \frac{\nu}{p} + O(p^{-2}) = \frac{\nu}{p-1} + O(p^{-2}).$$

The finite head is absorbed into D. This argument is formalized for an explicit prime-tail bound as well as a bound at every prime. A limiting reciprocal offset is not required for this RH transfer; that stronger hypothesis remains part of the reversal-spectrum theorem.

For integer $N \geq 1$, define

$$G_{\eta,N}(z) = \prod_{p \leq N} (1 + w_p z), \qquad a_{\eta,r}(N) = [z^r] G_{\eta,N}(z),$$
$$\mathcal{H}_\eta(z) = \prod_{p} (1 + w_p z)(1 - 1/p)^{\nu z}, \qquad C_{\eta,r}(u) = [z^r] \mathcal{H}_\eta(z) e^{uz}.$$

The complex powers use the real logarithm of the positive base $1 - 1/p$. Write

$$\kappa_\eta(z) = \log \mathcal{H}_\eta(z), \qquad \Gamma_\eta(z) = e^{\nu z \gamma} \mathcal{H}_\eta(z) \quad (z \geq 0).$$

These are exact convergent constants; exceptional primes remain in the products. If $v_p = 1/(p-1)$, then $w_p \leq (\nu + D) v_p$. The elementary bounds $0 \leq t - \log(1+t) \leq t^2/2$ give

$$|\log(1 + w_p z) - \nu z \log(1 + v_p)| \leq K_\eta(z) v_p^2,$$
$$K_\eta(z) = \frac{z^2(\nu + D)^2}{2} + zD + \frac{\nu z}{2}.$$

Since $\sum_{p > N} v_p^2 \leq 2/N$, we have the explicit bound

$$\left| \kappa_\eta(z) - \sum_{p \leq N} \big(\log(1 + w_p z) - \nu z \log(1 + v_p) \big) \right| \leq \frac{2K_\eta(z)}{N}. \tag{35}$$

Theorem 12.1 (Family product criteria)**.** *Under* (34)*, fix $z > 0$ and set*

$$E_\eta(N, z) = \nu z \big(\gamma + \log\log \theta(N) \big) + \kappa_\eta(z) - \log G_{\eta,N}(z).$$

For $\theta(N) > 1$,

$$|E_\eta(N, z) - \nu z F(N)| \leq 2K_\eta(z)/N.$$

RH is equivalent to each of the following eventual statements:

$$E_\eta(N, z) < 0, \qquad E_\eta(N, z) \leq A/N, \qquad \Gamma_\eta(z) (\log \theta(N))^{\nu z} < G_{\eta,N}(z),$$

where A is any fixed real number.

Proof. The finite identity $F(N) = \gamma + \log\log\theta(N) - \sum_{p\le N}\log(1+v_p)$ and (35) give the error bound. Under RH, the Nicolas negative margin dominates every fixed multiple of $1/N$. If RH fails, its positive integer-cutoff excursions exceed every such multiple. Multiplication by the fixed positive number νz preserves both conclusions. Positivity of all factors turns the logarithmic criterion into the displayed product inequality. These are applications of Nicolas's criterion and error estimates [8, (1.10), Proposition 2.1]. □

There is a broader sufficient qualification for the product theorem. For fixed nonnegative weights and fixed $\nu, z > 0$, it is enough that some real κ and $C \ge 0$ satisfy

$$\left|\kappa - \sum_{p\le N}\big(\log(1+w_p z) - \nu z\log(1+v_p)\big)\right| \le C/N \quad \text{eventually.} \tag{36}$$

The same proof gives the product criterion with constant $\exp(\nu z\gamma+\kappa)$. This qualification contains neither θ nor an RH sign assumption. It is formalized separately. We do not assert that either sufficient condition is a necessary classification of every family that might admit an RH criterion.

12.2 The coefficient ratio and deterministic rank shifts

Let

$$U_\eta(N) = \nu L(N), \qquad R_{\eta,r}(N) = \frac{a_{\eta,r-1}(N)}{a_{\eta,r}(N)}, \qquad \overline{R}_{\eta,r}(u) = \frac{C_{\eta,r-1}(u)}{C_{\eta,r}(u)}.$$

Here $L(N) = \gamma + \log\log\theta(N)$ as before.

Theorem 12.2 (Uniform family coefficient transfer)**.** *Assume* (34). *On every fixed positive compact band for $r/U_\eta(N)$, both denominators are positive for large N, r, and for a constant K depending on the law and band,*

$$\left|R_{\eta,r}(N) - \overline{R}_{\eta,r}(U_\eta(N)) - \frac{r}{U_\eta(N)^2}\nu F(N)\right| \le K\left(\frac{|\nu F(N)|}{U_\eta(N)^2} + \frac{1}{rN}\right). \tag{37}$$

For every fixed $\alpha > 0$, put $r(N) = \lfloor \alpha U_\eta(N)\rfloor$. Each of the eventual inequalities

$$R_{\eta,r(N)}(N) < \overline{R}_{\eta,r(N)}(U_\eta(N)), \qquad R_{\eta,r(N)}(N) \le \overline{R}_{\eta,r(N)}(U_\eta(N))$$

is equivalent to RH.

Proof. For $|z| \le R$, the normalized local factor differs from one by at most $K_R^{\mathbb{C}} v_p^2$, where one may take

$$K_R^{\mathbb{C}} = \big((D+\nu/2)R + \nu^2 R^2 e^{\nu R}\big) e^{\nu R}.$$

The product therefore defines an entire function and its finite prefix approximates it uniformly on each closed disk with error $O_R(N^{-1})$. The proof uses product differences, so zeros of individual factors on a disk cause no division problem. On $0 \le t \le b$,

$$\mathcal{H}_\eta(t) \ge \exp\left(-K_\eta(b)\sum_p v_p^2\right) > 0.$$

Cauchy's estimate therefore bounds the coefficient of degree j in the prefix error by $O_R(N^{-1}R^{-j})$.

Apply the factorial-polynomial argument of Section 10 to this entire profile. If $h_j = [z^j]\mathcal{H}_\eta(z)$, its normalized coefficient polynomial is

$$P_r(t) = \sum_{j=0}^{r} h_j t^j \frac{(r)_j}{r^j}.$$

The entire coefficient series and its derivatives give uniform approximations to $\mathcal{H}_\eta$ and $\mathcal{H}_\eta'$ on a compact interval. The displayed positive lower bound controls the denominator. The exact identity

$$\overline{R}_{\eta,r}(u) = t - \frac{t^2}{r}\frac{P_r'(t)}{P_r(t)}, \qquad t = r/u,$$

then gives a paired prefix/reference error $O((rN)^{-1})$ and reference slope $-r/u^2 + O(u^{-2})$. These are the same general polynomial estimates used for Theorem 10.1; their hypotheses have now been proved for $\mathcal{H}_\eta$.

The actual finite product is restored at $B_\eta(N) = \nu\sum_{p\le N}\log(1+v_p)$, and $U_\eta(N) - B_\eta(N) = \nu F(N)$ exactly. Integrating the slope over the interval between these two clocks proves (37). The floor-rank band has positive lower and upper bounds. The negative RH margin and the positive false-RH excursions dominate its inverse-cutoff error, giving both asserted equivalences. □

To interpret the ratio arithmetically, put

$$f_N = \#\{p \le N : \eta_p = 1\}, \qquad M_{\eta,N}(j) = [z^j]\prod_{p\le N}(1-\eta_p+\eta_p z).$$

For $r \geq 1$ the exact identity, with total division at a zero denominator, is

$$\frac{M_{\eta,N}(f_N + r - 1)}{M_{\eta,N}(f_N + r)} = R_{\eta,r}(N). \tag{38}$$

Indeed, replace each forced probability by zero, shift by its exact count, and factor out the strictly positive survival product of the remaining coordinates. Excluded coordinates contribute zero odds and no rank shift. Both the identity and its two RH consumers are formalized. At a prime prefix $N = p - 1$, the corresponding distinct-factor rank is $k = f_N + r + 1$.

Every reciprocal-smooth law qualifies by its proved odds estimate. Consequently every fixed affine prime-pair family in Section 8 has these product and mass-ratio RH criteria, with $\nu = m$ and its exact residue probabilities. No bounded-gap or short-interval hypothesis is needed for these equivalences. Arithmetic interpretation still takes the input-size limit first, with the finite prime prefix fixed. Sparse support families require a separate qualification check; the affine constructor makes no assertion about them.

12.3 Reference-scale constants and combinations of families

The same entire-profile estimates give a locally unique positive-level reference crossing with

$$\overline{R}_{\eta,r}(u_{r,s}) = s, \qquad u_{r,s} = \frac{r}{s} - \frac{\mathcal{H}'_\eta(s)}{\mathcal{H}_\eta(s)} + O_{\eta,s}(r^{-1}).$$

For fixed $g \geq 0$, take $s = 1 + g/\nu$ and $T_{\eta,r,g} = \exp(\exp(u_{r,s}/\nu - \gamma))$. Then

$$\log\log T_{\eta,r,g} = \frac{r}{g+\nu} - \gamma - \frac{1}{\nu}\frac{\mathcal{H}'_\eta(1+g/\nu)}{\mathcal{H}_\eta(1+g/\nu)} + O_{\eta,g}(r^{-1}). \tag{39}$$

This generalizes the reference constant, including every finite exception in $\mathcal{H}_\eta$. In a density application, the free rank r must still be replaced by $k - f_N - 1$. The theorem describes the reference crossing. Identifying the actual last ascent additionally requires the stated arithmetic supply and sufficient precision of the actual local gap threshold.

Finally, for a fixed finite collection of qualifying laws and fixed real coefficients a_i, let $z_i \geq 0$ and $\lambda = \sum_i a_i \nu_i z_i$. The exact family bounds imply

$$\left| \sum_i a_i E_{\eta_i}(N, z_i) - \lambda F(N) \right| \leq \frac{1}{N} \sum_i |a_i|\, 2K_{\eta_i}(z_i).$$

If $\lambda > 0$, eventual negativity of this combination is another RH criterion. If $\lambda = 0$, only the inverse-cutoff remainder remains. Thus fixed finite family averaging retains one common arithmetic signal. An infinite or cutoff-dependent collection needs a separate uniform summability argument; no such transfer is asserted here.

13 Reference crossings and RH ascent capacities

The preceding centered criterion concerns $R_r - \mathcal{R}_r(L)$. A raw ascent instead concerns $R_r - (g+1)$ for an actual gap g. We now keep both quantities in the comparison.

Theorem 13.1 (Sharp reference crossing)**.** *For every fixed $s > 0$ there are $A, C > 0$ and r_0 such that, for $r \geq r_0$, the equation $\mathcal{R}_r(u) = s$ has a unique solution $u_{r,s}$ in $[r/s - A, r/s + A]$, and*

$$u_{r,s} = \frac{r}{s} - \frac{\mathcal{H}'(s)}{\mathcal{H}(s)} + O_s(1/r). \tag{40}$$

Proof. Expanding the adjacent normalized coefficients as in (26) gives, for $t = r/u$ in a compact neighborhood of s,

$$\mathcal{R}_r(u) = t - \frac{t^2}{r}\frac{\mathcal{H}'(t)}{\mathcal{H}(t)} + O(r^{-2}).$$

Its values on either side of r/s at a sufficiently large fixed distance straddle s. The derivative (27) is strictly negative throughout that interval, giving existence and uniqueness. Substitute $u = r/s + v$, with bounded v, into the last display to obtain (40). □

For $s = h + 1$, put $X_{r,h} = \mathcal{E}(u_{r,h+1} - \gamma)$. The reference crossing admits a local spatial interpretation. Fix a comparison constant $D_0 \geq 1$. In the window

$$B(x) \in [r/(h+1) - A, r/(h+1) + A], \qquad \mathcal{C}(x) \leq D_0 x, \qquad X_{r,h} \leq D_0 x,$$

the maintained coefficient comparison gives, for sufficiently large r, x and a constant K depending on the fixed parameters,

$$\begin{aligned} X_{r,h} - \mathcal{C}(x) > K \log x &\implies R_r(x) > h + 1, \\ \mathcal{C}(x) - X_{r,h} > K \log x &\implies R_r(x) < h + 1. \end{aligned} \tag{41}$$

Indeed $R_r(x) - \mathcal{R}_r(B(x)) = O(1/(rx))$, whereas $|\mathcal{R}_r'| \gg 1/r$ in the window. The uncertainty in B is $O(1/x)$; the derivative of $\mathcal{E}(u-\gamma)$ is $\mathcal{E}(u-\gamma)\log\mathcal{E}(u-\gamma)$, giving the stated spatial margin. For an actual gap h, these are ascent/descent tests. Equation (32) locates their RH transition in the θ coordinate, with displacement $(2+W(x))\sqrt{x}+o(\sqrt{x})$.

Theorem 13.2 (RH endpoint and reversal-count envelope)**.** *Assume RH. Let P_k be the lower prime of the last ordinary ascent and $N_k = N_{\mathrm{ordinary}}(k)$. For all sufficiently large k, both are attained, and with $r = k-1$,*

$$\theta(P_k - 1) < \mathcal{E}(u_{r,3} - \gamma). \tag{42}$$

For every $\varepsilon > 0$, eventually

$$(1-\varepsilon)N_k \log N_k \le \theta(P_k - 1), \qquad N_k \le (1+\varepsilon)\frac{\mathcal{E}(u_{r,3}-\gamma)}{\log\mathcal{E}(u_{r,3}-\gamma)}.$$

There is a constant C such that

$$\log\log N_k \le \frac{k-1}{3} - \gamma - \frac{\mathcal{H}'(3)}{\mathcal{H}(3)} + \frac{C}{k-1} \tag{43}$$

for all sufficiently large k.

Proof. The unconditional lower reversal bound supplies a last ascent whose endpoint is at least double-exponential in a positive multiple of k. The unconditional upper cutoff puts it below $\mathcal{E}((1/3+\varepsilon)k)$ for every fixed $\varepsilon > 0$. PNT therefore places $r/L(P_k - 1)$ in a fixed positive compact band. Under RH, Theorem 10.1 and (20) give $R_r(P_k - 1) < \mathcal{R}_r(L(P_k - 1))$ uniformly in that band. The actual ascent requires $R_r > g + 1 \ge 3$.

Near $u_{r,3}$, the reference is strictly decreasing. Away from its fixed neighborhood the uniform leading expansion separates its value from 3. These two comparisons imply $L(P_k - 1) < u_{r,3}$, which is (42). Every separated witness uses a distinct ascent index, so $N_k \le \pi(P_k)$. PNT yields the first two count inequalities. Equivalently one can count all primes with $\theta(p-1)$ below the reference endpoint; this monotone capacity has asymptotic $T/\log T$ as $T \to \infty$.

Taking two logarithms of that capacity gives $u_{r,3} - \gamma + o(1/r)$: the correction from division by $\log T$ is negligible on this scale, and the multiplicative PNT error is bounded eventually. Equation (40) now proves (43). □

The same comparison keeps the actual gap of each ascent: in a fixed positive compact rank band, RH implies

$$g_i + 1 < \mathcal{R}_{k-1}(L(p_i - 1)).$$

Thus, on any sufficiently late finite band of prime indices, the number of ascents is at most the number of indices satisfying this reference inequality with their *actual* g_i. Adding the number of earlier primes bounds every reversal family. Combining this upper count with the unconditional reversal supply forces many reference-eligible actual gaps, but does not lower their maximum width without an additional distribution argument.

Neither the square-root clock displacement nor the $O(1/k)$ count correction changes a leading coefficient measured by $\log\log X/k$. The rank is fixed in the raw reversal count; it moves with x in Theorem 11.1.

14 Persistent excursions and positive-area criteria

The pointwise criterion also admits a one-sided integral formulation. This retains information lost by the leading reversal spectrum. Write $v_+ = \max(v, 0)$ and define

$$\begin{aligned} Q_F(X) &= \int_X^{2X} F(t)_+ \, dt, \\ S_\alpha(x) &= \frac{L(x)^2}{\lfloor \alpha L(x) \rfloor} \Delta_{\lfloor \alpha L(x) \rfloor}(x), \\ Q_\alpha(X) &= \int_X^{2X} (S_\alpha(t))_+ \, dt. \end{aligned}$$

All statements below concern sufficiently large X; the rank and coefficient denominators are then positive. Arbitrary definitions on the bounded initial range do not change an eventual criterion.

Lemma 14.1 (Persistence at a controlled peak)**.** *If RH is false, there is $0 < b < 1/2$ such that for every fixed $A > 0$ and $d < 1 - b/2$, arbitrarily large integers n satisfy*

$$F(y) > An^{-b} \qquad (n \le y \le n + n^d).$$

More precisely, controlled positive peaks with $K(n) > A_0 n^{-b}$ give intervals of length $c_{A_0} n^{1-b/2}/\sqrt{\log n}$ on which $F(y) \ge (A_0/2)n^{-b}$, for some fixed $c_{A_0} > 0$.

Proof. The two-sided peak construction in Theorem 11.2 supplies arbitrarily late integer peaks x with $K(x) > A_0 x^{-b}$ and $0 < \theta(x) - x \le (2A_0 + 1)x^{1-b}\log x$. For $0 < h \le x$ and $x \le y \le x + h$, distinct integer prime positions give

$$|\theta(y) - y| \le M := (2A_0 + 1)x^{1-b}\log x + 3(h+1)\log x$$

eventually. The kernel satisfies $\kappa(t) \le 2/(x^2 \log x)$ on this interval. Taylor's integral remainder for $\log\log t$, on arguments at least $x/2$, gives

$$F(y) \ge K(x) - \frac{2Mh + 8M^2}{x^2 \log x} \quad \text{when } M \le x/2.$$

The nonnegative tail $A(y)$ was discarded only in this lower bound. Take $h = \eta x^{1-b/2}/\sqrt{\log x}$, with $\eta = \sqrt{A_0}/100$. The term $2Mh$, after the displayed normalization and division by x^{-b}, tends to zero. Using $M^2 \le 2(2A_0 + 1)^2 x^{2-2b} \log^2 x + 18(h+1)^2 \log^2 x$, the remaining loss has limsup at most $144\eta^2 < A_0/2$. Also $M = o(x)$. This proves the more precise assertion. Choose $A_0 > 2A$; every fixed x^d with $d < 1 - b/2$ is eventually shorter than h. □

The maintained formal proof establishes every fixed-power version and the finite loss envelope. The logarithmic critical-length refinement is the ordinary calculation displayed above.

Theorem 14.2 (Positive-area RH criteria)**.** *For every fixed $\alpha > 0$, each of the following is equivalent to RH:*

$$Q_F(X) = O\big(X^{1/4}(\log X)^B\big) \quad \textit{for some fixed } B \ge 0,$$

$$Q_\alpha(X) = O\big(X^{1/4}(\log X)^B\big) \quad \textit{for some fixed } B \ge 0.$$

The bounds are required at every sufficiently large real X. For $1/4 \le a < 1$, either bound $Q(X) = O_\varepsilon(X^{a+\varepsilon})$ for every $\varepsilon > 0$ implies that every nontrivial zero satisfies $\operatorname{Re}\rho \le (1+2a)/3$. These are conditional zero restrictions, not unconditional strips.

Proof. The coefficient transfer gives, with a fixed C_α,

$$|S_\alpha(x) - F(x)| \le \frac{C_\alpha}{L(x)}|F(x)| + \frac{C_\alpha}{x}.$$

Once $C_\alpha/L(x) \le 1/2$, taking positive parts and integrating yields

$$Q_\alpha(X) \le \tfrac{3}{2}Q_F(X) + C_\alpha \log 2, \qquad Q_F(X) \le 2Q_\alpha(X) + 2C_\alpha \log 2.$$

Under RH both positive parts vanish eventually.

Under false RH, Lemma 14.1 gives $Q_F(n) \ge An^{d-b}$ for every fixed $d < 1 - b/2$ at arbitrarily large n. Since $b < 1/2$, choose d with $d - b > 1/4$. This defeats every fixed logarithmic factor at exponent $1/4$. The two area

comparisons transfer the contradiction. For a zero of real part $\sigma > (1+2a)/3$, choose fixed exponents with

$$1-\sigma < b < \frac{2(1-a)}{3}, \qquad a+b < d < 1-b/2.$$

A sufficiently small fixed $\varepsilon > 0$ gives the same contradiction. □

For an explicit sufficient estimate, put

$$Z(x) = \sqrt{x}\log x\, J(x), \qquad q(X) = 2\left\lceil \frac{\log X}{8\log 2} \right\rceil.$$

The estimate is the following unproved assertion:

$$\frac{1}{X}\int_X^{2X} |Z(t)|^{q(X)}\, dt \le C\, 2^{-q(X)} \quad \text{eventually, with fixed } C. \tag{44}$$

Indeed $P(x) \sim 2/(\sqrt{x}\log x)$, $D \ge 0$ and $A = O(1/x)$ give $F(x) \le (Z(x) - 1)/(\sqrt{x}\log x)$ eventually. For $q \ge 1$, $(u-1)_+ \le |u|^q$. Thus (44) implies $Q_F(X) \le CX^{1/4}/\log X$ and hence RH. A fixed second-moment envelope instead gives only $O(\sqrt{X}/\log X)$ by $(v-m)_+ \le v^2/(4m)$. This identifies the strength needed by that particular estimate; it does not prove that every second-moment argument must fail. No unconditional bound (44) is asserted.

14.1 A growing positive tilt

The exact identity (33) also permits a growing parameter, provided its tail is estimated directly. For fixed $c > 0$, set $z = c\sqrt{x}$. PNT and partial summation give

$$\sum_{p>x} (p-1)^{-2} \sim \frac{1}{x\log x}, \qquad \sum_{p>x} (p-1)^{-3} = O\left(\frac{1}{x^2\log x}\right).$$

The estimate needed here is uniform in the growing parameter. Put $N = \lfloor x \rfloor$ and $S_2(N) = \sum_{p>N} (p-1)^{-2}$. The elementary bounds $0 \le \log(1+v) - v + v^2/2 \le v^3/3$ for $v \ge 0$ give

$$\left| \log \mathcal{H}(z) - \log \mathcal{H}_N(z) + \frac{z^2 - z}{2} S_2(N) \right| \le \frac{2(z^3+z)}{3N^2} \qquad (N \ge 1,\ z \ge 0).$$

This weaker cubic bound already suffices: after division by $z = c\sqrt{x}$, its error is $O_c(x^{-1})$, which is $o_c(1/(\sqrt{x}\log x))$. PNT supplies $N \log N\, S_2(N) \to 1$, and $N/x \to 1$ retains the real cutoff. Consequently

$$\frac{T_{c\sqrt{x}}(x)}{c\sqrt{x}} = F(x) - \frac{c}{2\sqrt{x}\log x} + o_c\left(\frac{1}{\sqrt{x}\log x}\right). \tag{45}$$

It follows that RH is equivalent to eventual $T_{c\sqrt{x}}(x) < 0$ for each fixed $c > 0$. The RH direction follows from the negative source margin; the converse uses the positive x^{-b} excursions with $b < 1/2$, which dominate every fixed square-root-scale allowance. This argument does not apply a fixed-tilt error term uniformly without proof. Nor does a larger c establish the resulting product inequality. For comparison, a finite fixed combination satisfies $\sum_j a_j T_{z_j} = (\sum_j a_j z_j)F + O(1/x)$: canceling its leading coefficient cancels this RH signal as well.

15 RH detected by localized reversal counts

This section uses a specified collection of witnesses and a growing finite tilt grid. It does not replace the original maximal count $N_\eta(k)$ by a monotone proxy. In fact, ascent inclusion alone does not imply monotonicity of maximal separated reversal counts: the step signs $(-,+,+,-,+,+)$ allow two separated witnesses, while $(-,+,+,+,+,+)$ allow only one.

15.1 Actual tilted laws and exact rounding

For each $t > 0$ put $\eta_p(t) = t/(p-1+t)$. Its odds are $t/(p-1)$, so its degree-r elementary coefficient is exactly $t^r a_r$. At a supported prefix $x = p-1$ and a consecutive gap $g = q-p$, the rank-$(r+1)$ density therefore ascends exactly when

$$g + t < R_r(x);$$

it descends exactly when $R_r(x) < g + t$. These are statements about actual local-law densities. No unweighted arithmetic realization for every real t is assumed.

For $t_j = 1 + j/M$, $0 \le j \le HM$, let $A_M(R,g) = \#\{j : g + t_j < R\}$. If $M \ge 1$ and $1 \le R - g \le H + 1$, then

$$A_M(R,g) = \lceil M(R-g-1)\rceil.$$

Thus, when both $R-g$ and $U-g$ lie in this interval,

$$|A_M(R,g) - A_M(U,g) - M(R-U)| < 1. \tag{46}$$

The weak lower and strict upper endpoints follow from the strict ascent test; a threshold exactly on a grid point is not counted. For a nonempty finite set of pairs, summing this identity loses less than one unit per pair.

15.2 The sampling input and the specified cells

Alweiss–Luo [6, Corollary 1.2] prove that for each fixed $\delta \in [0.525, 1]$, some fixed width H and constants $c > 0, m$ give at least $ch/(\log Y)^m$ consecutive prime pairs of gap at most H in every sufficiently late $[Y - h, Y]$, whenever $Y^\delta \le h \le Y$. We use $\delta = 3/5$ and enlarge H to at least two. This H is not supplied by a theorem about global infinitude at a numerical width such as 186.

For integer X sufficiently large, set

$$h_X = \lceil 3X^{3/5} \rceil, \quad m_X = \left\lfloor \frac{X^{7/10}}{10h_X} \right\rfloor, \quad M_X = \lceil \sqrt{X} \rceil, \quad r_X = \lfloor (H+2)L(X) \rfloor.$$

The cell indexed by $0 \le i < m_X$ starts at $z_i = X + 1 + 10h_X i$. Its three sampling windows are

$$[z_i, z_i + h_X], \quad [z_i + 3h_X, z_i + 4h_X], \quad [z_i + 6h_X, z_i + 7h_X].$$

The explicit $+1$ ensures that every relevant prefix $p - 1$ is at least X. All prefixes used below are at most $X + X^{7/10}$. The source theorem applies to all three windows simultaneously, since their right endpoints are at most $2X$ and $h_X \ge (2X)^{3/5}$ eventually.

Fix $\ell = H + 3$. Every multiple of $\ell!$ starts a composite block at offsets $2, \dots, \ell$. Since $h_X \ge \ell! + 2$ eventually, a block can be placed between the left and middle windows. Choose the first qualifying consecutive bounded pair in each window and use its lower prime in the left and middle windows. A consecutive-prime step crossing the intervening block has length at least ℓ. Fix a choice of these crossings once and for all, using only their prime geometry. Denote its lower-prime index by d_i and that of the selected right pair by a_i. These deterministic choices use only primes and the fixed schedule.

The crossing lies between the selected left and middle primes, so $d_i < a_i$. The right endpoint in one cell is strictly before the next cell's descent lower prime; therefore $a_i + 1 < d_j$ for $i < j$. Finally $m_X \sim X^{1/10}/30$. These are genuine separated witnesses, not repeatedly counted copies of one reversal.

15.3 Two finite integer counts

Use the same grid $t_j = 1 + j/M_X$, $0 \le j \le HM_X$, in every cell. Define

$$C_H(X) = \sum_{i=0}^{m_X-1} \sum_{j=0}^{HM_X} \mathbf{1}\{d_i < a_i \text{ and the tilted rank-}(r_X+1) \text{ density descends at } d_i \text{ and ascends at } a_i\},$$

$$\overline{C}_H(X) = \sum_{i=0}^{m_X-1} \sum_{j=0}^{HM_X} \mathbf{1}\{g_{a_i} + t_j < \mathcal{R}_{r_X}(L(p_{a_i} - 1))\}.$$

Use index zero when a specified selection is unavailable on the bounded initial range. The finite formulas then define total counts; their initial values do not enter either eventual criterion. The reference uses the same actual prime pairs and their actual gaps.

Uniformly on the window, both actual and reference ratios are $H + 2 + o(1)$, with positive denominators. This follows from $R_r, \mathcal{R}_r = r/L + O(1/L)$ on compact rank bands and $|L(y) - L(X)| \le 1$ on $[X, 2X]$ eventually. Every chosen long step is consequently a strict descent for every grid tilt. For the short step, $2 \le g_{a_i} \le H$ puts both ratio-minus-gap thresholds inside $[1, H+1]$. Writing $\Delta_i = \Delta_{r_X}(p_{a_i} - 1)$, we obtain

$$C_H(X) - \overline{C}_H(X) = M_X \sum_{i=0}^{m_X-1} \Delta_i + e_X, \qquad |e_X| < m_X \tag{47}$$

whenever $m_X > 0$.

Theorem 15.1 (Localized separated-reversal criteria)**.** *For the preceding fixed width and deterministic construction, RH is equivalent to each of the following:*

$$C_H(X) \le \overline{C}_H(X) \quad \textit{eventually,} \tag{48}$$

$$C_H(X) - \overline{C}_H(X) \le CX^{1/10} \quad \textit{eventually for some fixed } C > 0. \tag{49}$$

Under false RH the signed difference divided by $X^{1/10}$ is unbounded above along integer X.

Proof. Under RH, the uniform coefficient sign comparison applies to r_X at every selected prefix. Actual ascent tests are included in reference tests, proving (48) and hence (49).

If RH is false, Lemma 14.1, followed by the uniform coefficient transfer, supplies arbitrarily large integer n such that

$$\Delta_r(y) > Tn^{-1/2} \qquad (n \le y \le n + n^{7/10})$$

for every prescribed fixed $T > 0$ and every rank in a fixed positive compact $r/L(y)$ band. Indeed its retained main term is $\gg n^{-b}/L(y)$ for some $b < 1/2$, and $n^{1/2-b}/\log\log n \to \infty$. All selected prefixes at $X = n$ lie in this interval, and the single rank r_n lies in the required band throughout. Since $M_n \ge \sqrt{n}$, (47) gives

$$C_H(n) - \overline{C}_H(n) > m_n(T-1).$$

Now $m_n \ge c_0 n^{1/10}$ eventually, with fixed $c_0 > 0$. Choosing $T > 1 + C/c_0$ contradicts any proposed eventual bound (49). This proves both converses. □

For each individual tilt, the selected witnesses are bounded by $N_{\eta(t)}(r_X + 1)$. The sum in the theorem ranges over a growing finite grid, so it is not the maximal count for one fixed law. Its reference also retains θ and the entire profile. An independent proof of either eventual count inequality would prove RH; the equivalence itself supplies no such estimate.

An ascent-only variant uses every consecutive gap at most the source width in $[X+1, X+1+\lceil 2X^{7/10}\rceil]$ and the same grid and rank. Let its actual and reference counts be $A_H(X), B_H(X)$. The source at $\delta = 3/5$, applied to this longer window, supplies $\gg X^{7/10}/(\log X)^m$ pairs. At the persistent false-RH peaks, (46) then gives

$$A_H(X) - B_H(X) \gg \frac{X^{6/5-b}}{(\log X)^m \log\log X}, \qquad b < 1/2.$$

Thus RH is also equivalent to eventual $A_H \le B_H$, and to an eventual one-sided bound $A_H - B_H = O(X^{7/10})$. The persistence interval may be taken with any fixed exponent between $7/10$ and $3/4$, which absorbs the shift and rounding in this ascent-only window.

16 Growing families and signed bounds

A fixed-family RH criterion does not by itself justify varying the family with the prime cutoff. We first give a transfer with explicit error intervals, then two arithmetic calculations in which the complete correction can be retained. Throughout this section, a finite-prime limiting law is taken before the cutoff or polynomial degree varies. We make no assertion uniform in a finite sampling box.

16.1 Correction intervals

Theorem 16.1 (Transfer through a correction interval)**.** *Let $E, \ell, u : \mathbb{N} \to \mathbb{R}$ and let $c > 0$ be fixed. Suppose that*

$$\ell(N) \le E(N) - cF(N) \le u(N) \quad \textit{for all sufficiently large } N.$$

Assume that, for every fixed $0 < b < 1/2$, there is a constant $C_b > 0$ such that $u(N) - \ell(N) \le C_b N^{-b}$ eventually. Then RH is equivalent to eventual $E(N) \le u(N)$, and also to eventual $E(N) < u(N)$.

Proof. Under RH, $F(N) < 0$ eventually, so the upper interval bound gives $E(N) < u(N)$. If RH fails, the integer-cutoff Nicolas excursion theorem gives a fixed $0 < b < 1/2$ and arbitrarily large N with $cF(N) > (C_b + 1)N^{-b}$. At these N, the lower interval bound implies $E(N) > u(N)$, a contradiction. □

The interval center need not be small or positive. What matters is the width and its proved arithmetic relation to the same E and F. Defining an endpoint from the unknown error would not provide an independent estimate. The maintained theorem has a fixed coefficient c; a moving coefficient requires a separate uniform argument.

16.2 An explicit polynomial-plus-prime family

For a positive integer d, put

$$P_d(a) = \prod_{j=1}^{d} (a - j), \qquad V_d(a, q) = P_d(a) + q,$$

where a ranges over integers and q over primes, with ordered representations retained. For each fixed finite set of primes, uniform integer residues, prime progression limits and the Chinese remainder theorem give independent local divisibility laws. The polynomial has exactly $\min(d, p)$ roots modulo p. Consequently

$$\eta_d(p) = \frac{p - \min(d, p)}{p(p-1)}. \tag{50}$$

In particular, every prime $p \le d$ is excluded. Use this exact law, including those exclusions, to form $E_d(N) = E_{\eta_d}(N, 1)$ as in Theorem 12.1. Set

$$Q_2(N) = \sum_{p > N} \frac{1}{(p-1)^2}, \qquad T(N, b) = \sum_{p > N} \log\left(1 + \frac{b}{(p-1)^2}\right) \quad (b \ge 0).$$

Theorem 16.2 (Exact correction and finite bounds)**.** *For $N \geq \max(d, 3)$,*

$$E_d(N) = F(N) - T(N, d-1). \tag{51}$$

For $N \geq 1$ and $b \geq 0$,

$$\left(b - \frac{b^2}{2N^2}\right) Q_2(N) \leq T(N, b) \leq bQ_2(N), \qquad 0 \leq Q_2(N) \leq \frac{2}{N}. \tag{52}$$

Moreover $N \log N \, Q_2(N) \to 1$.

Proof. For $p > d$, direct substitution into the odds gives

$$(1 + w_d(p))(1 - 1/p) = \left(1 + \frac{d-1}{(p-1)^2}\right)^{-1}.$$

The exact profile identity expresses $E_d - F$ as the sum of these logarithms over $p > N$. All excluded primes already lie in the finite prefix; their profile factors cancel exactly, proving (51). For $t \geq 0$, $t - t^2/2 \leq \log(1+t) \leq t$. Since $p > N$ implies $p - 1 \geq N$, summing these inequalities proves (52). Comparison with $\sum_{n \geq N} n^{-2} \leq 2/N$ gives the elementary upper bound. Partial summation of the prime number theorem gives the stated asymptotic for Q_2. □

Theorem 16.3 (Independent signs at logarithmically reduced degree)**.** *For every fixed integer $j \geq 1$, let*

$$d_j(N) = 1 + \left\lfloor \frac{N}{(\log N)^j} \right\rfloor.$$

Then, unconditionally,

$$(\log N)^{j+1} E_{d_j(N)}(N) \longrightarrow -1.$$

In particular $E_{d_j(N)}(N) < 0$ for every sufficiently large integer N. The diagonal degree $d = N$ also satisfies $E_N(N) \log N \to -1$.

Proof. For large N, $d_j(N) \leq N$, and $b_N = d_j(N) - 1 = N/(\log N)^j + O(1)$. The two bounds in (52), together with $b_N \leq N$, imply

$$(\log N)^{j+1} T(N, b_N) \longrightarrow 1.$$

For every fixed positive integer A, the quantitative prime number theorem gives the unconditional limit

$$F(N)(\log N)^A \longrightarrow 0.$$

Subtracting in (51) proves the result. For the diagonal, use $b = N - 1$ in the same bounds to obtain

$$T(N, N-1)\log N \longrightarrow 1. \qquad \square$$

These are complete independent signed estimates, but the family degree is $N^{1-o(1)}$. For every fixed $\alpha < 1$ and fixed j, $d_j(N)/N^\alpha \to \infty$. The theorem therefore supplies no fixed-power zero-free conclusion. Neither its onset nor its estimates are asserted to be uniform when j grows with N.

16.3 The effect of adding two primes

Let a base residue family have zero probability $r_p \in [0, 1]$ at each prime p. Add two independent uniform nonzero residues. Their exact zero probability is

$$\eta(p) = \frac{p - 2 + r_p}{(p-1)^2}.$$

Indeed there are $p - 1$ ordered nonzero pairs with sum zero and $p - 2$ with any prescribed nonzero sum. Thus the laws of $q_1 + q_2 + 1$ and $q_1 + q_2$ attain the two endpoints, independently of the complexity of the base family.

Theorem 16.4 (Attained two-prime bracket)**.** *For every such finite residue family, every $N \geq 3$, and $z \geq 0$,*

$$E_{q_1+q_2+1}(N, z) \leq E_{\text{base}+q_1+q_2}(N, z) \leq E_{q_1+q_2}(N, z).$$

At $z = 1$, define the convergent tails

$$T_3(N) = \sum_{p>N} \log\left(1 + \frac{1}{(p-1)^3}\right), \qquad U_2(N) = \sum_{p>N} -\log\left(1 - \frac{1}{(p-1)^2}\right).$$

Then the exact bracket is

$$F(N) - T_3(N) \leq E_{\text{base}+q_1+q_2}(N, 1) \leq F(N) + U_2(N).$$

Proof. For tail primes $p > N \geq 3$, all three probabilities are less than one. Both $\eta/(1-\eta)$ and $\log(1 + z\eta/(1-\eta))$ increase with η. Compare the convergent tail corrections to $zF(N)$, including their exact normalizations. Substitution of $r_p = 0, 1$ gives the two displayed tails at $z = 1$. $\square$

The favorable endpoint correction is asymptotic to $1/(2N^2 \log N)$ by partial summation. Thus adding arbitrary base complexity cannot enlarge this correction within the stated two-prime-addition class. This controls a

correction, not the sign of the complete expression containing $F(N)$. Additional independent prime additions can be included in the base distribution; this ordinary specialization does not assert uniform finite-box limits or a result for changing nonunit coefficients.

17 Power-sized degrees and zero-free regions

The distinction between a logarithmic saving and a fixed power saving can be made precise for the actual polynomial family of Section 16. The result below is a conditional implication. It does not assume a zero-free region, but it does assume the eventual sign of the complete arithmetic profile.

Theorem 17.1 (Power-degree sign criterion)**.** *Let $d : \mathbb{N} \to \mathbb{N}$ satisfy $d(N) \geq 1$. Fix real numbers $C > 0$ and $1/2 \leq \alpha < 1$. Suppose that, for every sufficiently large integer N,*

$$d(N) \leq CN^{\alpha}, \qquad E_{d(N)}(N) \leq 0, \tag{53}$$

where E_d is the complete profile of the consecutive-root polynomial $P_d(a)+q$, with all small-prime exceptions included. Then

$$\zeta(\rho) \neq 0 \qquad \text{whenever } \alpha < \operatorname{Re}\rho < 1.$$

The degree bound and the signed inequality refer to the same chosen sequence of families and both hold eventually, not merely along possibly different unbounded subsequences.

Proof. Since $\alpha < 1$, the degree bound implies $d(N) \leq N$ eventually. The exact identity (51) and (52) therefore give

$$F(N) \leq T(N, d(N)-1) \leq (d(N)-1)Q_2(N) \leq 2CN^{\alpha-1} \quad \text{eventually.} \tag{54}$$

Suppose $\zeta(\rho) = 0$ with $\beta = \operatorname{Re}\rho > \alpha$ and $\beta < 1$. Choose

$$b = \frac{(1-\beta)+(1-\alpha)}{2}.$$

Then $0 < 1-\beta < b < 1-\alpha \leq 1/2$. The zero-dependent integer Nicolas excursion argument from Section 11.1 gives arbitrarily large integers N with $F(N) > 2CN^{-b}$. But $\alpha - 1 < -b$, so for $N \geq 1$,

$$F(N) \leq 2CN^{\alpha-1} \leq 2CN^{-b}$$

by (54), a contradiction. This uses the already established integer excursions, including the nonlinear logarithm correction. It neither differentiates an asymptotic error nor assumes a new real-to-integer interpolation. □

Corollary 17.2 (Square-root degree)**.** *If $d(N) \geq 1$, $C > 0$, and eventually*

$$d(N) \leq C\sqrt{N}, \qquad E_{d(N)}(N) \leq 0,$$

then RH holds.

Proof. Theorem 17.1 excludes every nontrivial zero to the right of $1/2$. The functional equation then places every nontrivial zero on the critical line. □

The new content here is the exact passage from this arithmetic degree budget to the classical zero-dependent Nicolas obstruction; the underlying oscillation mechanism is classical [7, 8]. External priority for this polynomial-family formulation is not established.

The complete sign in (53) remains unproved independently for power-sized degrees. In contrast, Theorem 16.3 proves that sign at $d(N) = 1 + \lfloor N/(\log N)^j \rfloor$ for every fixed $j \geq 1$. Such a degree exceeds CN^α eventually for every fixed $C > 0$ and $\alpha < 1$. Thus that unconditional theorem cannot be substituted into (53). Increasing a fixed logarithmic saving does not reach any fixed power saving.

This criterion concerns the specified polynomial-plus-prime law. No degree-only assertion is made for arbitrary changing coefficients or sampling rules: those can introduce additional excluded primes and a different correction. Nor does an arbitrary sequence of individually qualifying fixed laws automatically satisfy the uniform estimate used here.

18 A generalized Robin critical window

The same signed prime-product calculation has a separate application to generalized divisor sums. It gives actual counterexamples to a proposed extension of a generalized Robin inequality. It does not supply an independent RH criterion.

For a real parameter $\kappa > 1$, define

$$S_\kappa(n) = \sum_{d|n} \mu(n/d)\sigma(d)^\kappa, \qquad \mathcal{G}_\kappa(n) = \frac{\zeta(\kappa)S_\kappa(n)}{(e^\gamma n \log\log n)^\kappa} \quad (n \geq 3).$$

Here κ is a real divisor-sum exponent, not the prime-factor rank k. These are the LCM-power functions of Fan–Kobayashi–Molnar [13, Definition 1.2]. For an integer $N \geq 8$, put

$$M_N = \operatorname{lcm}(1, \ldots, N) \prod_{\substack{p \leq N \\ N < p^2 \leq 2N}} p, \qquad K_N = \frac{3}{2} + \frac{2}{5 \log\log M_N}. \tag{55}$$

Theorem 18.1 (Uniform critical window)**.** *For every $X \in \mathbb{N}$, there is an integer $N > X$, $N \geq 8$, such that simultaneously for every real $1 < \kappa \leq K_N$,*

$$\log \mathcal{G}_\kappa(M_N) > \frac{1}{100\sqrt{N}\log N}. \tag{56}$$

Under RH, this holds for every sufficiently large integer N, with a common cutoff for the entire parameter window.

Corollary 18.2. *For every fixed real $1 < \kappa \leq 3/2$, there are arbitrarily large integers n for which $\mathcal{G}_\kappa(n) > 1$.*

18.1 Exact arithmetic and the full signed comparison

Let $a_p = v_p(M_N)$ for $p \leq N$, and define

$$u_p = p^{-a_p-1}, \qquad U_N = \sum_{p\leq N} u_p,$$
$$R_N = 4\sum_{p\leq N}\frac{u_p}{p} + 32\sum_{p\leq N} u_p^2, \qquad D_N = \log\log\log M_N - \log\log\theta(N).$$

All finite sums have the same prime support $p \leq N$. Write

$$\mathcal{T}_\kappa(N) = \sum_{p>N} -\log(1-p^{-\kappa}).$$

For $\kappa > 1$ this converges and equals the logarithm of the omitted part of the zeta Euler product.

The defining divisor sum is multiplicative, and at a prime power

$$S_\kappa(p^a) = \sigma(p^a)^\kappa - \sigma(p^{a-1})^\kappa = \left(\frac{p^a}{1-1/p}\right)^\kappa (1-p^{-\kappa})\, C_\kappa(p,a),$$

where

$$C_\kappa(p,a) = \frac{(1-p^{-a-1})^\kappa - p^{-\kappa}(1-p^{-a})^\kappa}{1-p^{-\kappa}} > 0.$$

Therefore the exact normalized identity is

$$\log \mathcal{G}_\kappa(M_N) = \mathcal{T}_\kappa(N) - \kappa\big(F(N) + D_N\big) + \sum_{p\leq N} \log C_\kappa(p,a_p). \tag{57}$$

This retains the actual integer height $\log M_N$.

Lemma 18.3 (Uniform finite lower bound)**.** *For $N \geq 8$ and $1 < \kappa \leq 2$,*

$$\log \mathcal{G}_\kappa(M_N) \geq \mathcal{T}_\kappa(N) - \kappa(F(N) + D_N + U_N) - R_N. \tag{58}$$

If $1 < \kappa \leq K \leq 2$, its right side may be replaced by

$$\mathcal{T}_K(N) - K \max(F(N) + D_N + U_N, 0) - R_N.$$

Proof. Put $u = p^{-a-1}$, $v = p^{-\kappa}$ and $t = p^{-a}$. For $1 \leq \kappa \leq 2$, $u \leq 1/8$ and $v \leq 1/2$, Bernoulli's inequality gives

$$C_\kappa(p, a) \geq 1 - \frac{\kappa u}{1 - v}.$$

The number $w = \kappa u/(1 - v)$ satisfies $0 \leq w \leq 4u \leq 1/2$ and $w \leq \kappa u + 4uv$. The bound $\log(1 - w) \geq -w - 2w^2$ yields

$$\log C_\kappa(p, a) \geq -\kappa u - 4uv - 32u^2 \geq -\kappa u - 4u/p - 32u^2.$$

For the exponents of M_N, $p^{a_p+1} > N$; hence $u_p \leq 1/8$ when $N \geq 8$. Sum the bound in (57). Finally $\mathcal{T}_\kappa \geq \mathcal{T}_K$, and for every real A, $-\kappa A \geq -K \max(A, 0)$. □

18.2 The two arithmetic costs and the positive tail

The ordinary prime number theorem gives

$$\sqrt{N} \log N \, D_N \longrightarrow \sqrt{2}, \qquad \sqrt{N} \log N \, U_N \longrightarrow \sqrt{2}, \qquad \sqrt{N} \log N \, R_N \longrightarrow 0. \tag{59}$$

Here is a direct explanation of all three limits. The exact height is

$$\log M_N = \psi(N) + \theta(\sqrt{2N}) - \theta(\sqrt{N}).$$

Since $\psi(N) - \theta(N) = \theta(\sqrt{N}) + O(N^{1/3} \log N)$, we have $\log M_N - \theta(N) \sim \sqrt{2N}$. The derivative of $\log \log t$ is $1/(t \log t)$, giving the first limit.

For $p > \sqrt{2N}$, $a_p = 1$, so the upper-band contribution is

$$\sum_{\sqrt{2N} < p \leq N} p^{-2}.$$

Its normalized limit is $\sqrt{2}$. Below $\sqrt{2N}$, split at $N^{1/3}$ to bound the remaining contribution by

$$N^{-1} \pi(N^{1/3}) + \sum_{p > N^{1/3}} p^{-3}.$$

Both terms are negligible on the displayed scale. Finally, $u_p < 1/N$ gives $R_N = O((1+\log N)/N)$ and proves the third limit.

We need only a finite lower comparison for the moving zeta tail. For $q > 1$, $m \in \mathbb{N}$, and $1 < K \le 2$, put

$$\begin{aligned} A_2(x) &= \sum_{p>x} p^{-2}, \\ L_m(N,q,K) &= N^{2-K} A_2(N) \\ &\quad + \sum_{j=0}^{m-1} \big((Nq^{j+1})^{2-K} - (Nq^j)^{2-K}\big) A_2(Nq^{j+1}). \end{aligned} \tag{60}$$

Then $\mathcal{T}_K(N) \ge L_m(N,q,K)$. To see this, first use $-\log(1-p^{-K}) \ge p^{-K}$, and then bound the increasing function p^{2-K} from below by its step approximation at the displayed thresholds. Every increment is nonnegative, so summation preserves the bound.

Fix $s > 1$ and m before taking $N \to \infty$, and put $q = s^2$. Because $x \log x\, A_2(x) \to 1$ and $\log\log M_N / \log N \to 1$, the finitely many terms give

$$\begin{aligned} \sqrt{N} \log N\, L_m\left(N, s^2, \frac{3}{2} + \frac{c}{\log\log M_N}\right) &\longrightarrow C_m(s) e^{-c}, \\ C_m(s) &= 1 + \frac{1}{s} - \frac{1}{s^{m+1}}. \end{aligned} \tag{61}$$

Only fixed-dilation prime-square-tail limits are used. In particular, we have not assumed an asymptotic for a prime sum uniform in a moving exponent.

Choose $s = 1001/1000$ and $m = 10^6$. Bernoulli's inequality gives $C_m(s) > 999/500$. With $\beta_0 \le 1/20$ from the classical Nicolas constant, the limiting complete RH lower bound exceeds

$$C_m(s) e^{-2/5} - 3(\sqrt{2}-1) - \frac{3}{2}\beta_0 > \frac{7}{600}. \tag{62}$$

For example, $\log(3/2) > 2/5$ and $\sqrt{2} < 99/70$ show that the left side is greater than $333/250 - 87/70 - 3/40 = 99/7000 > 7/600$. The finite choice $m = 10^6$ is a symbolic parameter choice; no million-term numerical verification is used.

18.3 Proof in the two RH cases

Proof of Theorem 18.1. Under RH, (20) gives

$$\limsup_{N\to\infty} \sqrt{N} \log N\, F(N) \le \beta_0 - 2.$$

Insert this and (59)–(61) into Lemma 18.3, with $K = K_N$. For large N, $K_N < 2$. The limiting lower bound for the common scaled comparison is at least the quantity in (62). Since $1/100 < 7/600$, one cutoff works for every $1 < \kappa \le K_N$.

If RH fails, negative Nicolas excursions give arbitrarily large integer N with

$$F(N)\sqrt{N} < -1. \tag{63}$$

For completeness, take real excursions $F(x) < -2x^{-b}$ with $0 < b < 1/2$ and $x > 3$, and let $N = \lfloor x \rfloor$. The prime products and θ are constant on $[N, N+1)$, so $F(N) = F(x)$. Also $x < N+1 \le 4N$. Thus $F(N) < -2/\sqrt{x} \le -1/\sqrt{N}$. Only F is rounded in this argument; no floor invariance of M_N or W is used.

By (59), the positive arithmetic costs satisfy $(D_N + U_N)\sqrt{N}\log N \to 2\sqrt{2}$, whereas (63) gives $F(N)\sqrt{N}\log N < -\log N$. Hence $F(N) + D_N + U_N < 0$ at arbitrarily large integer cutoffs. At those cutoffs the maximum in Lemma 18.3 vanishes. Equations (59) and (61) then give the stronger positive limiting lower bound $C_m(s)e^{-2/5}$. Taking cutoffs beyond the common asymptotic thresholds proves (56). The same explicit integer family therefore works in both logical cases. □

Proof of Corollary 18.2. We have $\log M_N \ge \theta(N) \to \infty$, so $M_N \to \infty$. Also $K_N > 3/2$. Apply Theorem 18.1 beyond any prescribed integer-height threshold and use positivity of $\mathcal{G}_\kappa$ to exponentiate its positive logarithm. □

The theorem concerns integer cutoffs and supplies no effective onset or frequency of the exceptional cutoffs without RH. It establishes neither optimality of $2/5$ nor that M_N is colossally abundant. Every fixed $\kappa > 3/2$ eventually lies outside the shrinking window, so this result is consistent with the fixed-parameter upper theorem of [13]. In particular, the counterexamples do not violate ordinary Robin's inequality and do not prove or disprove RH.

19 Implications and scope

The spectrum gives a quantitative dictionary. Bounded-gap counts yield reversal counts through the finite selection lemma; the upper comparison shows that no other late ascents can improve the rate. Finite changes in an admissible law preserve that rate. Many different affine prime-pair expressions consequently share a spectrum despite different forced primes and small ranks.

Location and frequency encode different information. Corollary 9.2 depends on the least recurring gap. The rate $\Lambda(\nu)$ depends on how frequently bounded gap classes occur. Infinitude alone need not give $\gamma_H > 0$: a hypothetical count growing as a power of $\log X$ has $\gamma_H = 0$. This is why bare infinitude cannot replace the input to Corollary 5.2.

For Goldbach-type expressions, the arithmetic theorem describes factorization in the full representation-weighted ensemble. It gives no lower bound for representations of each fixed even integer. Fixed-modulus independence likewise does not supply uniform estimates when a prime cutoff grows with input size. Moving-tail distribution laws require additional uniformity.

Repeated raw reversals and centered moving-rank excursions have different quantifiers. At each fixed rank there is a final decreasing tail. The RH criterion lets the rank grow with the endpoint and compares its ratio with the full reference profile. It is the transfer in Theorem 10.1 that relates this second question to Nicolas's error. Theorem 12.2 extends the centered comparison to the qualifying local laws and the actual affine families. An independently proved centered inequality in any one such family would therefore imply RH. A coarse reversal rate or a numerical prime-gap bound does not itself provide that sign.

The exact raw comparison can be written as

$$R_r(x) - (g+1) = \big(\mathcal{R}_r(L(x)) - (g+1)\big) + \Delta_r(x).$$

RH controls the second term. The first term and the occurrence of the actual gap remain necessary for an ascent. In particular, for fixed $1/5 < b < c < 1/3$ and all sufficiently large k, an ascent at a prime with $bk \le \mathcal{L}(p-1) \le ck$ occurs exactly when the next gap is two. This follows directly from the uniform ratio margins. Proving positive ascent supply in this window independently would prove twins; the identity itself is not such a supply proof.

The Hardy–Littlewood input in Theorem 9.3 provides actual twins approaching the eligibility cutoff. RH instead refines that cutoff's reference error and gives Theorem 13.2. The global twin asymptotic allows a distance $o(X_k)$ from the cutoff, which can exceed $\sqrt{X_k}$; it cannot by itself transfer the RH square-root clock correction to the location of the last actual ascent. No implication from RH to twin-prime infinitude is established by these results.

The relation to Robin and Lagarias is through their established equivalences with RH [12]. Thus the eventual inequality in Theorem 11.1, if proved independently, would imply the corresponding universal divisor-sum inequalities. A valid violation of either criterion would imply the false-RH

coefficient excursions. These are logical consequences of known equivalences, not new bounds for colossally abundant numbers. Indeed, for $n = \prod p^{a_p}$,

$$\frac{\sigma(n)}{n} = \prod_{p|n} \frac{1 - p^{-(a_p+1)}}{1 - p^{-1}},$$

and optimizing the exponent-dependent numerator factors together with $\log\log n$ remains a separate problem.

The classical inputs include Nicolas's criterion and spectral formula, Landau oscillation, prime-factor generating products and saddle-point estimates. The potentially new contributions are the exact reversal-frequency spectrum, its affine realization and the sign-preserving moving-coefficient transfer with its applications. These attributions are provisional: proof verification does not establish priority. Speculative RH and independent small-gap-supply investigations are excluded from the theorem claims of this paper.

19.1 Finite smaller gaps and the complete spectrum

Suppose, for a fixed even $h \geq 2$, that all sufficiently late consecutive gaps are at least h. Every fixed initial prime prefix has finite support in the rank variable. Consequently, for all sufficiently large ranks, none of its steps can be an ascent. The RH argument in Theorem 13.2 then uses the threshold $h + 1$ throughout and gives

$$\theta(P_k - 1) < T_{k-1,h} := \mathcal{E}(u_{k-1,h+1} - \gamma), \qquad N_k \leq (1 + \varepsilon)\frac{T_{k-1,h}}{\log T_{k-1,h}}$$

eventually, and hence

$$\log\log N_k \leq \frac{k-1}{h+1} - \gamma - \frac{\mathcal{H}'(h+1)}{\mathcal{H}(h+1)} + \frac{C_h}{k-1}. \tag{64}$$

Constants and onsets depend on the fixed h. There is no fixed-width sign transition: $\mathcal{H}(s) > 0$ at every positive s, and the RH source multiplier $-(2 + W)$ stays negative in every fixed positive rank band.

Finite twins give $h = 4$ as an eventual lower bound and $\gamma_2 = \gamma_3 = 0$. If additionally $\Lambda(1) = 1/5$, the spectrum forces $\gamma_4 = 1$: every $H \geq 5$ contributes at most $1/6$, while the only remaining possible maximizing term is $\gamma_4/5$. It follows that

$$\Lambda(\nu) = \frac{1}{4 + \nu} \qquad (\nu > 0).$$

Indeed the full-frequency width-four term dominates every larger width, and smaller widths have zero exponent. The RH ceiling (64) now has the same leading coefficient $1/5$. The matching leading values yield no opposing signed inequality. Nor does $\gamma_H = 1$ specify an actual density: both $G_H(X) \sim X^{\delta}$, with any fixed $\delta > 0$, and $G_H(X) \sim cX/\log^2 X$ have that exponent.

19.2 When local frequency reveals the RH displacement

A stronger frequency hypothesis makes the reference displacement visible in actual last-ascent locations. Fix even $h \geq 2$, assume smaller gaps occur only finitely often, and assume every sufficiently late interval

$$[X, X + \ell(X)], \qquad \ell(X) = \frac{\sqrt{X}}{(\log X)^2},$$

contains the lower member of a consecutive gap exactly h. Let V_k be the last supported prime eligible for a hypothetical gap-h ascent, and P_k the last actual ascent prime. Then

$$P_k/V_k \longrightarrow 1, \qquad T_{k-1,h} - \mathcal{C}(P_k - 1) = o(\sqrt{P_k}). \tag{65}$$

Under RH this gives

$$\frac{T_{k-1,h} - \theta(P_k - 1)}{\sqrt{P_k}} = 2 + W(P_k - 1) + o(1).$$

Here is the precision argument. Monotonicity of the supported ordinary coefficient ratio makes eligible primes an initial segment. Ordinary prime existence provides a supported seed below $V_k/2$. Apply local coverage at $X = V_k - 2\ell(V_k)$. For large V_k, $X \geq V_k/2$ and $\ell(X) \leq 2\ell(V_k)$, so the supplied gap-h pair has lower prime in $(V_k - 2\ell(V_k), V_k]$. Support propagates from the seed, and monotonicity makes that step an actual ascent. Beyond V_k, the eventual lower gap bound h makes any ascent eligible for the hypothetical gap-h test, contradicting the definition of V_k. Thus all finite exceptions are accounted for and $V_k - 2\ell(V_k) < P_k \leq V_k$.

At $x = V_k - 1$ and $y = w - 1$, with w the next prime after V_k, the finite-product comparison gives $R_r(x) = \mathcal{R}_r(B(x)) + O_h(1/(rx))$, and likewise at y. Only the prime V_k is added between these prefixes, so $B(y) - B(x) = \log(V_k/(V_k - 1)) = O(1/V_k)$. Ordinary prime existence and the eventual cut-off bound first place $r/B(x)$, $r/B(y)$ and $r/u_{r,h+1}$ in a common fixed positive compact interval. The uniform signed coefficient comparison then applies on both connecting segments. The strict eligible sign at x, the noneligible sign

at y, and the displayed one-prime jump give $|B(x) - u_{r,h+1}| = O_h(1/V_k)$. No prior spatial closeness of the two clocks is assumed.

The quantitative PNT-strengthened Mertens estimate gives $e^{-\gamma}G_x(1) - \log x = o(1)$, and hence $\mathcal{C}(x)/x \to 1$. The weaker relation $\log \mathcal{C}(x) \sim \log x$ would not suffice. Exponentiating the proved clock difference now gives $|\mathcal{C}(V_k - 1) - T_{r,h}| = O_h(\log V_k)$. Each prime-clock jump in the remaining interval is $O(\log V_k)$. There are at most $2\ell(V_k) + 1$ integer sites, so their total change is $O(\sqrt{V_k}/\log V_k) = o(\sqrt{V_k})$. This proves (65); add (32) at the same actual endpoint for the RH assertion.

A global pair-count asymptotic alone gives only proportional coverage. For the exact-gap count $G_h^{=}(X)$, the maintained sufficient condition is

$$G_h^{=}(X) = c_h \operatorname{Li}_2(X) + E(X), \qquad c_h > 0,$$
$$E(X) = o(\sqrt{X}/\log^4 X), \qquad \operatorname{Li}_2(X) = \int_2^X \frac{dt}{(\log t)^2}.$$

Subtract this estimate at X and $X + \ell(X)$. The main increment is of order $\sqrt{X}/\log^4 X$ and dominates both endpoint errors. The full integral main term is essential at this precision; $c_h X/\log^2 X$ cannot replace it. A square-root-plus-epsilon global error is insufficient for this subtraction. This is a sufficient route to local coverage, not its necessary strength.

Finally, on any of the finite supports in Section 15, the signed excess is exactly the number of grid thresholds in $[U_i, R_i)$ minus the number in $[R_i, U_i)$, where $U_i = \mathcal{R}_r(L(p_i - 1))$. A positive excess arbitrarily far out contradicts RH. Neither finite twins nor a positive main term for another gap class has been shown to force that signed excess. These frequency statements therefore identify additional information needed by this route; they establish neither RH nor twin-prime infinitude.

20 Formalization and reproducibility

The principal maintained owners are:

- `Assembly/ReciprocalSmoothSpectrum.lean`: all-rank finite maxima and Theorem 3.2;
- `Assembly/Headline.lean`: Theorem 3.3;
- `Assembly/AffineSpectrum.lean`: Theorem 3.4 and arithmetic phase recovery;
- `Assembly/GapSpectrumPhase.lean`: Theorem 7.1;

- `Assembly/GapRecurrence.lean` and `Proof/Upper/GapScaleCutoff.lean`: Theorem 9.1, Corollary 9.2, and the conditional inverse twin-prime transfer;
- `Helpers/PrimeProfileCanonicalLinearization.lean` and `Helpers/PrimeProfileUniformSign.lean`: the full-profile coefficient comparison and uniform sign transfer;
- `Assembly/FamilyRHResults.lean`: Theorems 12.1 and 12.2, the general probability-law constructor, exact forced-rank mass criteria, actual affine-family consumers, and the generalized reference constant;
- `Helpers/FamilyIntervalTransfer.lean` and `Helpers/DoublePrimeAdditionCriterion.lean`: Theorems 16.1 and 16.4;
- `Helpers/ConsecutiveRootProbability.lean`, `Helpers/ConsecutiveRootCriterion.lean`, `Helpers/PrimeAddedRootBudget.lean` and `Helpers/LogarithmicRootCriterion.lean`: the exact root law, finite correction bounds, and Theorem 16.3;
- `Helpers/ConsecutiveRootZeroFree.lean`: Theorem 17.1 and Corollary 17.2, with the degree and complete-sign premises explicit;
- `Helpers/LcmDivisorArithmetic.lean`, `Helpers/LcmNormalizedComparison.lean` and the modified-LCM height, defect and finite-tail modules: the actual divisor sum and the complete finite and asymptotic bounds used in Section 18;
- `Helpers/ModifiedLcmRHWindow.lean`, `Helpers/ModifiedLcmNegativeSupply.lean` and `Helpers/ModifiedLcmCriticalWindow.lean`: Theorem 18.1 and Corollary 18.2, with separate RH and false-RH intermediate statements and unconditional maintained endpoints;
- `Helpers/NicolasPowerPeak.lean` and `Helpers/PrimeProfilePowerExcursions.lean`: the corrected logarithm and actual moving-rank power excursions;
- `Helpers/NicolasLogSpectral.lean`, `Helpers/NicolasSpatialClock.lean` and the positive-tilt modules: the classical spectral inputs and their clock/integral consumers;
- `Assembly/GeneralRHSharpCapacity.lean`: attained last ascents and sharp count capacities under any eventual lower gap, including every finite exception;

- `Helpers/NicolasPowerPersistence.lean`, `Helpers/NicolasCriticalPersistence.lean`, `Helpers/NicolasPositiveArea.lean` and `Helpers/PrimeProfilePositiveArea.lean`: power and critical logarithmic-length positive intervals and both positive-area RH criteria;

- `Helpers/NicolasMomentConsumer.lean` and `Helpers/MomentSecondArea.lean`: the growing-moment sufficient condition and the fixed-second-moment area envelope. Their proposed moment hypotheses remain unproved;

- `Helpers/TiltedDensity.lean`, `Helpers/TiltedReversalCount.lean`, `Helpers/TiltedCountRHUpper.lean`, `Helpers/TiltedCountExcursion.lean` and `Helpers/PrimeProfileWindowBands.lean`: actual tilted densities, finite signed counts, RH inclusion, false-RH interval margins and uniform fixed-window rank bands;

- `Helpers/LocalizedReversalCriterion.lean` and `Helpers/LocalReversalSampling.lean`: both localized reversal-count equivalences, explicit cell selection and the all-short-interval arithmetic input boundary;

- `Helpers/LocalizedAscentCriterion.lean` and `Helpers/LocalAscentSampling.lean`: both ascent-only criteria, using all actual pairs and an explicit quantitative sampling premise;

- `Helpers/TiltedSignedExcess.lean` and `Helpers/FixedTiltCombinations.lean`: exact signed counts, aggregate rounding and fixed-tilt cancellation;

- `Helpers/GapFrequencyCoverage.lean`: the global Li_2 error implication for actual local exact-gap occurrence;

- `Assembly/ConditionalLocation/FineLocationRH.lean`: the actual last-ascent sandwich, relative limit, square-root clock tracking, and full RH wave under the stated local coverage hypothesis;

- `Helpers/NicolasQuantitativeClock.lean`: unconditional $F(x)\log x \to 0$ and $\mathcal{C}(x)/x \to 1$, with sufficient precision before exponentiation;

- `Helpers/NicolasGrowingTilt.lean`, `Helpers/NicolasGrowingTiltAsymptotic.lean`, `Helpers/FiniteTwinsSpectrum.lean` and `Helpers/ReversalCountCounterexample.lean`: growing-tilt RH criteria and the sharp growing-tilt correction, the conditional finite-twins spectrum consequence, and the generic count-monotonicity counterexample.

Paths are relative to `PrimeFactorOscillations/`. The theorem ledger maps intermediate arguments to declarations. Ordinary prose sometimes combines several compiled lemmas; these combinations are distinguished from individually named declarations.

The analytic inputs are the reciprocal-prime estimate, fixed-modulus prime progression limits and (11). Narrow leaves of the pinned Erdős 690 development supply ordinary density and symmetric bounds; its conditional classification is not an input. The width-600 count is extracted from a separately pinned prime-cluster development. The RH part additionally reuses the Robin1984 analytic development and the verified explicit-formula and complex-analytic dependencies. Section 15 additionally uses the published Alweiss–Luo all-short-interval input. This input is not a discharged Lean dependency. The complete ordinary proof above must be distinguished from formal count theorems conditional on explicit arithmetic sampling hypotheses. Those hypotheses concern actual prime positions, gap widths, separation and the number of cells, and contain no density sign or RH conclusion. No project axiom asserts the source theorem. Classical analytic results in the maintained closed development are retained as proved dependencies, not research-target axioms. The paper provides the coefficient-transfer argument and its consumers, while citing the underlying classical prime-distribution and Nicolas theorems.

The relative-location implication in Theorem 9.3 is maintained in `Assembly/ConditionalLocation/TwinCountAsymptoticBridge.lean`. Its cutoff, order and last-ascent dependency closure has passed the maintained build and a public foundational-axiom audit. The positive twin-count asymptotic is explicit in its theorem type. The resulting logarithmic location and count limits are ordinary consequences of that proof and the maintained transfers. Likewise (28) combines verified estimates, but is not advertised as a separately compiled normalized-wave declaration. These qualifications concern formal integration, not an assumption that Hardy–Littlewood or RH has been proved.

The public Challenge states two headline theorems over Mathlib primitives. Solution contains their actual proofs. The two deliberate Challenge placeholders are absent from the Solution closures. The 5 October proof audit checked the two public compiled statement pairs and 539 selected theorem closures, finding only `propext`, `Classical.choice`, and `Quot.sound`. A fresh 6 October audit, importing the maintained family owners and Solution, checks 18 exact theorem statements and dependency closures for the additions in Sections 16– 18. Every axiom set is exactly the same three standard foundations. This includes all four power-degree declarations and the un-

conditional modified-LCM endpoints. The power-degree proof was already maintained before this editorial revision; it was independently rechecked, not replaced by an assumption wrapper. Its eventual signed premise remains explicit. The RH statements have maintained consumers in Solution with their RH, false-RH or zero hypotheses explicit. An axiom audit does not remove any of these hypotheses. The two Challenge statements concern the original reversal and affine-spectrum headlines; they are not claimed to be a Challenge comparison for every later RH declaration.

The dependency record and build guide specify Windows Lean 4.34.0, source revisions and artifact hashes. The project reuses the Erdős dependency's published cache without rebuilding or redistributing it. Additional analytic compatibility artifacts still require an existing matching view; a complete fresh-machine bootstrap is not claimed. The verification record separates the initial publication, the 4 October analytic development, the completed 5 October post-paper formalization, and the PDF checks. Branch links provide mutable navigation; the records retain checked source hashes.

No numerical experiment proves an asymptotic step here. The exact inequality $1/602 < 1/601$ supplies the canonical constant's strict margin; no effective onset rank is given. Official Comparator/NanoDa replay and independent expert review have not been claimed. Kernel checking verifies proof terms for the stated definitions under the reported foundations. Source fidelity, attribution, novelty and significance require scrutiny beyond the kernel. AI assistance was used for research, proof development and exposition.

21 Second moments, simplicity defects, and prime-profile rigidity

The statements in this section separate three kinds of information: a conditional spectral identity, an unconditional inverse theorem for local probabilities, and an unconditional limit on what the normalized profile can identify. No improved zero-free boundary is asserted.

21.1 The full second moment and a simplicity defect

Assume RH. Let Γ be the set of distinct ordinates of the nontrivial zeros, including both signs, write m_γ for their analytic multiplicities, and put

$$c_\gamma = (1/4 + \gamma^2)^{-1}, \qquad C = \gamma_{\mathrm{E}} + 2 - \log(4\pi).$$

The existing zero-wave formula is absolutely convergent and has the form $W(x) = \sum_{\gamma\in\Gamma} m_\gamma c_\gamma x^{i\gamma}$. Its reality follows from conjugate pairing, and $|W(x)| \le C$.

Theorem 21.1 (Universal second moment and excess multiplicity)**.** *For any two fixed qualifying quadratic prime laws F, G and any fixed positive rank parameters α, α', let W_F, W_G denote the normalized moving-rank observables already compared to W. Under RH,*

$$\lim_{M\to\infty} \frac{1}{\log M} \sum_{n=2}^{M} \frac{W_F(n^4)W_G(n^4)}{n} = V := \sum_{\gamma\in\Gamma} m_\gamma^2 c_\gamma^2.$$

Define

$$B = \sum_{\gamma\in\Gamma} m_\gamma c_\gamma^2, \qquad \mathcal{D} = V - B.$$

Then

$$0 \le \mathcal{D} = \sum_{\gamma\in\Gamma} m_\gamma(m_\gamma - 1)c_\gamma^2 \le C^2 - B.$$

In particular $\mathcal{D} = 0$ if and only if all these zeros are simple. For every finite set $S \subset \Gamma$,

$$\sum_{\gamma\in S} m_\gamma(m_\gamma - 1)c_\gamma^2 \le \mathcal{D}, \qquad \frac{m_\gamma(m_\gamma - 1)}{(1/4 + \gamma^2)^2} \le C^2 - B.$$

Proof. For every real u, $(\log M)^{-1} \sum_{n=2}^{M} n^{iu}/n$ tends to 1 for $u = 0$ and to 0 otherwise. Absolute summability permits termwise extraction from the product of two spectra. All pairs of frequencies whose sum is zero remain; conjugate symmetry therefore gives $\sum_\gamma m_\gamma^2 c_\gamma^2$, with no simplicity assumption. The established comparison gives $W_F(n^4) - W(n^4) \to 0$ and $W_G(n^4) - W(n^4) \to 0$. Since W is bounded, their product minus $W(n^4)^2$ tends to zero. The normalized harmonic average of that error tends to zero as well. Thus the same limit holds for both probability observables. The pointwise bound $W(n^4)^2 \le C^2$ gives $V \le C^2$. Subtracting B produces the nonnegative series displayed above. Every summand is zero precisely when its positive integer multiplicity is one. Finite subsums and a single complete frequency fiber give the remaining inequalities. □

Here n^4 is a divisor cutoff for each fixed family; it is not an argument substituted into a two-variable polynomial family. Finite exceptional primes and forced-rank shifts are included in the exact reference used to define each normalized observable. These bounds do not prove simplicity, RH, or a better unconditional multiplicity estimate than those already available.

21.2 What the entire Bernoulli profile determines

For a law $\eta(p) \in [0,1]$ with $\sum_p \eta(p)^2 < \infty$, define

$$\Phi_\eta(z) = \prod_p (1 + (z-1)\eta(p)) \exp(-(z-1)\eta(p)).$$

This is a locally uniformly convergent entire product. Its zeros, with multiplicity, are exactly $1 - \eta(p)^{-1}$ for $\eta(p) > 0$. Thus the profile recovers the unordered positive probabilities. This standard canonical-product observation alone does not label them by primes.

The analytic input now has a maintained Lean proof in `Mathlib/Analysis/SpecialFunctions/Log/BernoulliProfileZeros.lean` under the project prefix. For $w = \eta(p)(z-1)$,

$$(1+w)e^{-w} - 1 = -(e^w - 1 - w)e^{-w}.$$

Consequently, on $|z-1| \le R$, each factor's defect from one is at most $R^2 e^{2R} \eta(p)^2$. Summing this bound gives locally uniform product convergence and entire continuation. At a point where no linear factor vanishes, the summable-defect nonvanishing theorem for infinite products makes the product nonzero. Conversely a zero factor makes the product zero. Thus the compiled zero-set statement is exactly

$$\Phi_\eta(z) = 0 \quad \Longleftrightarrow \quad \exists p : \ \eta(p) > 0, \quad z = 1 - \eta(p)^{-1}.$$

The same module proves invariance under every bijective relabelling. The companion maintained module `Mathlib/Analysis/SpecialFunctions/Log/BernoulliProfileMultiplicity.lean` also proves the full multiplicity identity

$$\operatorname{ord}_{z=1-1/c} \Phi_\eta = m_c, \qquad m_c = \#\{p : \eta(p) = c\}, \quad c > 0.$$

Indeed square summability makes this fibre finite. Splitting off the *entire* fibre writes the profile as the m_c-th power of $(1 + c(z-1))e^{-c(z-1)}$ times the profile of the remaining coordinates. The displayed factor has a simple zero at $1 - 1/c$; the remaining product is entire and nonzero there by the proved zero-set theorem. Orders add, giving the identity. It follows that equal profiles recover every positive probability with its complete multiplicity. The case $c = 1$, giving a zero at $z = 0$, is included. Zero probabilities contribute the unit factor and are invisible to the profile itself. The maintained companion `Mathlib/Analysis/SpecialFunctions/`

`Log/BernoulliProfileReconstruction.lean` now proves the complete unlabelled inverse statement: two admissible, square-summable arrays have equal entire profiles if and only if there is a bijection between their positive coordinates preserving every probability value. To construct it, match each finite positive-value fibre using the proved multiplicity identity, then assemble these fibre bijections over all positive values. Conversely, delete the zero coordinates, whose factors are one, and reindex the remaining product by the value-preserving bijection. No hypothesis about the Riemann hypothesis or prime distributions enters this unlabelled reconstruction. The next maintained module, `Mathlib/Analysis/SpecialFunctions/Log/BernoulliProfileTailMatching.lean`, also proves the ordered-tail matching step. If a_n, b_n have eventually strictly decreasing positive tails, $a_n \to 0$, and the same set of positive values, then there are fixed indices A, B such that

$$a_{A+n} = b_{B+n} \qquad (n \geq 0).$$

To exclude finite exceptions, take a tail value a_A smaller than every positive exceptional value in either sequence. Its matching value b_B lies in the ordered tail. Every positive value at most a_A then lies after A and after B, respectively. These two tails have the same range. Their mutual value-matching maps are strictly increasing maps of the natural numbers and are inverse to one another; each strictly increasing self-map satisfies $n \leq f(n)$, so both maps must be the identity. The Lean equal-profile corollary consumes the positive-coordinate bijection above. The maintained assembly `Assembly/PrimeProfileInverse.lean` now proves the prime-labelled tail conclusion as well: equal profiles and reciprocal asymptotics $a_n^{-1} = p_n/\nu_1 + b_1 + o(1)$ and $b_n^{-1} = p_n/\nu_2 + b_2 + o(1)$, with $\nu_i > 0$, force $\nu_1 = \nu_2$, $b_1 = b_2$, and $a_n = b_n$ eventually. Here p_n is the increasing enumeration of primes. Its proof derives tail positivity, convergence to zero and strict ordering from the reciprocal remainder, then consumes the compiled matching theorem. The existing unconditional theta PNT and exact prime-prefix identity give $p_{n+b}/p_{n+a} \to 1$ for every fixed a, b. This identifies the leading scales. The remaining integer prime difference converges to $\nu(b_1 - b_2)$, hence is eventually constant. A positive constant would translate every sufficiently large prime to a prime, contradicting the maintained prime-translation theorem. These complete consumers are implemented in `Helpers/PrimeProfilePrimeRatio.lean`, `Helpers/PrimeProfileInverseAsymptotics.lean`, and `Helpers/PrimeProfileShiftRigidity.lean`. The finite-exception conclusion is now compiled as well. The main-

tained modules `Mathlib/Analysis/SpecialFunctions/Log/BernoulliProfileIdentity.lean` and `Mathlib/Analysis/SpecialFunctions/Log/BernoulliProfileFinitePermutation.lean` split off the common tail and cancel it near $z = 1$, where it has value one. The analytic identity theorem makes the remaining finite profiles equal everywhere. Their positive coordinates admit a value-preserving bijection; the equal total number of coordinates then gives equally many zeros. Joining the two bijections gives a permutation of the exceptional prefix, extended by the identity on its entire complement. Probabilities equal to one are included. The assembled theorem `prime_profile_eq_iff_finite_permutation` proves both directions of this classification together with equality of the scale and intercept. The general finite-accumulation version is also compiled, with equality expressed as frequent agreement in a punctured neighborhood of the accumulation point.

Theorem 21.2 (Prime-coordinate rigidity)**.** *Suppose two such laws satisfy, outside finite sets,*

$$0 < \eta_i(p) < 1, \qquad \eta_i(p)^{-1} = p/\nu_i + b_i + o(1), \quad \nu_i > 0.$$

If $\Phi_{\eta_1} = \Phi_{\eta_2}$, then $\nu_1 = \nu_2$, $b_1 = b_2$, and $\eta_1(p) = \eta_2(p)$ for every sufficiently large prime. The complete laws differ only by a permutation of finitely many prime coordinates. Conversely every such finite permutation preserves the profile. Equality on a real interval, or on any set with a finite accumulation point, suffices.

Proof. Write the reciprocal remainders as $r_i(p) \to 0$. For sufficiently large consecutive primes the reciprocal increment is at least $1/\nu_i - 2\sup|r_i| > 0$. Each tail is therefore strictly ordered. Equality of entire functions gives equality of their zero multisets. After removing finitely many exceptional probabilities, the two ordered tails consequently match with one fixed prime-index shift: $\eta_1(p_{j+a}) = \eta_2(p_{j+b})$ for all sufficiently large j. The prime number theorem gives $p_{j+a}/p_{j+b} \to 1$, so comparison of the leading reciprocal terms forces $\nu_1 = \nu_2 = \nu$. Now

$$p_{j+b} - p_{j+a} = \nu(b_1 - b_2) + o(1).$$

An integer sequence with a real limit is eventually constant. If this constant were nonzero, one direction of the matching would give a fixed positive integer c such that every sufficiently large prime p has $p + c$ prime. Iteration would then make $p + pc = p(1+c)$ prime, a contradiction. Hence the constant and the shift are zero, and $b_1 = b_2$. The prime-labelled tails coincide.

Cancel the common tail product in a neighborhood of $z = 1$. The remaining finite products have the same positive-probability multiset. Enlarging to one common finite prime set gives the same number of zero probabilities too, hence a finite permutation. This also retains probabilities equal to one, whose zeros lie at $z = 0$. The converse follows by reordering finitely many factors. The identity theorem gives the last assertion. □

21.3 Sharp cubic camouflage outside the rigid class

Theorem 21.3 (Identical profiles with optimally small finite-cutoff defects)**.** *There are continuum many admissible laws $\eta(p) = p^{-1} + O(p^{-2})$, pairwise distinct on infinitely many primes, whose entire profiles are identical. They can be chosen so that for each fixed $T \geq 0$, every $N \geq 1200$ and $0 \leq t \leq T$,*

$$0 \leq L_\eta(N,t) - L_0(N,t) \leq 4800\, T^2 N^{-3},$$
$$L_\eta(N,t) = \sum_{p \leq N} \{\log(1 + t\eta(p)) - t\eta(p)\}.$$

For every nontrivial construction with infinitely many swaps and every fixed $t > 0$, the difference is not $o(N^{-3})$. These laws need not come from a fixed polynomial arithmetic family.

Proof. The established bounded-gap theorem supplies disjoint pairs $p_j < q_j \leq p_j + H$, where $H = 600$, $p_j > 2H$, and $q_j < p_{j+1}$. Swap the probabilities $1/p_j, 1/q_j$ on any chosen subset of these pairs. The resulting law obeys $|\eta(p) - 1/p| \leq 2H/p^2$. The entire product is unchanged, because its locally uniform absolute convergence permits permutation.

Every cutoff meets at most one incomplete pair. Completed pairs cancel exactly. If $p \leq N < q$ is an incomplete selected pair, the complete signed defect is

$$d_t(p,q) = \log(1 + t/q) - \log(1 + t/p) + t(1/p - 1/q).$$

Applying $\log u \leq u - 1$ to both reciprocal ratios gives

$$\frac{t^2(q-p)}{pq(q+t)} \leq d_t(p,q) \leq \frac{t^2(q-p)}{pq(p+t)} \leq \frac{t^2(q-p)}{p^3}.$$

All denominators are positive. Since $p > N - H \geq N/2$, this proves the asserted bound. At the selected cutoffs $N = p_j$ the two-sided bounds imply $p_j^3 d_t(p_j, q_j)/(q_j - p_j) \to t^2$. As $q_j - p_j \geq 1$, the error cannot be $o(N^{-3})$. Partition the pair indices into countably many infinite blocks and swap all

indices in a block according to one binary choice. This gives continuum many laws, any two differing on infinitely many coordinates. Finally their reciprocal deviations $\eta(p)^{-1} - p$ take positive and negative integer values at infinitely many swapped pairs; they have no limiting constant. Thus they lie outside the preceding rigidity hypothesis, demonstrating the significance of its $o(1)$ remainder. □

The canonical product is classical mod-Poisson machinery; the prime-coordinate rigidity and arithmetic swap comparisons are the specific conclusions considered here, without a claim of literature priority. The prime-coordinate inverse, an infinite swap construction and its exact optimal constant are now Lean-formalized. The continuum-cardinality assertion retains its ordinary proof. Additional maintained owners for completed surrounding inputs, relative to `PrimeFactorOscillations/`, are:

- `Helpers/XiWaveMoments.lean` and `Helpers/XiMultiplicityBudget.lean`;
- `Mathlib/Analysis/Fourier/LogarithmicBilinear.lean`;
- `Mathlib/Analysis/SpecialFunctions/Log/CompensatedReciprocal.lean`;
- `Mathlib/NumberTheory/Primes/Translation.lean`;
- `Assembly/PrimeProfileRigidity.lean`.

The latest promotion audit is separate from the older verification totals elsewhere in this manuscript; those historical totals do not certify this new batch or an improved analytic exponent.

21.4 Universal cutoff rigidity and recurring prime gaps

The swap construction admits a converse without a reciprocal-intercept assumption. Write $p_0 = 2 < p_1 < \cdots$, and suppose $0 \le a_j \le 1$, $\sum_j a_j^2 < \infty$, and $\Psi_a = \Psi_{(1/p_j)}$. Define the complete signed defect

$$\Delta_{a,n}(t) = \sum_{j=0}^{n} \{\log(1 + ta_j) - ta_j - \log(1 + t/p_j) + t/p_j\}, \qquad t > 0.$$

The term $t/p_j - ta_j$ is part of the observable and is retained throughout.

Theorem 21.4 (Universal adaptive cutoff inverse)**.** *For every positive probe sequence* (t_n)*,*

$$\frac{p_n^2(p_n+t_n)}{t_n^2}\Delta_{a,n}(t_n)\longrightarrow 0$$

if and only if $a_j = 1/p_{\sigma(j)}$ *for a permutation* σ *fixing every sufficiently large index. In particular, for each fixed* $t>0$*,*

$$\Delta_{a,n}(t)=o(p_n^{-3}) \quad\Longleftrightarrow\quad a \text{ is a finite permutation of } (1/p_j).$$

Proof. Multiplicity recovery shows that the positive coordinates of a are exactly the distinct numbers $1/p_j$. First suppose no coordinate is zero, so $a_j = 1/p_{\sigma(j)}$ for a permutation σ. Put $H_t(j)=\log(1+t/p_j)-t/p_j$, which is strictly increasing and negative.

For $B_n=\{0,\dots,n\}$ and $S_n=\sigma(B_n)$, cancel the common coordinates in the signed sum. If $m_n=|S_n\setminus B_n|=|B_n\setminus S_n|$, every entering weight is at least $H_t(n+1)$ and every leaving weight is at most $H_t(n)$. Consequently

$$\Delta_{a,n}(t)\ge m_n\{H_t(n+1)-H_t(n)\}\ge 0. \tag{21.1}$$

Our signed scalar comparison gives, with $p=p_n$ and $q=p_{n+1}$,

$$H_t(n+1)-H_t(n)\ge\frac{t^2(q-p)}{pq(q+t)}.$$

Eventually $q\le 2p$, by the unconditional adjacent-prime ratio limit. Since $q-p\ge 1$, uniformly for every $t>0$,

$$\frac{p^2(p+t)}{t^2}\{H_t(n+1)-H_t(n)\}\ge\frac{p(p+t)}{q(q+t)}\ge\frac14. \tag{21.2}$$

A vanishing normalized defect therefore forces $m_n=0$ at every sufficiently large index. Two successive invariant prefixes imply $\sigma(n)=n$: the image of n belongs to B_n but cannot belong to the invariant prefix B_{n-1}. Conversely, a permutation fixing a tail preserves every sufficiently large prefix, whose defect is exactly zero.

It remains to rule out a zero coordinate. After the first zero has entered B_n, the positive probabilities in that prefix correspond to a set S_n of at most n distinct prime indices. If $r=|S_n\setminus B_n|$ and $s=|B_n\setminus S_n|$, then $s\ge r+1$. The same common-coordinate cancellation, now using the boundary weight $H_t(n)$ on both sides, yields

$$\Delta_{a,n}(t)\ge(r-s)H_t(n)\ge -H_t(n).$$

For $y > 0$,

$$y - \log(1+y) \geq \frac{y^2}{2(1+y)}.$$

Indeed the difference vanishes at zero and its derivative is $y^2/(2(1+y)^2) \geq 0$. Taking $y = t/p_n$ gives

$$\frac{p_n^2(p_n+t)}{t^2}\Delta_{a,n}(t) \geq \frac{p_n}{2} \longrightarrow \infty, \tag{21.3}$$

uniformly over all positive probes. Thus a zero coordinate cannot satisfy the proposed accuracy condition. For fixed t, the normalizing factor is asymptotic to p_n^3/t^2, proving the final equivalence. □

Theorem 21.5 (Optimal distortion records the least recurring prime gap)**.** *Let G be the least positive integer occurring infinitely often as $p_{n+1} - p_n$. This integer exists by bounded-gap supply and satisfies $2 \leq G \leq 600$. Every same-profile law a that is not a finite permutation satisfies, for every positive probe sequence (t_n),*

$$\limsup_{n\to\infty} \frac{p_n^2(p_n+t_n)}{t_n^2}\Delta_{a,n}(t_n) \geq G.$$

There is one infinitely supported permutation law attaining equality for every such probe sequence. It satisfies $a(p) = 1/p + O(p^{-2})$.

Proof. Every gap smaller than G occurs only finitely often; there are only finitely many such integer gaps. Hence eventually $q - p \geq G$. An infinite permutation moves infinitely many prefixes, and at each such sufficiently large prefix (21.1) gives normalized defect at least

$$G\frac{p(p+t)}{q(q+t)} \geq G\left(\frac{p}{q}\right)^2 \longrightarrow G,$$

uniformly in $t > 0$. Zero coordinates instead give the divergent lower bound (21.3).

For attainment, choose disjoint adjacent prime pairs $p < q = p + G$ at arbitrarily large heights, and swap each pair. At any cutoff at most one pair is incomplete; at its lower endpoint the exact defect is $d_t(p,q)$. At every other prime cutoff it is zero. The two-sided signed comparison gives

$$G\left(\frac{p}{q}\right)^2 \leq \frac{p^2(p+t)}{t^2}d_t(p,q) \leq G\frac{p}{q} \leq G.$$

Both endpoints tend to G uniformly in $t > 0$, proving the claimed limsup for every probe sequence, including probes tending to zero or growing without bound. The bounded-displacement estimate proves the stated local-law error. □

Thus the optimal normalized cost among all infinite rearrangements is exactly G. In particular, cost 2 is equivalent to infinitely many twin primes. This is an inverse characterization, not a new bound on G or a zero-free region.

Corollary 21.6 (The same constant governs late ascents)**.** *Let C denote the optimal distortion constant in the preceding theorem. The critical late-ascent scale for the ordinary and generic odd laws is*

$$b_* = \frac{1}{1+C} = \frac{1}{1+G}.$$

Every $0 < b < b_*$ admits arbitrarily late simultaneous ascents in both laws; every $b > b_*$ has eventual strict descent beyond the corresponding moving-rank scale. The endpoint $b = b_*$ is left open.

Proof. Apply the maintained recurring-gap/late-ascent critical-scale theorem to the actual least recurring gap G. The preceding optimality theorem identifies $C = G$. □

The full adaptive inverse is maintained in `Assembly/PrimeProfileZeroHoles.lean`. The exact optimality theorem, including the definition and bounds $2 \le G \le 600$, its universal lower threshold for every law *not a finite permutation*, and a single witness for all positive probe sequences, is maintained in `Assembly/PrimeProfileMinimax.lean`. The formal statements use the equivalent threshold formulation: every witness defect is at most G, and for every $\varepsilon > 0$ and N some $n \ge N$ has defect at least $G - \varepsilon$. The witness also satisfies the explicit bound $|a_n - 1/p_n| \le 2G/p_n^2$ whenever $p_n \ge 2G$, retaining all small-prime exceptions. Its signed construction is in `Assembly/PrimeProfileAttainment.lean` and `Mathlib/Combinatorics/AdjacentSwapPrefix.lean`. The late-ascent corollary has its own compiled declaration. The stronger zero-hole inequality (21.3) retains its ordinary proof; the formal inverse uses the sufficient uniform bound $p_n/4$. Finite permutations have eventual zero defect and are excluded from every positive lower-distortion assertion. No literature priority is claimed.

21.5 Two-coordinate cutoff transport

For a prime-coordinate permutation σ, define the crossing width

$$m_n = \big|\sigma(\{0,\dots,n\}) \setminus \{0,\dots,n\}\big|,$$
$$D_n(t) = \frac{p_n^2(p_n+t)}{t^2}\Delta_{a,n}(t), \qquad a_j = 1/p_{\sigma(j)}.$$

Put $g_j = p_{j+1} - p_j$. The full signed defect detects more than whether a prefix moves: it distinguishes simultaneous crossings.

Lemma 21.7 (Complete two-crossing cost)**.** *For every $n \geq 1$ with $m_n \geq 2$, and every $t > 0$,*

$$D_n(t) \geq T_n \left(\frac{p_n}{p_{n+2}}\right)^2, \qquad T_n = g_{n-1} + 2g_n + g_{n+1}.$$

Proof. Write $W_t(j) = \log(1 + t/p_j) - t/p_j$, an increasing weight. Cancel common coordinates from the two prefixes. There are m_n entering and m_n leaving coordinates. Every entering coordinate has index at least $n+1$, and at least one has index at least $n+2$. Every leaving coordinate has index at most n, and at least one has index at most $n-1$. Consequently,

$$\begin{aligned}\Delta_{a,n}(t) &\geq m_n(W_t(n+1) - W_t(n)) \\ &\quad + W_t(n+2) - W_t(n+1) + W_t(n) - W_t(n-1) \\ &\geq W_t(n+1) + W_t(n+2) - W_t(n) - W_t(n-1).\end{aligned}$$

The remaining crossing contributions have been retained in this inequality; no unsigned coefficient bound is used.

The maintained scalar comparison, with a middle reference denominator, states that for $t > 0$, $0 < p \leq q$ and $p \leq r$,

$$\frac{r^2(r+t)}{t^2}\left\{\log(1+t/q) - t/q - \log(1+t/p) + t/p\right\} \geq (q-p)\left(\frac{r}{q}\right)^2.$$

Apply it to $(p,q,r) = (p_{n-1}, p_{n+1}, p_n)$ and (p_n, p_{n+2}, p_n). Replace the first ratio by the smaller common ratio p_n/p_{n+2} and add the two signed costs. Their denominator displacements sum to T_n. □

Lemma 21.8 (The second prime-geometry threshold)**.** *For $n \geq 3$, $T_n \geq 12$. Equality holds exactly when*

$$(g_{n-1}, g_n, g_{n+1}) = (2,4,2) \quad \text{or} \quad (4,2,4).$$

The exceptional initial values are $T_1 = 7$ and $T_2 = 10$; a double crossing is impossible at $n = 0$.

Proof. The four primes start beyond 3, so all three gaps are even and at least 2. Two consecutive gaps cannot both be 2: among $p, p+2, p+4$, one is divisible by 3 and exceeds 3. If the central gap is at least 4, the weighted sum is at least $2 + 2 \cdot 4 + 2 = 12$, with equality only for $(2, 4, 2)$. If the central gap is 2, both neighbours are at least 4, giving $4+2\cdot 2+4 = 12$, with equality only for $(4, 2, 4)$. The initial values use $2, 3, 5, 7, 11$ directly. □

Corollary 21.9 (Double-crossing rigidity)**.** *Fix any positive probe sequence* (t_n)*. For a law that is* not a finite permutation *and has* $m_n \geq 2$ *at arbitrarily large indices,*

$$\limsup_{n\to\infty} D_n(t_n) \geq 12.$$

Equivalently, an upper limit strictly below 12 *forces* $m_n \leq 1$ *eventually. If the upper limit is strictly below* 14 *and double crossings persist, one of the two displayed four-prime patterns occurs arbitrarily far out.*

Proof. The fixed-shift prime ratio p_n/p_{n+2} tends to 1. The first two lemmas therefore give the lower threshold uniformly over every positive probe sequence. If the integer T_n is not 12 at a sufficiently large double crossing, its parity gives $T_n \geq 14$. Such crossings would force an upper limit at least 14, contradicting the stated ceiling. Thus every sufficiently late double crossing has one of the two equality patterns. Persistent double crossings give their recurrence. □

Theorem 21.10 (A sharp probability-profile equivalence)**.** *The following statements are equivalent:*

1. *Arbitrarily late consecutive four-prime windows, starting beyond* 3*, have offsets* $(0, 2, 6, 8)$ *or* $(0, 4, 6, 10)$*.*

2. *There is one permutation* σ *with* $m_n \geq 2$ *at arbitrarily large cutoffs and*
$$\forall \varepsilon > 0 \ \exists N \ \forall n \geq N \ \forall t > 0, \qquad D_n(t) \leq 12 + \varepsilon.$$

For the law $a_j = 1/p_{\sigma(j)}$ *in the construction,* $0 < a_j \leq 1$*,* $\sum_j a_j^2 < \infty$*, and the entire canonical profile equals that of* $u_j = 1/p_j$*. It is* not a finite permutation*. For every positive probe sequence, chosen after the law,*

$$\limsup_{n\to\infty} D_n(t_n) = 12.$$

The cutoff in the upper estimate is independent of the positive probe.

Proof. Suppose the windows recur. Greedily choose an unbounded set A of their starting indices with $a + 4 \le b$ whenever $a < b$ in A. This is possible by selecting the next start beyond the previous start plus four. Set $B = A \cup (A + 1)$. No j and $j + 2$ both belong to B: such a pair would put two selected starts at distance less than four. Define

$$\sigma(j) = \begin{cases} j + 2, & j \in B, \\ j - 2, & j \notin B,\ j \ge 2,\ j - 2 \in B, \\ j, & \text{otherwise.} \end{cases}$$

The absence of two-step chains makes this an involution. In each selected block it swaps the first with the third coordinate and the second with the fourth. Both prime displacements equal six, for either offset pattern. At cutoff $n = a+1$ the distinct exterior coordinates $a+2, a+3$ enter, so double crossings recur.

For an arbitrary real weight W, completed pairs cancel exactly:

$$\sum_{j=0}^{k+1} \big(W(\sigma(j)) - W(j)\big) = \mathbf{1}_B(k+1)\big(W(k+3) - W(k+1)\big) + \mathbf{1}_B(k)\big(W(k+2) - W(k)\big).$$

This identity includes every finite-prefix term. It follows either by pairing completed swaps or by induction, retaining the two pairs that can cross the boundary.

We also need the upper comparison at a middle reference. Write $\delta_t(p, q) = \log(1 + t/q) - t/q - \log(1 + t/p) + t/p$. For $t > 0$ and $0 < p \le r \le q$,

$$\frac{r^2(r+t)}{t^2}\delta_t(p, q) \le (q - p)\left(\frac{r}{p}\right)^2.$$

Indeed, $\delta_t(p, q) = \int_p^q t^2/(x^2(x+t))\,dx \le t^2(q-p)/(pq(p+t))$. After normalization, the desired comparison follows from the nonnegative identity

$$\left\{(q - p)(r/p)^2 - \frac{(q - p)r^2(r + t)}{pq(p + t)}\right\} p^2q(p + t) = (q - p)r^2\{p(q - r) + t(q - p)\} \ge 0.$$

All denominators are positive.

Apply the exact prefix identity to $W = W_t$ and this upper comparison with reference $r = p_{k+1}$. Each present pair has displacement six. The pair

starting at $k+1$ costs at most 6, and the pair starting at k costs at most $6(p_{k+1}/p_k)^2$. Absent pairs contribute zero. Thus, for every $k \geq 0$ and $t > 0$,

$$D_{k+1}(t) \leq 6(p_{k+1}/p_k)^2 + 6.$$

The right side tends to 12, proving the upper estimate with a cutoff uniform in t. The previous lower threshold at recurring double crossings supplies arbitrarily late values at least $12 - \varepsilon$ for every positive probe sequence. This proves the exact upper limit, including the rank shift $n = k + 1$.

Reindexing preserves square summability and the canonical entire product, so the constructed sequence has the asserted probability properties and profile. A finite permutation fixes all sufficiently late coordinates and preserves all later prefixes; it cannot have recurring double crossings. The selected blocks start beyond 3, so the exceptional prime coordinates $2, 3$ are fixed. The first cutoff $n = 0$ plays no role in the eventual estimate.

Conversely, use the constant probe $t = 1$ and $\varepsilon = 1$ in statement (ii). The eventual ceiling is $13 < 14$. The compiled inverse criterion therefore forces an equality pattern at every sufficiently late double crossing. Their recurrence gives statement (i). □

The complete signed finite-set inequality is maintained in `Mathlib/Combinatorics/TwoBoundaryTransport.lean`; the actual prime-profile specialization and its middle-prime normalization are maintained in `Helpers/PrimeProfileDoubleCrossing.lean`. The prime-congruence classification is maintained in `Mathlib/NumberTheory/Primes/WeightedQuadruplet.lean`. The quantified uniform lower threshold and arbitrarily late witnesses are compiled there, as is the eventual width bound under a ceiling $C < 12$. Their limsup formulations are mathematical consequences of these quantified statements. The stronger inverse under an eventual ceiling $C < 14$ is maintained in `Helpers/PrimeProfileQuadrupletInverse.lean`. The exceptional initial values $7, 10$ retain their elementary checks.

The converse is maintained in `Assembly/PrimeProfileQuadrupletAttainment.lean`. It formally proves the actual probability-profile equivalence, with one law before every positive probe, as well as positivity, the unit bound, square summability, full profile equality, arbitrarily late lower witnesses and exclusion of finite permutation representations. Its compiled upper formulation has one cutoff for all positive probes. The sparse block selection and the complete uniform upper estimate are maintained in `Helpers/PrimeQuadrupletSupport.lean` and `Helpers/PrimeProfileTwoStepTransport.lean`; the distance-two involution, exact

signed prefix identity and double-crossing witness are reusable owners in `Mathlib/Combinatorics/TwoStepSwapPrefix.lean`. The matching middle-reference upper comparison is maintained in `Mathlib/Analysis/SpecialFunctions/Log/CompensatedReciprocalNormalization.lean`. No infinitude of either four-prime pattern is asserted.

21.6 Quantized distortion and probe independence

The whole finite transport controls the distortion of every bounded profile. For a permutation σ, put

$$S_n = \sum_{j=0}^{n} p_{\sigma(j)} - \sum_{j=0}^{n} p_j.$$

This is the exact signed prime-denominator transport. It is nonnegative, because the first $n+1$ primes have the least possible sum.

Lemma 21.11 (Localization and a uniform transport error)**.** *Let $r = p_n$, $C \geq 0$, $C < r/4$, and suppose that $D_n(t) \leq C$ for one $t > 0$. Then every entering prime is at most $r + 4C$, and every leaving prime is at least $r - C$. Moreover,*

$$0 \leq S_n \leq 4C, \qquad S_n \left(\frac{r}{r+4C}\right)^2 \leq D_n(v) \leq S_n \left(\frac{r}{r-C}\right)^2 \quad (v > 0),$$

and, uniformly over every new positive probe,

$$|D_n(v) - S_n| \leq \frac{32C^2}{p_n}.$$

Proof. Let E and L be the finite sets of entering and leaving prime values, after common coordinates cancel. They have equal cardinalities. For $\delta_t(p,q) = \log(1 + t/q) - t/q - \log(1 + t/p) + t/p$ and $N_r(t) = r^2(r+t)/t^2$, exact insertion of the common reference gives

$$D_n(t) = N_r(t) \left\{ \sum_{q \in E} \delta_t(r,q) + \sum_{p \in L} \delta_t(p,r) \right\},$$
$$S_n = \sum_{q \in E} (q - r) + \sum_{p \in L} (r - p).$$

Every retained summand is nonnegative. Thus each reference cost is at most the full cost C. The middle-reference lower comparison gives $N_r(t)\delta_t(p,r) \geq$

$r-p$ for a leaving prime. If an entering prime satisfies $q \geq 2r$, monotonicity in q gives

$$N_r(t)\delta_t(r,q) \geq N_r(t)\delta_t(r,2r) \geq r/4,$$

contrary to $C < r/4$. Hence $q < 2r$, and the lower comparison gives $N_r(t)\delta_t(r,q) \geq (q-r)(r/q)^2 \geq (q-r)/4$. This proves both localization bounds for every transported coordinate.

For any $v > 0$ and $0 < \ell \leq p \leq r \leq q \leq u$, the two scalar comparisons already proved give

$$(q-p)(r/u)^2 \leq N_r(v)\delta_v(p,q) \leq (q-p)(r/\ell)^2.$$

Use $\ell = r - C > 0$ and $u = r + 4C$, and sum this inequality on the *same* two reference transports. This proves the displayed two-sided comparison for $D_n(v)$ and S_n. Applying its lower bound at the original probe t, with $r/(r+4C) > 1/2$, gives $S_n \leq 4C$.

Write $\lambda = (r/(r+4C))^2$ and $\mu = (r/(r-C))^2$. The following nonnegative identities justify the remaining constants:

$$\{\lambda - (1 - 8C/r)\}\, r(r+4C)^2 = 48rC^2 + 128C^3,$$
$$\{1 + 4C/r - \mu\}\, r(r-C)^2 = C\{2r(r-4C) + rC + 4C^2\}.$$

Thus $1 - 8C/r \leq \lambda$ and $\mu \leq 1 + 4C/r$. Retaining the signed two-sided comparison and $0 \leq S_n \leq 4C$ gives

$$S_n - D_n(v) \leq 8CS_n/r \leq 32C^2/r,$$
$$D_n(v) - S_n \leq 4CS_n/r \leq 16C^2/r.$$

The asserted absolute bound concerns this isolated error after the complete signed transport has been compared. □

Theorem 21.12 (A probe-independent even distortion level)**.** *Fix a permutation σ. Suppose that one positive probe sequence (t_n) has eventually bounded normalized distortion: $D_n(t_n) \leq C$ for all sufficiently large n, with one fixed $C \geq 0$. Then there is an integer $K \geq 0$, independent of subsequent probes, such that*

$$\limsup_{n\to\infty} D_n(v_n) = 2K$$

for every positive sequence (v_n). More strongly, for every $\varepsilon > 0$ one cutoff gives $D_n(v) \leq 2K + \varepsilon$ for all later n and all $v > 0$; arbitrarily late cutoffs satisfy $D_n(v) \geq 2K - \varepsilon$ for all $v > 0$ *simultaneously. The level is zero exactly for a finite permutation. Consequently a law that is* not a finite permutation *has a positive even level whenever one probe has eventually bounded cost.*

Proof. Apply the lemma at every sufficiently large cutoff using the original probe t_n. Since $p_n \to \infty$, it gives a single eventual bound $0 \le S_n \le 4C$ and an error tending to zero uniformly in every new positive probe. This step only uses $p_n \ge n$, not a quantitative prime number theorem.

Once $p_n - C > 2$, every entering and leaving prime is odd. The two transported sets have equal cardinalities, so their sums have the same parity. Hence S_n is a nonnegative even integer. Only finitely many such values lie in $[0, 4C]$. At least one occurs arbitrarily late; let $2K$ be the largest value that does. Every larger value then disappears after one cutoff, so $S_n \le 2K$ eventually, whereas $S_n = 2K$ at arbitrarily late cutoffs. The error estimate in the lemma gives both uniform assertions, with the same cutoffs for all positive probes, and therefore every displayed limsup.

A finite permutation has zero defect at every sufficiently late prefix, so its level is zero. Conversely, level zero and nonnegativity give $D_n(1) \to 0$. The maintained adaptive cutoff inverse then forces the permutation to be finite. □

The localization theorem is maintained in `Helpers/PrimeProfileCostLocalization.lean`. The complete cross-probe comparison, nonnegative transport, budget $S_n \le 4C$ and explicit error $32C^2/p_n$ are maintained in `Helpers/PrimeProfileTransportComparison.lean`. The even-lattice and uniform-tail owners are `Helpers/PrimeProfileTransportParity.lean` and `Helpers/PrimeProfileTransportTail.lean`. The quantified uniform even-level theorem is maintained in `Helpers/PrimeProfileQuantization.lean`. Its single cutoff is selected before every new positive probe, and its lower witnesses are shared by all positive probes. The limsup wording and zero-level characterization above are mathematical consequences of that compiled formulation and the previously maintained cutoff inverse; they are not advertised as separate named Lean declarations. The generic finite reference transport and parity lemmas are maintained in `Mathlib/Combinatorics/FinitePrefixDisplacement.lean`; the bounded natural-sequence recurrence lemma is maintained in `Mathlib/Order/BoundedNatSequence.lean`. No independent arithmetic bound on an RH detector is inferred from this profile classification.

Theorem 21.13 (Complete shared-profile distortion alternatives)**.** *Let $a_j \in [0,1]$, $\sum_j a_j^2 < \infty$, and suppose its entire canonical profile equals the*

reciprocal-prime profile. Write

$$D_{a,n}(t) = \frac{p_n^2(p_n + t)}{t^2} \sum_{j=0}^{n} \{\log(1 + ta_j) - ta_j - \log(1 + t/p_j) + t/p_j\}.$$

Exactly one of the following alternatives holds:

1. *There is an integer $K \geq 0$ such that, for every $\varepsilon > 0$, one cutoff gives $D_{a,n}(t) \leq 2K + \varepsilon$ for all later n and every $t > 0$, while arbitrarily late cutoffs satisfy $D_{a,n}(t) \geq 2K - \varepsilon$ for every $t > 0$ simultaneously.*

2. *For every real C and every N, some $n \geq N$ satisfies $D_{a,n}(t) > C$ for every $t > 0$ simultaneously.*

Thus every positive adaptive probe sequence has the same upper limit, belonging to $\{0, 2, 4, \ldots\} \cup \{+\infty\}$. The upper limit is zero exactly when a is a finite coordinate permutation of the reciprocal-prime law. If even one coordinate of a is zero, the stronger conclusion

$$\forall C \in \mathbb{R}\ \exists M\ \forall n \geq M\ \forall t > 0 : \quad D_{a,n}(t) > C$$

holds.

Proof. First suppose that all coordinates are positive. The maintained entire-profile reconstruction gives one permutation σ with $a_j = 1/p_{\sigma(j)}$. Substituting this equality into the complete signed prefix sum gives $D_{a,n}(t) = D_n(t)$ exactly.

If the constant unit probe $D_n(1)$ is eventually bounded, the preceding theorem gives the first alternative. Otherwise, fix $C \in \mathbb{R}$ and N, and put $A = \max(C, 0)$. There are arbitrarily late cutoffs with $D_n(1) > 12A$; choose one beyond N with $p_n > 4A$. If any positive probe v had $D_n(v) \leq A$, the localization lemma, applied with v as the original probe, would give

$$D_n(1) \leq S_n + \frac{32A^2}{p_n} \leq 4A + 8A = 12A,$$

a contradiction. Thus every positive probe exceeds $A \geq C$ at this same cutoff. This proves the second alternative without selecting different cutoffs for different probes.

If $a_i = 0$, the maintained zero-hole estimate instead gives $D_{a,n}(t) \geq p_n/4$ for every $n \geq i$ and every $t > 0$. Since $p_n \geq n$, it gives the asserted eventual uniform divergence. Nonnegativity of the coordinates makes these two cases exhaustive.

The alternatives are disjoint: in the first, use $\varepsilon = 1$; in the second, use $C = 2K + 2$ and a cutoff beyond that upper envelope. Their quantified bounds give the stated common upper limit. A finite permutation has eventual zero defect. Conversely, the zero level is incompatible with a zero coordinate, so reconstruction reduces it to the positive permutation case, where the preceding cutoff inverse applies. In particular, a law that is *not a finite permutation* has a positive even level or infinite upper limit. □

The quantitative implication $D_n(v) \le C \Rightarrow D_n(1) \le 12C$ for $p_n > 4C$, the shared unbounded witnesses, and the permutation dichotomy are compiled in `Helpers/PrimeProfileUniformDichotomy.lean`. The general-law dichotomy and eventual uniform zero-hole divergence are compiled in `Assembly/PrimeProfileDistortionClassification.lean`. The exact coordinate-substitution identity is maintained in `Assembly/PrimeProfileCutoffInverse.lean`. The formal alternatives use the displayed quantified upper and lower envelopes; their upper-limit wording, exclusivity and zero-level characterization are mathematical consequences, not additional named Lean declarations. The prime 2 is retained in every finite prefix; evenness is used only after the localization window excludes it. No new prime-pattern infinitude or improvement of the certified 7/8 RH boundary follows from this classification.

Theorem 21.14 (Exact finite-accuracy transport readout)**.** *Fix a prime-coordinate permutation σ and a cutoff with $r = p_n \ge 3$. Suppose $C \ge 0$, $D_n(t) \le C$ for one $t > 0$, and $32C^2 < r$. Then there is one integer $K \ge 0$ such that*

$$S_n = 2K, \qquad |D_n(v) - 2K| < 1 \quad \textit{for every } v > 0.$$

For each positive probe v separately, K is the unique nonnegative integer j with $|D_n(v) - 2j| < 1$.

Proof. The precision hypothesis implies $C < r/4$: otherwise $32C^2 \ge 2r^2 \ge 6r > r$. Hence $r - C > 3r/4 \ge 9/4 > 2$. Localization excludes the exceptional prime 2 from the transported sets, so their exact signed sum is $S_n = 2K$ for some $K \ge 0$. The uniform error estimate gives $|D_n(v) - 2K| \le 32C^2/r < 1$ for every $v > 0$. Two distinct even integers have distance at least 2; they cannot both lie at distance strictly less than 1 from the same observation. This proves uniqueness for each probe, with the same recovered value for every probe. □

The theorem has no global boundedness assumption. For example, suppose one positive adaptive probe has eventual cost at most $C_n \geq 0$, where $C_n^2/p_n \to 0$. Then, at every sufficiently late cutoff, *every* positive probe determines the exact moving transport S_n by its unique nearest even integer. Moreover the uniform readout error is at most $32C_n^2/p_n \to 0$. The levels may grow with n; no constant distortion level is assumed. The finite theorem, including per-probe uniqueness, is compiled in `Helpers/PrimeProfileTransportReadout.lean`. The growing-budget statement is a mathematical consequence of that finite certificate and the maintained uniform error estimate, not a separately named limit declaration.

Theorem 21.15 (Pairwise recovery and discrete separation)**.** *Let σ, τ be two permutations of the prime coordinates. Suppose one positive adaptive probe sequence (t_n) and one $C \geq 0$ satisfy*

$$D_n^\sigma(t_n) \leq C, \qquad D_n^\tau(t_n) \leq C$$

for all sufficiently large n. Then

$$|D_n^\sigma(t_n) - D_n^\tau(t_n)| \longrightarrow 0 \quad \Longleftrightarrow \quad \sigma(n) = \tau(n) \text{ for all sufficiently large } n.$$

More quantitatively, if the two permutations do not agree eventually, then for every $\varepsilon > 0$ and every N there is $n \geq N$ such that

$$|D_n^\sigma(v) - D_n^\tau(w)| \geq 2 - \varepsilon \qquad \text{for every } v, w > 0$$

simultaneously. The cutoff is shared by all pairs of probes. The hypothesis of nonagreement excludes pairs differing only finitely; in particular, at least one permutation is not a finite permutation*.*

Proof. At a cutoff with $p_n > 4C$ and $p_n - C > 2$, let $S_n^\sigma = 2K$ and $S_n^\tau = 2L$ be the exact signed transports. For any two positive probes, the two previously proved error bounds give

$$|S_n^\sigma - S_n^\tau| \leq |D_n^\sigma(v) - D_n^\tau(w)| + \frac{64C^2}{p_n}.$$

Consequently

$$|D_n^\sigma(v) - D_n^\tau(w)| + \frac{64C^2}{p_n} < 2 \quad \Longrightarrow \quad S_n^\sigma = S_n^\tau,$$

because distinct even integers have distance at least 2.

Exact consecutive-prefix subtraction gives

$$S^{\sigma}_{n+1} - S^{\sigma}_{n} = p_{\sigma(n+1)} - p_{n+1},$$

and the same identity holds for τ. Thus eventual equality of the signed transport sequences forces eventual equality of the prime coordinates, hence of their indices. If σ and τ do not agree eventually, the transports therefore differ at arbitrarily late cutoffs. Choose one beyond N, both cost cutoffs, and a threshold where

$$p_n > \max(4C,\, 2 + C,\, 64C^2/\varepsilon).$$

Its two even transports differ by at least 2. The displayed error comparison then proves the claimed $2 - \varepsilon$ separation for every pair of positive probes at this one cutoff.

In particular, asymptotically equal observations are incompatible with eventual nonagreement: use separation with $\varepsilon = 1$ and the original probes. Conversely, if the permutations agree past a cutoff, their images of every later finite prefix are identical. Indeed those images have the same complement, namely the common image of the remaining tail. All weighted prefix sums, and therefore the normalized defects at each common probe, then agree exactly. □

The finite comparison, shared separation witnesses, and pairwise inverse are compiled in `Helpers/PrimeProfilePairwiseRigidity.lean`. The formal inverse states convergence by its explicit ε–cutoff condition. The reusable consecutive-defect identity, two-prefix coordinate recovery, equal-tail prefix-image identity and eventual exact-defect equivalence are compiled in `Mathlib/Combinatorics/FinitePrefixDisplacement.lean`. This recovery compares two possibly infinite rearrangements; it does not assume either is the reference order. It identifies their tails even though their entire canonical profiles are equal. The separation constant 2 is proved, with no claim here that it is attained by two distinct classes.

21.7 Two-moment compensation and a larger exact-readout range

The preceding readout uses one logarithmic probe. Two finite probability moments cancel its first nonlinear transport error. Write $I_n = \{0, \dots, n\}$, $A_n = \sigma(I_n)$, $r = p_n$, and

$$E_n = A_n \setminus I_n, \qquad L_n = I_n \setminus A_n.$$

Both sets are finite and have equal cardinality. Define

$$\Delta_j = \sum_{i=0}^{n} \left(p_i^{-j} - p_{\sigma(i)}^{-j} \right), \qquad Q_n = 3r^2\Delta_1 - r^3\Delta_2.$$

Thus Q_n uses only the first two finite power sums of the actual Bernoulli probabilities and their reference probabilities. These are finite-prefix observations; equality of the entire canonical profiles does not assert equality of these labelled finite sums.

Theorem 21.16 (Cubic compensated transport and exact readout)**.** *Suppose $D_n(t) \le C$ for one $t > 0$, where $C \ge 0$ and $C < r/4$. Then the full signed transport satisfies*

$$0 \le S_n - Q_n \le \frac{64C^3}{r^2}.$$

If instead one assumes $r \ge 3$ and $32C^3 < r^2$, the condition $C < r/4$ follows automatically, and $S_n = 2K$ is the unique nonnegative even integer in $[Q_n, Q_n + 2)$. In particular, K is determined by the compensated observation without knowing S_n.

Proof. Set

$$\Phi_r(p) = -\frac{3r^2}{p} + \frac{r^3}{p^2}.$$

For positive p, r, direct algebra gives the exact identity

$$\Phi_r(p) - \Phi_r(r) = (p - r) - \frac{(p - r)^3}{p^2}.$$

By the definition of the finite moments, $Q_n = \sum_{i=0}^{n} \Phi_r(p_{\sigma(i)}) - \sum_{i=0}^{n} \Phi_r(p_i)$. Cancel the common coordinates and insert the common reference $\Phi_r(r)$. Equal cardinalities cancel its constant contribution. The analogous reference decomposition of S_n therefore gives

$$S_n - Q_n = \sum_{i \in E_n} \frac{(p_i - r)^3}{p_i^2} + \sum_{i \in L_n} \frac{(r - p_i)^3}{p_i^2}.$$

This retains the compensation and the unit coefficient of S_n; both displayed remainders are nonnegative.

The proved one-probe localization gives $0 \le p_i - r \le 4C$ for $i \in E_n$ and $0 \le r - p_i \le C$ for $i \in L_n$. On the entering set, $p_i \ge r$. On the leaving set,

$p_i \geq r - C > 3r/4$. Hence, on either set, the displacement divided by p_i is at most $4C/r$. It follows that

$$\begin{aligned} S_n - Q_n &\leq \frac{16C^2}{r^2}\left(\sum_{i\in E_n}(p_i - r) + \sum_{i\in L_n}(r - p_i)\right) \\ &= \frac{16C^2}{r^2}S_n \leq \frac{64C^3}{r^2}, \end{aligned}$$

using the maintained bound $S_n \leq 4C$.

For the readout statement, if $C \geq r/4$ then $32C^3 \geq r^3/2 \geq 3r^2/2 > r^2$, a contradiction. Thus $r - C > 3r/4 \geq 9/4 > 2$. Localization excludes the exceptional prime 2 from both transported sets, but retains it in the original prefixes. Their equal cardinalities force $S_n = 2K$, $K \geq 0$. The error bound and the precision hypothesis give $0 \leq 2K - Q_n < 2$. An interval of the form $[Q_n, Q_n + 2)$ cannot contain two distinct even integers. This proves the asserted recovery. □

For growing budgets, $C_n^3/p_n^2 \to 0$ gives eventual exact recovery with error tending to zero. In particular, the sufficient scale is now $C_n = o(p_n^{2/3})$, compared with $o(p_n^{1/2})$ for the preceding single-probe readout. This is a stronger observation theorem. It supplies no new arithmetic upper bound for the probe cost itself, so it changes no RH boundary.

The exact scalar identity and signed entering/leaving comparisons are compiled in `Mathlib/Analysis/SpecialFunctions/Log/CubicReciprocalTransport.lean`. The two-moment identity, complete signed comparison, cubic error and unique finite even readout are compiled in `Helpers/PrimeProfileCubicReadout.lean`. The growing-budget statement is a mathematical consequence of the finite certificates, rather than a separately named limit declaration. The coefficient cancellation is elementary; no literature-priority claim is made for this quantitative application to prime-profile transport.

21.8 Geometric compensation and stable logarithmic-moment readout

For an integer $m \geq 0$, define the centered reciprocal polynomial

$$\Psi_{m,r}(p) = r\sum_{j=1}^{m}(1-r/p)^j, \qquad P_{m,n} = \sum_{i=0}^{n}\big(\Psi_{m,r}(p_{\sigma(i)}) - \Psi_{m,r}(p_i)\big), \quad r = p_n.$$

The empty sum is zero. Equivalently,

$$P_{m,n} = r\sum_{j=1}^{m} M_{j,n}, \qquad M_{j,n} = \sum_{i=0}^{n}\big((1-r/p_{\sigma(i)})^j - (1-r/p_i)^j\big).$$

These are finite centered power-sum differences of the probabilities. Binomial expansion expresses $M_{j,n}$ using their ordinary power sums of degrees at most j; the constant terms cancel. For $m = 2$, the polynomial is $\Phi_r(p) + 2r$, so $P_{2,n} = Q_n$.

Theorem 21.17 (All even orders preserve the exact transport)**.** *Suppose one positive probe satisfies $D_n(t) \le C$, with $0 \le C < r/4$. For every even integer $m = 2k \ge 0$,*

$$0 \le S_n - P_{m,n} \le 4C(4C/r)^m.$$

If $r \ge 3$ and $4C(4C/r)^m < 2$, the exact transport $S_n = 2K$ is the unique nonnegative even integer in $[P_{m,n}, P_{m,n} + 2)$.

Proof. The finite geometric identity gives, for every $p > 0$,

$$\Psi_{m,r}(p) = (p-r) - \frac{(p-r)^{m+1}}{p^m}, \qquad \Psi_{m,r}(r) = 0.$$

One may verify the identity by induction on m; the successor adds $r(1 - r/p)^{m+1}$ and cancels the preceding remainder. After the same common-coordinate cancellation and equal-cardinality reference decomposition as before, evenness of m gives exactly

$$S_n - P_{m,n} = \sum_{i\in E_n} \frac{(p_i - r)^{m+1}}{p_i^m} + \sum_{i\in L_n} \frac{(r-p_i)^{m+1}}{p_i^m}.$$

Every summand is nonnegative. The proved localization implies that the displacement divided by p_i is at most $4C/r$ on both sets. Each summand is therefore at most $(4C/r)^m$ times its displacement. Their displacement sum is $S_n \le 4C$, proving the bound. This argument includes $m = 0$: both sides reduce to the existing transport bound. For the readout statement, $r - C > 2$ excludes the exceptional prime from the transported sets, and the error interval of width less than two identifies the even integer uniquely. □

Theorem 21.18 (Stable recovery from logarithmically many centered moments)**.** *Suppose $r = p_n \ge 3$ and one positive probe satisfies $D_n(t) \le r/8$. Put*

$$k = \lfloor \log_4 r \rfloor + 1, \qquad m = 2k.$$

There is an integer $K \geq 0$ with $S_n = 2K$ such that, simultaneously for every real observation y,

$$|y - P_{m,n}| < \tfrac{1}{2} \quad \Longrightarrow \quad |y - 2K| < 1.$$

For each such y, K is the unique nonnegative integer j with $|y - 2j| < 1$. Thus nearest-even rounding recovers the exact transport despite an observation error of almost one half.

Proof. Since $r < 4^k$, the preceding signed estimate with $C = r/8$ gives

$$0 \leq S_n - P_{m,n} \leq \frac{r}{2}(1/2)^{2k} = \frac{r}{2 \cdot 4^k} < \frac{1}{2}.$$

The localization window satisfies $r - C = 7r/8 \geq 21/8 > 2$, so $S_n = 2K$ for some $K \geq 0$. If $|y - P_{m,n}| < 1/2$, the retained one-sided error gives $-1 < y - 2K < 1/2$, hence $|y - 2K| < 1$. Two distinct even integers cannot both be at distance less than one from y, proving uniqueness with the same K for every observation. □

The number of centered moments is at most $\log_2 r + 2$. The noiseless readout needs only $m = 2\lfloor \log_4 r \rfloor$, using the interval of width two; the two extra moments above leave a fixed observation-error allowance. For example, if each centered difference $M_{j,n}$ is supplied with absolute error at most $1/(4rm)$, the combined observation $y = r \sum_{j=1}^{m} \widehat{M}_{j,n}$ has error at most $1/4$, so the stable readout applies. This last statement follows by the finite sum inequality. Its precision concerns the centered differences; expanding them into raw powers requires retaining the binomial coefficient amplification.

The finite geometric identity, reference value and arbitrary-order remainder are compiled in `Mathlib/Analysis/SpecialFunctions/Log/ReciprocalMomentCore.lean`. The signed coordinate comparisons are compiled in `Mathlib/Analysis/SpecialFunctions/Log/ReciprocalMomentTransport.lean`. The complete signed estimate, finite readout, explicit degree 2 $\mathrm{Nat.log}_4(p_n)$ and uniform noisy readout with degree $2(\mathrm{Nat.log}_4(p_n) + 1)$ are compiled in `Helpers/PrimeProfileMomentReadout.lean`. Here the natural-number logarithm is the integer base-four logarithm. The real-logarithm moment-count inequality and the per-moment measurement-error consequence are ordinary mathematical consequences, not separate named Lean declarations. The error theorem retains $r = 2$; the stated even-integer

recovery certificates explicitly require $r \geq 3$. The construction improves finite observation accuracy under an assumed arithmetic cost bound, and leaves the certified RH boundary unchanged.

21.9 Uniform recovery on the full localization window

The actual coordinate denominators sharpen the preceding geometric estimate. No separation from the endpoint $C = r/4$ is needed.

Theorem 21.19 (Full-window signed accuracy and stable readout)**.** *Suppose one positive probe satisfies $D_n(t) \leq C$, where $0 \leq C < r/4$ and $r = p_n$. For every even order $m = 2k \geq 0$,*

$$0 \leq S_n - P_{m,n} \leq 2^{-m} S_n \leq 4C\, 2^{-m}.$$

If $r \geq 3$, put

$$k = \lfloor \log_4(2r) \rfloor + 1, \qquad m = 2k.$$

Then there is one integer $K \geq 0$, with $S_n = 2K$, such that every observation y satisfying $|y - P_{m,n}| < 1/2$ has $2K$ as its unique nonnegative even integer at distance less than one. The degree is independent of C throughout $0 \leq C < r/4$.

Proof. For an entering coordinate q, localization gives $q - r \leq 4C < r$, hence $q < 2r$ and

$$\frac{q-r}{q} < \frac{1}{2}.$$

For a leaving coordinate p, it gives $r - p \leq C < r/4$, hence $p > 3r/4$ and

$$\frac{r-p}{p} < \frac{1}{3} < \frac{1}{2}.$$

The exact even-order remainder from the preceding theorem therefore satisfies, with its two signs retained,

$$\begin{aligned} S_n - P_{m,n} &= \sum_{i \in E_n} (p_i - r) \left(\frac{p_i - r}{p_i} \right)^m + \sum_{i \in L_n} (r - p_i) \left(\frac{r - p_i}{p_i} \right)^m \\ &\leq 2^{-m} \left(\sum_{i \in E_n} (p_i - r) + \sum_{i \in L_n} (r - p_i) \right) = 2^{-m} S_n. \end{aligned}$$

The maintained bound $S_n \leq 4C$ completes the uniform error estimate, including $m = 0$.

For the chosen degree, $2r < 4^k$, so

$$0 \le S_n - P_{m,n} \le \frac{4C}{4^k} < \frac{r}{4^k} < \frac{1}{2}.$$

Since $r - C > 2$, the same exceptional-prime argument gives $S_n = 2K$. Combining the retained one-sided error with $|y - P_{m,n}| < 1/2$ yields $-1 < y - 2K < 1/2$. Thus $|y - 2K| < 1$, and distinct even integers again prove uniqueness with the same K before all possible observations are chosen. □

The degree is at most $\log_2 r + 3$. This covers every budget in the full localization window, including budgets approaching its upper endpoint, with no deterioration of the moment count or the observation noise allowance. The previously stated per-centered-moment accuracy condition applies with this new value of m.

The arbitrary relative-ratio remainder and the two half-window coordinate comparisons are compiled in `Mathlib/Analysis/SpecialFunctions/Log/ReciprocalMomentWindow.lean`. The complete signed sandwich, retained $2^{-m}S_n$ error, and uniform noisy readout are compiled in `Helpers/PrimeProfileFullWindowReadout.lean`. The formal degree uses $\mathrm{Nat.}\log_4(2p_n)$; the real-logarithm upper bound on the degree is an ordinary consequence. As before, the error estimate includes the first prime and the even-integer readout explicitly assumes $p_n \ge 3$. This is a stronger observation certificate under the same localization hypothesis; it supplies no new arithmetic estimate of that hypothesis.

21.10 Polynomial square filters preserve the full transport

The geometric polynomial is one choice in a larger class of finite probability observables. This gives freedom to suppress the complete remainder while retaining the principal transport coefficient exactly.

Theorem 21.20 (Arbitrary normalized polynomial filters)**.** *Let $H \in \mathbb{R}[X]$ satisfy $H(0) = 1$, and suppose*

$$|H(x)| \le \eta \qquad (1/2 \le x \le 4/3).$$

There is a polynomial G with $\deg G \le 2 \deg H$ such that, for every $p > 0$,

$$\Phi_{H,r}(p) := (p - r)\big(1 - H(r/p)^2\big) = rG(r/p).$$

Define the complete signed observable

$$A_{H,n} = \sum_{i=0}^{n} \Phi_{H,r}(p_{e(i)}) - \sum_{i=0}^{n} \Phi_{H,r}(p_i), \qquad r = p_n.$$

Under the same hypothesis $D_n(t) \le C$ at one positive probe, with $0 \le C < r/4$,

$$0 \le S_n - A_{H,n} \le \eta^2 S_n \le 4C\eta^2.$$

Thus $A_{H,n}$ is a finite linear combination of probability moments of orders at most $2 \deg H$, with principal coefficient one.

Proof. Put $R = 1 - H^2$. Since $R(0) = 0$, write $R = XQ$ and set $G = (1 - X)Q$. Then $\deg G \le 2 \deg H$; the zero polynomial case is immediate. For $p \neq 0$,

$$rG(r/p) = r(1 - r/p)Q(r/p) = (p - r)R(r/p) = \Phi_{H,r}(p).$$

The constant coefficient of G cancels between the two equal-size prefixes, so this is genuinely a finite probability-moment observable.

Cancel the common coordinates before estimating anything. On E_n, localization gives $r \le p_i < 2r$; on L_n, it gives $3r/4 < p_i \le r$. Consequently $r/p_i \in [1/2, 4/3]$ on both retained supports. No such assertion is needed for a common small-prime coordinate. Equal cardinalities give the exact signed identity

$$\begin{aligned} S_n - A_{H,n} = &\sum_{i \in E_n} (p_i - r) H(r/p_i)^2 \\ &+ \sum_{i \in L_n} (r - p_i) H(r/p_i)^2. \end{aligned}$$

Every displayed displacement is nonnegative. Applying the filter bound to this complete remainder gives

$$0 \le S_n - A_{H,n} \le \eta^2 \left(\sum_{i \in E_n} (p_i - r) + \sum_{i \in L_n} (r - p_i) \right) = \eta^2 S_n.$$

Finally use the already proved $S_n \le 4C$. The principal term has not been discarded or bounded separately. □

The polynomial realization and degree bound are compiled in
`Mathlib/Analysis/SpecialFunctions/Log/`

`PolynomialReciprocalTransport.lean`. The actual profile polynomial representation, complete signed comparison and retained $\eta^2 S_n$ error are compiled in `Helpers/PrimeProfileSquareFilter.lean`. These statements allow the identity profile and finite permutations; the recovered transport may be zero.

21.11 An accelerated quadratic filter and fewer moments

Consider the explicit polynomial

$$B(x) = 1 - \frac{528}{217}x + \frac{288}{217}x^2 = \frac{2(12x-11)^2 - 25}{217}.$$

It satisfies $B(0) = 1$. This is the normalized shifted quadratic Chebyshev polynomial; the construction below uses its explicit formula.

Theorem 21.21 (Accelerated stable moment readout)**.** *Set $H_k = B^k$ and $A_{k,n} = A_{H_k,n}$, for each integer $k \geq 0$. Under $D_n(t) \leq C$ at one positive probe, $0 \leq C < r/4$, the observable uses probability moments of degree at most $4k$, and*

$$0 \leq S_n - A_{k,n} \leq \left(\frac{25}{217}\right)^{2k} S_n \leq 64^{-k} S_n \leq 4C\, 64^{-k}.$$

If $r \geq 3$, choose

$$k = \lfloor \log_{64}(2r) \rfloor + 1.$$

There is a single $K \geq 0$ with $S_n = 2K$ such that every observation y with $|y - A_{k,n}| < 1/2$ satisfies $|y - 2K| < 1$, and no other nonnegative even integer has distance less than one from y.

Proof. For $1/2 \leq x \leq 4/3$, the quantity $12x - 11$ lies in $[-5, 5]$. Hence

$$-25 \leq 2(12x-11)^2 - 25 \leq 25, \qquad |B(x)| \leq \frac{25}{217}.$$

It follows that $H_k(0) = 1$, $\deg H_k \leq 2k$, and $|H_k(x)| \leq (25/217)^k$ throughout the retained window. The preceding theorem therefore gives the exact-rate error. The rational inequality $(25/217)^2 < 1/64$ gives the stated convenient geometric bound, also valid at $k = 0$.

For the specified k, we have $2r < 64^k$, and consequently

$$0 \leq S_n - A_{k,n} \leq \frac{4C}{64^k} < \frac{r}{64^k} < \frac{1}{2}.$$

The localization inequality $r - C > 2$ excludes the prime 2 from the transported coordinates. Equal entering and leaving cardinalities then imply $S_n = 2K$ for some $K \geq 0$, as in the earlier parity proof. Combining the one-sided error with $|y - A_{k,n}| < 1/2$ gives $-1 < y - 2K < 1/2$. Distinct even integers are separated by two, which proves uniqueness uniformly over all observations. □

At the same degree $m = 4k$, the previous geometric certificate gave $2^{-m}S_n = 16^{-k}S_n$; the quadratic filter gives $64^{-k}S_n$. Thus the certified remainder is smaller by a factor 4^{-k}. The explicit degree is at most

$$4\big(\lfloor \log_{64}(2r) \rfloor + 1\big) \leq \frac{2}{3}\log_2 r + \frac{14}{3},$$

compared with the earlier $\log_2 r + 3$ bound. The leading coefficient in the logarithmic moment count is reduced by one third. This real-logarithm comparison is an ordinary consequence of the compiled integer-logarithm certificate; it is not a separate Lean theorem.

The fixed noise allowance concerns the combined observable. It does not assert that all raw probability moments can be measured with the same accuracy. If $G(x) = \sum_{j=0}^{m} a_j x^j$ and the measured raw moment differences have errors at most ϵ_j, their induced observable error is bounded by

$$r \sum_{j=1}^{m} |a_j| r^j \epsilon_j.$$

Requiring this quantity to be less than $1/2$ suffices. The constant moment cancels exactly.

The explicit quadratic normalization, degree and all-power window bounds are compiled in `Mathlib/Analysis/SpecialFunctions/Log/QuadraticReciprocalFilter.lean`. The degree-$4k$ profile representation, full $64^{-k}S_n$ comparison, and explicit noisy even-integer recovery are compiled in `Helpers/PrimeProfileQuadraticFilterReadout.lean`. The sharper $(25/217)^{2k}$ error is a consequence of the compiled general filter theorem and compiled norm bound. The error statements retain the first prime; the even-integer readout assumes $p_n \geq 3$ explicitly. No minimax optimality is asserted. This improves observation accuracy under the same arithmetic cost hypothesis. It does not establish a stronger bound on that cost or change the certified RH boundary.

21.12 All-order filters and a halved logarithmic moment count

The quadratic can be replaced by a polynomial of every order. The required uniform bound and nonzero denominator follow from an elementary coupled recurrence; no trigonometric continuation is needed.

Define polynomials T_k, U_k by

$$T_0 = 1, \quad U_0 = 0, \qquad \begin{cases} T_{k+1} = XT_k - (1 - X^2)U_k, \\ U_{k+1} = T_k + XU_k. \end{cases}$$

Here U_k is the shifted second-kind coordinate, so the initial zero polynomial is part of the definition. Induction gives $\deg T_k \le k$ and $\deg U_k \le k - 1$ for $k \ge 1$, with $U_0 = 0$ handled separately.

Lemma 21.22 (Unit-window invariant and positive normalizer)**.** *For every $k \ge 0$ and real x,*

$$T_k(x)^2 + (1 - x^2)U_k(x)^2 = 1.$$

Consequently $|T_k(x)| \le 1$ on $[-1, 1]$. At the exterior point $11/5$, the normalizer $a_k = T_k(11/5)$ satisfies

$$a_k > 0, \qquad 4^k \le 2a_k.$$

Proof. Writing $d = 1 - x^2$, the recursion preserves the quadratic form because

$$(xT - dU)^2 + d(T + xU)^2 = T^2 + dU^2.$$

The initial value is one. For $|x| \le 1$ its second term is nonnegative, so $T_k(x)^2 \le 1$, including both endpoints.

For the exterior estimate, put $b_k = U_k(11/5)$. The exact recursion is

$$a_{k+1} = \frac{11}{5}a_k + \frac{96}{25}b_k, \qquad b_{k+1} = a_k + \frac{11}{5}b_k.$$

We retain the coupled inequalities

$$b_k \ge 0, \qquad a_k \ge \frac{9}{5}b_k, \qquad V_k := a_k + \frac{9}{5}b_k \ge 4^k.$$

They hold at $k = 0$. Assuming them at k, the next b is nonnegative, and

$$a_{k+1} - \frac{9}{5}b_{k+1} = \frac{2}{5}a_k - \frac{3}{25}b_k \ge \frac{3}{5}b_k \ge 0.$$

Moreover

$$V_{k+1} = 4V_k + \frac{3}{5}b_k \geq 4^{k+1}.$$

Thus all three inequalities hold by induction. Since $V_k \leq 2a_k$, we obtain $4^k \leq 2a_k$ and $a_k > 0$. The normalizing denominator has therefore been controlled before division. □

Define

$$H_k(x) = \frac{T_k((11-12x)/5)}{a_k}, \qquad \mathcal{A}_{k,n} = A_{H_k,n}.$$

Then $H_k(0) = 1$, $\deg H_k \leq k$, and the affine map sends $[1/2, 4/3]$ into $[-1, 1]$. The preceding lemma gives

$$|H_k(x)| \leq \frac{1}{a_k} \leq 2\,4^{-k} \qquad (1/2 \leq x \leq 4/3).$$

Theorem 21.23 (All-order stable readout)**.** *Under the same one-positive-probe hypothesis $D_n(t) \leq C$, where $0 \leq C < r/4$ and $r = p_n$, the observable $\mathcal{A}_{k,n}$ uses probability moments of degree at most $2k$, and*

$$0 \leq S_n - \mathcal{A}_{k,n} \leq 4\,16^{-k} S_n \leq 16C\,16^{-k} \qquad (k \geq 0).$$

If $r \geq 3$, choose

$$k = \lfloor \log_{16}(8r) \rfloor + 1.$$

There is a single $K \geq 0$ with $S_n = 2K$ such that every observation y satisfying $|y - \mathcal{A}_{k,n}| < 1/2$ has $2K$ as its unique nonnegative even integer at distance less than one.

Proof. Apply the complete signed square-filter theorem with $\eta = 2\,4^{-k}$. The principal coefficient remains one, and the actual error is still

$$\sum_{i \in E_n} (p_i - r) H_k(r/p_i)^2 + \sum_{i \in L_n} (r - p_i) H_k(r/p_i)^2.$$

The squared norm is $4\,16^{-k}$, giving the stated error bound. The polynomial representation has degree at most $2 \deg H_k \leq 2k$.

For the displayed choice of k, $8r < 16^k$. Hence

$$0 \leq S_n - \mathcal{A}_{k,n} \leq \frac{16C}{16^k} < \frac{4r}{16^k} < \frac{1}{2}.$$

As before $r - C > 2$, so the retained entering and leaving primes are odd and equal cardinalities imply $S_n = 2K$ with $K \geq 0$. The combined error bounds give $-1 < y - 2K < 1/2$. Separation of even integers proves uniqueness for every observation with this same K. □

The explicit degree now satisfies

$$m \le 2\big(\lfloor \log_{16}(8r) \rfloor + 1\big) \le \frac{1}{2}\log_2 r + \frac{7}{2}.$$

Thus its leading logarithmic coefficient is half the original full-window geometric certificate's coefficient. At a matched even degree $m = 2k$, the new complete error bound is $4\,4^{-m}S_n$, compared with $2^{-m}S_n$ previously; it is strictly smaller for every $m > 2$. These real-logarithm and matched-degree comparisons are ordinary consequences of the compiled formulas. The observation-noise allowance remains a condition on the combined observable. The coefficient-weighted raw-moment accuracy requirement from the preceding subsection still applies.

The recurrence, degree bounds, quadratic invariant, unit-window norm and rational growth packet are compiled in `Mathlib/Analysis/SpecialFunctions/Log/ReciprocalChebyshevCore.lean`. The positive-denominator normalization, affine degree bound and uniform filter norm are compiled in `Mathlib/Analysis/SpecialFunctions/Log/ReciprocalChebyshevFilter.lean`. The actual probability-moment representation, complete signed error and explicit integer-logarithm noisy recovery are compiled in `Helpers/PrimeProfileChebyshevReadout.lean`. All readout certificates are exported together by `Assembly/PrimeProfileReadouts.lean`. The error theorem includes $p_n = 2$; even-integer recovery assumes $p_n \ge 3$. Identity profiles and finite permutations remain allowed. No minimax optimality or external-priority claim is made. The gain concerns finite observation accuracy under an assumed arithmetic cost bound, and supplies no new RH boundary.

21.13 Exact spectral error and cost-adaptive recovery

The coupled argument admits an exact refinement. This keeps both characteristic branches of the normalizer and gives a sufficient recovery condition in terms of the available cost budget.

For a real exterior point $a \ge 1$, choose $0 \le c \le a$ with $c^2 \le a^2 - 1$, and put $\delta = a^2 - 1 - c^2 \ge 0$. Writing $A = T_k(a)$ and $B = U_k(a)$, the exact step identities are

$$T_{k+1}(a) - cU_{k+1}(a) = (a - c)(A - cB) + \delta B,$$
$$T_{k+1}(a) + cU_{k+1}(a) = (a + c)(A + cB) + \delta B.$$

They preserve the cone $B \ge 0$, $A - cB \ge 0$ and imply $(a + c)^k \le A + cB \le 2A$. In particular the exterior normalizer is positive. On the characteristic

boundary $\delta = 0$, both linear forms have exact powers:

$$T_k(a) + cU_k(a) = (a + c)^k, \qquad T_k(a) - cU_k(a) = (a - c)^k.$$

Indeed the displayed step identities multiply the two forms by $a + c$ and $a - c$, respectively, and both start at one.

Take $a = 11/5$ and $c = 4\sqrt{6}/5$, and define

$$\lambda = \frac{11 + 4\sqrt{6}}{5}, \qquad \mu = \frac{11 - 4\sqrt{6}}{5}, \qquad W_k = \lambda^k + \mu^k.$$

Here $\lambda > 4$, $\mu > 0$, and $\lambda\mu = 1$, so

$$W_k = \lambda^k + \lambda^{-k} = 2T_k(11/5) > 0.$$

These sign and nonvanishing conditions hold for every $k \geq 0$, including $W_0 = 2$. The preceding normalized filter therefore satisfies

$$|H_k(x)| \leq \frac{2}{W_k} \qquad (1/2 \leq x \leq 4/3).$$

Theorem 21.24 (Exact spectral certificate)**.** *For the preceding observable $\mathcal{A}_{k,n}$, suppose $D_n(t) \leq C$ at one positive probe, $0 \leq C < r/4$, $r = p_n$. Then, for every $k \geq 0$,*

$$0 \leq S_n - \mathcal{A}_{k,n} \leq \frac{4S_n}{W_k^2} \leq \frac{16C}{W_k^2}.$$

The degree remains at most $2k$. If $r \geq 3$ and

$$32C < W_k^2,$$

there is a single $K \geq 0$ with $S_n = 2K$ such that every observation y with $|y - \mathcal{A}_{k,n}| < 1/2$ has $2K$ as its unique nonnegative even integer at distance less than one.

Proof. The exact characteristic identities prove the formula for W_k. The numerator of H_k has absolute value at most one on the retained window, so division by its positive normalizer $W_k/2$ gives the stated filter norm. Apply the same complete signed comparison, with its unchanged remainder

$$\sum_{i \in E_n} (p_i - r) H_k(r/p_i)^2 + \sum_{i \in L_n} (r - p_i) H_k(r/p_i)^2.$$

This proves $0 \leq S_n - \mathcal{A}_{k,n} \leq 4S_n/W_k^2$. The maintained estimate $S_n \leq 4C$ gives the last inequality.

Under $32C < W_k^2$, positivity of W_k^2 gives $16C/W_k^2 < 1/2$. The exceptional-prime argument is unchanged: $r - C > 2$, so $S_n = 2K$ for some $K \geq 0$. Combining the one-sided error with the observation noise gives $-1 < y - 2K < 1/2$, and separation of even integers proves uniqueness. The same K works before any observation is chosen. □

This is a condition on C and the chosen order, rather than a requirement to use the earlier worst-case order depending only on r. Since $\mu > 0$ and $\lambda > 4$, we have $W_k > 4^k$ for every $k \geq 0$. Thus the exact error coefficient $4/W_k^2$ is strictly smaller than the previously used $4\,16^{-k}$. Also $W_k^2 \geq \lambda^{2k}$, so a sufficient condition at an even degree $m = 2k$ is

$$\lambda^m > 32C.$$

For $C > 0$, any nonnegative even $m > \log(32C)/\log\lambda$ suffices. Uniformly over $C < r/4$, one can instead choose an even degree with

$$m > \frac{\log(8r)}{\log\lambda}, \qquad m \leq \frac{\log(8r)}{\log\lambda} + 2.$$

Its leading coefficient relative to $\log_2 r$ is $\log 2/\log\lambda < 1/2$. These real-logarithm choices and coefficient comparisons are ordinary consequences of the compiled exact formula and spectral signs. No minimum-degree or minimax assertion is made. The noise condition still concerns the combined observable; raw-moment measurement errors retain their coefficient weights.

The quantified exterior cone and both characteristic identities are compiled in `Mathlib/Analysis/SpecialFunctions/Log/ReciprocalChebyshevCore.lean`. The exact two-branch denominator, reciprocal-root identities, positivity, and exact filter norm are compiled in `Mathlib/Analysis/SpecialFunctions/Log/ReciprocalChebyshevFilter.lean`. The full signed spectral bound and the cost-adaptive noisy readout are compiled in `Helpers/PrimeProfileChebyshevReadout.lean`. The error estimate includes the first prime, whereas even-integer recovery explicitly assumes $p_n \geq 3$. Finite permutations and zero transport remain included. This proves a stronger finite observation certificate under the same arithmetic hypotheses, without a new arithmetic cost estimate or an improved RH boundary.

21.14 Nonlinear coercivity and the half-reference window

The preceding localization used a tangent estimate for each entering coordinate. Integrating the exact compensated kernel gives a stronger bound on the whole signed transport. Fix a cutoff $n \geq 0$ and put $r = p_n$, with the prefix containing exactly $n+1$ primes. The probe $t > 0$ is arbitrary and may be chosen independently at each cutoff. Write

$$B_t(x) = \log(1 + t/x) - t/x,$$
$$N_t(r; p, q) = \frac{r^2(r+t)}{t^2}\big(B_t(q) - B_t(p)\big),$$
$$f_r(d) = \frac{r}{2}\left(1 - \frac{r^2}{(r+d)^2}\right).$$

In the gauge f_r we require $r > 0$ and $d \geq 0$.

Theorem 21.25 (Full signed nonlinear coercivity)**.** *For every prime-coordinate permutation, every cutoff, and every positive probe, the complete displacement and normalized cost satisfy*

$$0 \leq S_n, \qquad f_r(S_n) \leq D_n(t).$$

Consequently, if $D_n(t) \leq C < r/2$, then

$$S_n \leq r\left(\frac{1}{\sqrt{1 - 2C/r}} - 1\right).$$

The hypotheses imply $C \geq 0$. Every denominator is positive on the stated domain.

Proof. The exact kernel is

$$B_t'(x) = \frac{t^2}{x^2(x+t)} \qquad (x > 0).$$

For an entering coordinate $q \geq r$, the function

$$F(x) = \frac{r^2(r+t)}{t^2}B_t(x) + \frac{r^3}{2x^2}$$

has derivative

$$F'(x) = \frac{r^2 t(x-r)}{x^3(x+t)} \geq 0 \qquad (r \leq x \leq q).$$

Comparing its endpoint values gives the probe-uniform inequality

$$N_t(r; r, q) \geq \frac{r}{2}\left(1 - \frac{r^2}{q^2}\right) = f_r(q - r).$$

For a leaving coordinate $0 < p \leq r$, the maintained exact lower comparison gives

$$N_t(r; p, r) \geq r\left(\frac{r}{p} - 1\right) \geq r - p \geq f_r(r - p).$$

The middle inequality follows from $r(r/p - 1) - (r - p) = (r - p)^2/p \geq 0$. For the last inequality, the exact identity

$$d - f_r(d) = \frac{d^2(3r + 2d)}{2(r + d)^2} \geq 0$$

holds for every $d \geq 0$; also $f_r(d) \geq 0$.

The gauge is subadditive. To see this without changing any signed main term, set

$$a = \frac{r}{r + x}, \quad b = \frac{r}{r + y}, \quad c = \frac{r}{r + x + y} \qquad (x, y \geq 0).$$

Then $0 \leq a, b, c \leq 1$ and

$$c - ab = \frac{rxy}{(r + x)(r + y)(r + x + y)} \geq 0.$$

Hence $c^2 \geq a^2b^2$, and

$$1 - c^2 \leq 1 - a^2b^2 = (1 - a^2) + (1 - b^2) - (1 - a^2)(1 - b^2) \leq (1 - a^2) + (1 - b^2).$$

Multiplying by $r/2$ proves $f_r(x + y) \leq f_r(x) + f_r(y)$; induction gives its finite-sum form.

Let

$$E_n = e(\{0, \ldots, n\}) \setminus \{0, \ldots, n\}, \qquad L_n = \{0, \ldots, n\} \setminus e(\{0, \ldots, n\}).$$

They have equal cardinality. Cancelling common coordinates and retaining the boundary reference gives the exact identities

$$S_n = \sum_{j \in E_n} (p_j - r) + \sum_{j \in L_n} (r - p_j),$$
$$D_n(t) = \sum_{j \in E_n} N_t(r; r, p_j) + \sum_{j \in L_n} N_t(r; p_j, r).$$

Each displacement is nonnegative. Apply finite subadditivity to the first identity and the scalar bounds to each reference cost in the second. This proves $0 \le f_r(S_n) \le D_n(t)$ and $S_n \ge 0$. No estimate of the magnitudes of the original signed terms is used.

Finally, $f_r(S_n) \le C < r/2$ implies

$$0 < 1 - \frac{2C}{r} \le \left(\frac{r}{r + S_n}\right)^2.$$

Both square roots are nonnegative, and $r + S_n > 0$, so

$$\sqrt{1 - 2C/r}\,(r + S_n) \le r.$$

Division by the strictly positive square root proves the displayed localization. Nonnegativity of the gauge also gives $C \ge 0$. □

The finite-set theorem is more general than the prime specialization: it applies to any two equal-cardinality finite sets whose retained coordinates lie on the stated sides of a positive reference. Only the leaving coordinates need an explicit positivity hypothesis; the entering coordinates are automatically positive.

This extends the available localization domain from $C < r/4$ to $C < r/2$. Within the former domain the new upper bound approaches $r(\sqrt{2} - 1)$ as C increases to $r/4$, whereas the old bound $S_n \le 4C$ approaches r. The new theorem includes the first prime $r = 2$, every finite permutation, and zero displacement. It makes no even-integer readout assertion at the exceptional first prime and does not assume an eventual prime-ratio estimate. The boundary prime is p_n throughout, while both finite prefixes have length $n + 1$; there is no shift to p_{n+1} in the cost.

The exact derivative, integrated reference costs, subadditive gauge, finite signed assembly and inverse bound are compiled in `Mathlib/Analysis/SpecialFunctions/Log/CompensatedReciprocalCoercivity.lean`. The prime-profile coercivity and half-reference localization are compiled in `Helpers/PrimeProfileNonlinearCoercivity.lean`. The public export is `Assembly/ProbabilityProfiles.lean`. This is a stronger estimate of the finite probability-profile cost. It does not estimate the outstanding signed Möbius packet, and the certified unconditional RH boundary remains 7/8.

21.15 Rational characteristic acceleration from the integrated window

The integrated reference costs also sharpen the retained coordinate window. Keep $D_n(t) \le C$, $0 \le C < r/4$, and $r = p_n$. For each entering prime q, its individual reference cost is at most the full cost, so

$$\frac{r}{2}\left(1 - \frac{r^2}{q^2}\right) \le C < \frac{r}{4}, \qquad \left(\frac{r}{q}\right)^2 > \frac{1}{2}.$$

For each leaving prime p,

$$r\left(\frac{r}{p} - 1\right) \le C < \frac{r}{4}, \qquad \frac{r}{p} < \frac{5}{4}.$$

The entering ratios are at most one, and the leaving ratios are at least one. Since $(45/64)^2 = 2025/4096 < 1/2$, every retained ratio therefore belongs to

$$J = [45/64, 5/4].$$

This assertion concerns precisely $E_n \cup L_n$ after cancellation. It makes no claim about a common small prime.

The rational left endpoint is chosen to give exact characteristic roots. The affine map and its exterior parameters are

$$z(x) = \frac{125 - 128x}{35}, \qquad a = z(0) = \frac{25}{7}, \qquad c = \frac{24}{7}.$$

The map sends J to $[-1, 1]$, while $c^2 = a^2 - 1$, $a + c = 7$, and $a - c = 1/7$. Define

$$V_k = 7^k + 7^{-k}, \qquad H_k(x) = \frac{2T_k(z(x))}{V_k} \qquad (k \ge 0).$$

The characteristic identities already proved above give $V_k = 2T_k(25/7) > 0$. In particular $H_k(0) = 1$ and $\deg H_k \le k$. The retained-window bound is

$$|H_k(x)| \le \frac{2}{V_k} \le 2\,7^{-k} \qquad (x \in J).$$

Both branches are retained; $V_0 = 2$ and $H_0 = 1$.

Use the same signed observable

$$A_{k,n} = \mathrm{Def}_{e,n+1}\big((p - r)(1 - H_k(r/p)^2)\big),$$

where $\mathrm{Def}_{e,n+1}$ denotes the permuted finite prefix sum minus the original finite prefix sum.

Theorem 21.26 (Rational characteristic readout)**.** *Under the stated quarter-budget hypotheses, for every $k \geq 0$ the observable $A_{k,n}$ is a probability-moment combination of degree at most $2k$, and*

$$0 \leq S_n - A_{k,n} \leq \frac{4S_n}{V_k^2} \leq \frac{16C}{V_k^2},$$
$$S_n - A_{k,n} \leq 4\,49^{-k} S_n \leq 16C\,49^{-k}.$$

If $r \geq 3$ and $32C < V_k^2$, there is a single $K \geq 0$ with $S_n = 2K$ such that every observation satisfying $|y - A_{k,n}| < 1/2$ has $2K$ as its unique nonnegative even integer at distance less than one. A uniform sufficient integer order is

$$k = \lfloor \log_{49}(8p_n) \rfloor + 1.$$

The Lean statement uses the natural-number logarithm at the positive integer argument $8p_n$.

Proof. The exact kernel comparisons prove the window as just shown. The coupled rotation invariant bounds $|T_k(z)|$ by one for $-1 \leq z \leq 1$, and the exterior characteristic identity gives the strictly positive denominator V_k. This proves the stated polynomial norm and normalization at zero.

The normalized square-filter representation supplies a polynomial G_k of degree at most $2k$ with

$$(p - r)(1 - H_k(r/p)^2) = rG_k(r/p).$$

Thus $A_{k,n} = r\,\mathrm{Def}_{e,n+1}(G_k(r/p))$. Only the retained coordinates are used in estimating its error:

$$S_n - A_{k,n} = \sum_{j \in E_n} (p_j - r) H_k(r/p_j)^2 + \sum_{j \in L_n} (r - p_j) H_k(r/p_j)^2.$$

Every displayed displacement is nonnegative, and the principal coefficient remains one. The support-only comparison therefore gives $0 \leq S_n - A_{k,n} \leq 4S_n/V_k^2$. The previously compiled full signed estimate $S_n \leq 4C$ proves the first cost bound. Since $V_k \geq 7^k$, squaring gives $V_k^2 \geq 49^k$ and proves both geometric bounds.

For recovery, $r \geq 3$ and $C < r/4$ give $r - C > 2$. All retained primes are odd, so the maintained parity theorem gives $S_n = 2K$ for some nonnegative integer K. The condition $32C < V_k^2$ makes the full signed observation error less than $1/2$. Combining it with $|y - A_{k,n}| < 1/2$ puts y at distance less than one from $2K$. Two different even integers cannot both be at distance

less than one from the same y, proving uniqueness with the same K for all admissible observations.

Finally the chosen natural logarithm gives $49^k > 8r > 32C$, hence $V_k^2 > 32C$. This proves the sufficient order without a finite sampling or growing-cutoff assumption. □

At matched degree $m = 2k$, the geometric error factor changes from $4\,16^{-k}$ to $4\,49^{-k}$. The explicit sufficient moment degree has the ordinary real-logarithm bound

$$m \le \frac{\log r}{\log 7} + \frac{3\log 2}{\log 7} + 2.$$

Its leading coefficient relative to $\log_2 r$ is $\log 2/\log 7$, compared with $1/2$ for the earlier geometric certificate. These degree comparisons are ordinary consequences of the compiled natural-logarithm choice; no minimum-degree or minimax claim is made. The exact formula retains the contracting branch and can require less than this uniform sufficient order.

The integrated scalar window is compiled in `Mathlib/Analysis/SpecialFunctions/Log/CompensatedReciprocalWindow.lean`; its two-support prime specialization is in `Helpers/PrimeProfileReciprocalWindow.lean`. The exact rational filter and all normalization and norm certificates are in `Mathlib/Analysis/SpecialFunctions/Log/ReciprocalSevenFilter.lean`. The reusable signed comparison requiring a norm only on the retained supports is in `Helpers/PrimeProfileSupportedFilter.lean`. The degree, exact and geometric full error, cost-adaptive recovery, and explicit logarithmic recovery are compiled in `Helpers/PrimeProfileSevenReadout.lean`, publicly exported by `Assembly/PrimeProfileReadouts.lean`. The error theorem includes $r = 2$; the even-integer recovery explicitly requires $r \ge 3$. Every finite permutation and zero transport remain included. The noise allowance concerns the combined observable, with the previously stated coefficient weights on raw moment errors. This proves a stronger finite observation theorem under the same cost hypothesis. It supplies no independent bound on the open signed Möbius packet; the unconditional RH boundary remains 7/8.

21.16 The exact locality threshold for one fixed probe

The universal half-reference estimate can be complemented by an exact threshold when the positive probe itself is fixed. Fix $n \geq 0$, $r = p_n$, and $t > 0$. Define

$$c_t(r) = \frac{r^2(r+t)}{t^2},$$
$$K_t(r) = c_t(r)\left(\frac{t}{r} - \log(1 + t/r)\right).$$

The threshold is strictly positive. Throughout this subsection n and t are held fixed while the coordinate permutation varies.

Theorem 21.27 (Exact fixed-probe locality threshold)**.** *For every real budget C, the following are equivalent:*

$$\textit{There is an integer } M \textit{ such that, for every permutation } e,$$
$$D_n(t) \leq C \implies e(\{0, \dots, n\}) \subset \{0, \dots, M-1\};$$
$$C < K_t(r).$$

More quantitatively, if $D_n(t) \leq C < K_t(r)$, every entering prime $q = p_j$ satisfies

$$(K_t(r) - C)\, q(q+t) \leq r^2(r+t).$$

At every budget $C \geq K_t(r)$ the failure is already witnessed by finite transpositions: for every real B and integer M, a swap of n with some $j > \max(n, M)$ has

$$S_n > B, \qquad D_n(t) < K_t(r) \leq C.$$

The same fixed t works for all these witnesses.

Proof. For $u > 0$, the strict logarithmic tangent inequality and its reciprocal application give

$$0 < u - \log(1+u) \leq \frac{u^2}{1+u}.$$

Indeed $\log(1+u) < u$. Also $\log(1/(1+u)) \leq 1/(1+u) - 1$, hence $\log(1+u) \geq u/(1+u)$, which proves the upper bound. Taking $u = t/r$ proves $K_t(r) > 0$.

For any positive coordinate q, exact cancellation gives

$$K_t(r) - N_t(r; r, q) = c_t(r)\left(\frac{t}{q} - \log(1 + t/q)\right),$$
$$0 < K_t(r) - N_t(r; r, q) \leq \frac{r^2(r+t)}{q(q+t)}.$$

This keeps the full compensated logarithmic expression. For an entering coordinate, the maintained signed reference decomposition gives $N_t(r; r, q) \leq D_n(t) \leq C$. Put $\delta = K_t(r) - C > 0$. Then

$$\delta \leq K_t(r) - N_t(r; r, q) \leq \frac{r^2(r+t)}{q(q+t)}.$$

Multiplying by the strictly positive denominator proves the quantitative coordinate constraint.

Let $A = r^2(r+t)$. Since a prime $q \geq 2$ and $t > 0$ have $q + t > 1$, the same constraint gives $\delta q \leq A$ and thus $q \leq A/\delta$. The zero-based prime enumeration satisfies $j \leq p_j = q$. Choose an integer

$$M > \max\left(n+1, \frac{A}{\delta}\right).$$

An image coordinate common to the original prefix has index at most $n < M$. Every other image coordinate is entering, and its index satisfies $j \leq q \leq A/\delta < M$. This proves the finite ambient prefix assertion, uniformly over all permutations satisfying the same budget. The argument also covers negative C, when applicable cost constraints may be vacuous.

For the converse, fix any $B \in \mathbb{R}$ and $M \geq 0$. Choose an integer $Q > B + r$ and put $j = Q + n + M + 1$. Then $j > \max(n, M)$ and $p_j \geq j \geq Q$. The permutation exchanging exactly n and j has the exact crossing-prefix identities

$$S_n = p_j - r > B, \qquad D_n(t) = N_t(r; r, p_j).$$

The positive residual already proved gives $N_t(r; r, p_j) < K_t(r)$. This constructs arbitrarily large displacement and entering index with the same fixed probe. If $C \geq K_t(r)$, each such swap satisfies $D_n(t) \leq C$. Its image prefix contains $j > M$, contradicting any proposed uniform ambient prefix of length M. Thus finite uniform localization implies $C < K_t(r)$. □

The residual estimate also shows, as an ordinary consequence, that the swap costs approach $K_t(r)$ from below with error at most $r^2(r+t)/(p_j(p_j + t))$, while their displacements tend to infinity. The sharpness assertion concerns the budget value for one fixed probe and cutoff. It does not change the probe with the escaping coordinate. In the converse, the counterexamples are finite permutations; no assertion restricted to permutations that are not finite is used.

This gives an exact criterion for reducing reconstruction from a single cost inequality to a finite search over possible prefix images. The complete

probability profile and its existing inverse theorems supply additional information beyond that one inequality. All statements here include the first prime $r = 2$, use a prefix of length $n + 1$, and keep the boundary reference p_n.

The strict logarithmic deficit, positive saturation, complete rational residual and coordinate constraint are compiled in `Mathlib/Analysis/SpecialFunctions/Log/CompensatedReciprocalSaturation.lean`. The crossing transposition identity is compiled in `Mathlib/Combinatorics/FinitePrefixTransposition.lean`. The full-cost incoming bound, uniform ambient-prefix theorem, unbounded finite-swap witnesses and exact equivalence are compiled in `Helpers/PrimeProfileSaturationThreshold.lean`, exported by `Assembly/ProbabilityProfiles.lean`. The displayed asymptotic consequence follows from the compiled residual and unbounded prime enumeration; it is not a separately compiled limit declaration. No assertion of external priority or optimal moment degree is made. The outstanding signed arithmetic estimate and certified RH boundary 7/8 are unchanged.

21.17 The sharp locality threshold uniformly over positive probes

The order of quantifiers changes when a single ambient prefix must work for every positive probe. The preceding fixed-probe theorem keeps t fixed; the next theorem requires the same M for all $t > 0$. Again $n \geq 0$ is fixed, $r = p_n$, and the observed prefix has length $n + 1$.

Theorem 21.28 (Exact probe-uniform locality threshold)**.** *For every real C, the following are equivalent:*

There is an integer M such that, for every $t > 0$ and permutation e,

$$D_n(t) \leq C \implies e(\{0, \dots, n\}) \subset \{0, \dots, M-1\};$$

$$C < \frac{r}{2}.$$

In the subcritical case, every entering prime q satisfies the probe-independent quadratic constraint

$$(r - 2C)q^2 \leq r^3.$$

At the critical budget itself, finite transpositions already give unbounded displacement. More precisely, for every real B and integer M, choose $j > \max(n, M)$ with $q = p_j$ and $q - r > B$. The positive probe

$$t_{r,q} = \frac{r^3}{2(q^2 - r^2)}$$

and the swap of n with j satisfy

$$S_n = q - r > B,$$

$$D_n(t_{r,q}) \leq \frac{r}{2} - \frac{r^3}{4q^2} < \frac{r}{2}.$$

Thus the failure at every $C \geq r/2$ uses finite permutations and strictly positive probes, with no limiting probe admitted as an observation.

Proof. We first prove the upper bound complementary to the maintained integrated lower bound. For $t > 0$ and $q \geq r > 0$, put

$$c = \frac{r^2(r+t)}{t^2}, \qquad a = \frac{r^2(r+t)}{2}, \qquad F(x) = -cB_t(x) - \frac{a}{x^2}.$$

The exact derivative $B_t'(x) = t^2/(x^2(x+t))$ gives

$$F'(x) = \frac{r^2(r+t)t}{x^3(x+t)} > 0 \qquad (x > 0).$$

Consequently $F(r) \le F(q)$, which rearranges to

$$\begin{aligned} N_t(r;r,q) &\le \frac{a}{r^2} - \frac{a}{q^2} \\ &= \frac{r+t}{2}\left(1 - \frac{r^2}{q^2}\right). \end{aligned}$$

Together with the integrated lower bound this gives the two-sided, fully compensated estimate

$$\frac{r}{2}\left(1 - \frac{r^2}{q^2}\right) \le N_t(r;r,q) \le \frac{r+t}{2}\left(1 - \frac{r^2}{q^2}\right).$$

Suppose first that $D_n(t) \le C < r/2$. In the exact signed reference decomposition,

$$D_n(t) = \sum_{j \in E} N_t(r;r,p_j) + \sum_{i \in L} N_t(r;p_i,r),$$

the common coordinates have canceled, and every retained reference cost is nonnegative. Thus an entering q has $N_t(r;r,q) \le C$. The lower integrated bound yields

$$\frac{r}{2}\left(1 - \frac{r^2}{q^2}\right) \le C, \qquad (r-2C)q^2 \le r^3.$$

All denominators multiplied here are positive. Write $\delta = r - 2C > 0$. Since $q \ge 2$, we have $q \le q^2$, so $q \le r^3/\delta$. Choose once and for all an integer

$$M > \max\left(n+1, \frac{r^3}{r-2C}\right).$$

A coordinate retained from the original prefix has index at most $n < M$. Every entering coordinate has $j \le p_j = q \le r^3/\delta < M$. This M depends only on n and C, and therefore works simultaneously for every $t > 0$ and every admissible permutation.

For the converse, let B and M be arbitrary. The unbounded prime enumeration supplies $j > \max(n, M)$ with $q = p_j$ and $q - r > B$. In particular $q > r$, so $t_{r,q} > 0$. Its exact substitution into the upper bound gives

$$\begin{aligned} N_{t_{r,q}}(r;r,q) &\le \frac{r + t_{r,q}}{2}\left(1 - \frac{r^2}{q^2}\right) \\ &= \frac{r}{2} - \frac{r^3}{4q^2} < \frac{r}{2}. \end{aligned}$$

The crossing transposition has exactly this normalized cost and displacement $q - r$. For $C \geq r/2$ it is budget-admissible, but its image prefix contains $j > M$. Since M was arbitrary, no common finite ambient prefix exists. This also treats the critical equality $C = r/2$. □

The theorem proves that the earlier half-reference condition for localization is sharp when the positive probe is allowed to vary. In a fixed-probe experiment the exact threshold remains $K_t(r)$ from the preceding subsection. Both results concern the information supplied by one cost-budget inequality; the complete profile inverse theorem has additional information. The small prime $r = 2$, cutoff $n = 0$, and finite permutations are included without parity restrictions or a rank shift.

The integrated upper bound, positivity of the explicit escaping probe, and quantitative half-reference margin are compiled in `Mathlib/Analysis/SpecialFunctions/Log/CompensatedReciprocalUpperCost.lean`. The probe-independent coordinate bound, common finite-prefix theorem, unbounded finite-swap witnesses and exact equivalence are compiled in `Helpers/PrimeProfileUniformProbeLocality.lean`, exported by `Assembly/ProbabilityProfiles.lean`. The earlier integrated lower bound and exact fixed-probe theorem are consumed directly. No external-priority claim is made. The unresolved signed arithmetic estimate and certified RH boundary 7/8 are unchanged.

21.18 Rational residue-class laws: a stronger inverse theorem

The preceding inverse theorem extends beyond a common leading coefficient. Fix an integer $Q \geq 1$. Suppose that outside finitely many primes the laws have the form $\eta_1(p) = R_a(p)$ and $\eta_2(p) = S_a(p)$ for $p \equiv a \pmod{Q}$, where a runs over the units modulo Q. Each R_a, S_a is a rational function with rational coefficients. Assume each is either identically zero or has an expansion $\nu/x + \kappa/x^2 + O(x^{-3})$ with rational $\nu > 0$. The actual probabilities are in $[0, 1]$.

Theorem 21.29 (Residue-rational profile rigidity)**.** *Equality of the entire profiles implies $R_a = S_a$ as rational functions for every unit class a. The complete laws then differ by a finite permutation of prime coordinates. More precisely, if one active R_a differs from S_a, there is a unit progression $r \pmod{M}$, refining $a \pmod{Q}$, such that for every sufficiently large prime*

$p \equiv r \pmod{M}$ *the value* $R_a(p)$ *is absent from the entire second law. Consequently the first profile has infinitely many zeros absent from the second. Using the prime number theorem in a fixed progression, their number in* $[-T, 0]$ *is at least* $(\nu/\varphi(M) + o(1))T/\log T$*. If only the second law is active in a discrepant class, interchange the two laws.*

Proof. Write $R_a(x) = \nu/x + \kappa/x^2 + O(x^{-3})$. For each active target class b, write $S_b(x) = \mu_b/x + \lambda_b/x^2 + O(x^{-3})$. If $R_a(p) = S_b(q)$ and p is large, then q is large and comparable to p: finitely many small target primes have probabilities bounded away from the positive values tending to zero under consideration. Taking reciprocals gives

$$q = \frac{\mu_b}{\nu} p + \left(\frac{\lambda_b}{\mu_b} - \frac{\mu_b \kappa}{\nu^2} \right) + O(p^{-1}).$$

Both displayed coefficients are rational. The error belongs to a fixed rational lattice because p, q are integers, so it is eventually zero. Thus every large equality satisfies an exact affine equation $C_b q = A_b p + B_b$, with positive integers A_b, C_b and integer B_b.

If $B_b = 0$, primeness and sufficiently large p, q force $q = p$ and $A_b = C_b$. Then $b = a$, and $R_a(p) = S_a(p)$ has only finitely many solutions unless the two rational functions are identical. For each remaining equation, $B_b \neq 0$. Choose a distinct auxiliary prime ℓ_b not dividing $QA_bB_bC_b$, and impose $p \equiv -B_b A_b^{-1} \pmod{\ell_b}$. This is a nonzero residue. Together with $p \equiv a \pmod{Q}$, the Chinese remainder theorem gives one unit class modulo $M = Q \prod_b \ell_b$. Each affine equality on this class would make $\ell_b \mid q$, impossible once the prime $q > \ell_b$. This excludes every target class simultaneously; the finite exceptional target primes were already removed.

Dirichlet's theorem supplies infinitely many primes in the chosen unit progression. Their probabilities are eventually distinct, since $R_a(p)^{-1} = p/\nu - \kappa/\nu^2 + O(p^{-1})$ is strictly increasing on large prime arguments. They therefore give distinct exclusive zeros. Their cutoff is $p \leq \nu T + O(1)$, which yields the stated count from the fixed-progression prime number theorem. Equal profiles cannot have such exclusive zeros. Hence all rational functions agree, the labelled tails agree, and cancellation of the common tail gives the finite-permutation conclusion. □

This is an ordinary proof; its finite affine-exclusion and residue-rational classification have not yet been formalized in Lean. It applies only after a family's rational local laws and exceptional prime set have been established. It does not assert such a transfer for every polynomial, marked family, or Dirichlet L-function.

21.19 The completed signed-packet calculus and its open consumer

For a finite $S \subset \{1, \dots, N\}$ and complex coefficients a, b, put $A_d = \sum_{n \in S,\ d \mid n} a_n$ and $B_d = \sum_{n \in S,\ d \mid n} b_n$. The maintained signed covariance identity is exactly

$$\sum_{\substack{n,m \in S \\ (n,m)=1}} (a_n - cb_n)\overline{(a_m - cb_m)} = \sum_{d=1}^{N} \mu(d) |A_d - cB_d|^2.$$

Indeed $\sum_{d \mid (n,m)} \mu(d)$ is the coprimality indicator, and finite interchange of sums proves the identity. Each divisor row retains the full expansion

$$|A_d - cB_d|^2 = |A_d|^2 - cB_d\overline{A_d} - \overline{c}A_d\overline{B_d} + |c|^2 |B_d|^2.$$

In particular, if b is supported on primes larger than D, then $B_d = 0$ for every $2 \le d \le D$. Such a prime-only compensator changes none of these low rows, regardless of the complex coefficient c. This rules out that specific compensation mechanism on those rows; it is not an obstruction to other coupled compensators.

There is also an exact partition using every prime allowed by the finite support, rather than a chosen finite set. For $v_1 = 0$, define, for $1 \le n \le N$ and prime $p \le N$,

$$v_p(n) = \begin{cases} \mu(n) v_n \dfrac{\log p}{\log n}, & p \mid n,\ p \text{ prime}, \\ 0, & \text{otherwise}. \end{cases}$$

Only $n > 1$ enters the first branch. Since $\mu(n) = 0$ off the squarefree numbers, $\sum_{p \le N} v_p(n) = \mu(n) v_n$ for $1 \le n \le N$. The coprime form of two blocks with the same prime index vanishes exactly: both supported arguments are divisible by that prime. Writing $\mathcal{C}(f, g) = \sum_{n,m \in S,\,(n,m)=1} f_n \overline{g_m}$, this gives the exact identity

$$\mathcal{C}(\mu v, \mu v) = \sum_{\substack{p,q \le N \\ p,q \text{ prime},\ p \ne q}} \mathcal{C}(v_p, v_q).$$

The coefficient v may itself retain a full compensation, so this identity does not erase its cross terms. It does not estimate their joint cancellation.

The reconciliation inventory `paper/completed-research-inventory.json` and the public theorem-audit inventory record the maintained owners for these covariance identities, full-prime partitions, Gaussian principal

and Poisson inputs, Euler-continuation results, detector adapters, and quantitative tail lemmas. Their public export is `PrimeFactorOscillations/Assembly/CompletedDetectorPackets.lean`. A copied source is not certified by this list: the inventory separately records the fresh compilation and axiom-audit status. An earlier 20 October reconciliation checked 1,139 owned Lean modules and 4,843 selected theorem closures with standard foundations only; both the public and canonical QRH statement comparisons passed. That reconciliation identified 301 maintained owners and transferred all 132 proved private graph nodes to maintained sources. A later 1,150-module promotion covered the identity, finite-permutation and prime-coordinate inverse consumers. Subsequent maintained owners add the full adaptive cutoff inverse and exact optimal-distortion theorem, with separate build and axiom receipts. The residue-rational classification and continuum-cardinality assertion remain ordinary proofs. The signed arithmetic estimate for an improved RH boundary remains open.

The unfinished analytic consumer remains explicitly open. In the fixed-mask notation of that programme it asks, for each $\varepsilon > 0$ and each fixed admissible G, for constants $C_{\varepsilon,G}, D_{0,\varepsilon,G}$ such that

$$|I_{D,G}| \le C_{\varepsilon,G} D^{3/2+\varepsilon} \quad (D \ge D_{0,\varepsilon,G}).$$

No uniformity for a growing G is implicit. The identities above do not prove this estimate. Detector precision, zero-signal preservation and smaller separate remainders are recorded as structural results; the certified unconditional boundary remains 7/8. Failed absolute-value reductions and restricted-range attempts retain their failure records and are not counted as improvements of that boundary.

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